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(54) **WEARABLE FACIAL MOVEMENT
TRACKING DEVICES**

(58) **Field of Classification Search**

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(71) Applicant: **CORNELL UNIVERSITY**, Ithaca, NY
(US)

(56)

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(72) Inventors: **Cheng Zhang**, Ithaca, NY (US);
Tuochao Chen, Ithaca, NY (US);
Benjamin Steeper, Ithaca, NY (US);
Francois Guimbretiere, Ithaca, NY
(US); **Ke Li**, Suzhou (CN); **Ruidong
Zhang**, Lianyungang (CN); **Hyunchul
Lim**, Ithaca, NY (US)

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(73) Assignee: **Cornell University**, Ithaca, NY (US)

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Primary Examiner — Nimesh Patel

(74) *Attorney, Agent, or Firm* — Ryan, Mason & Lewis,
LLP

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ABSTRACT

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This technology provides systems and methods for tracking facial movements and reconstructing facial expressions by learning skin deformation patterns and facial features. Frontal view images of a user making a variety of facial expressions are acquired to create a data training set for use in a machine-learning process. Head-mounted or neck-mounted wearable devices are equipped with one or more camera(s) or acoustic device(s) in communication with a data processing system. The cameras capture images of contours of the users face from either the cheekbone or the chin profile of the user. The acoustic devices transmit and receive signals to calculate a representation of the skin deformation. A data processing system uses the images, the profile of the contours, or skin deformation to track facial movement or to reconstruct facial expressions of the user based on the data training set.

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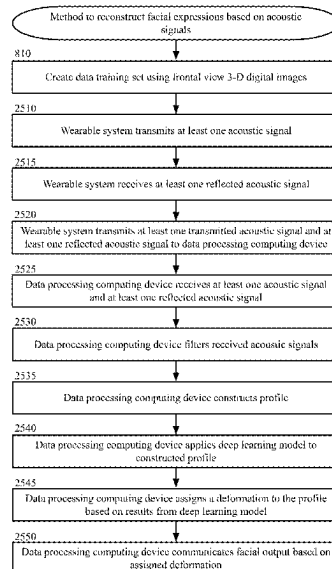
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See application file for complete search history.

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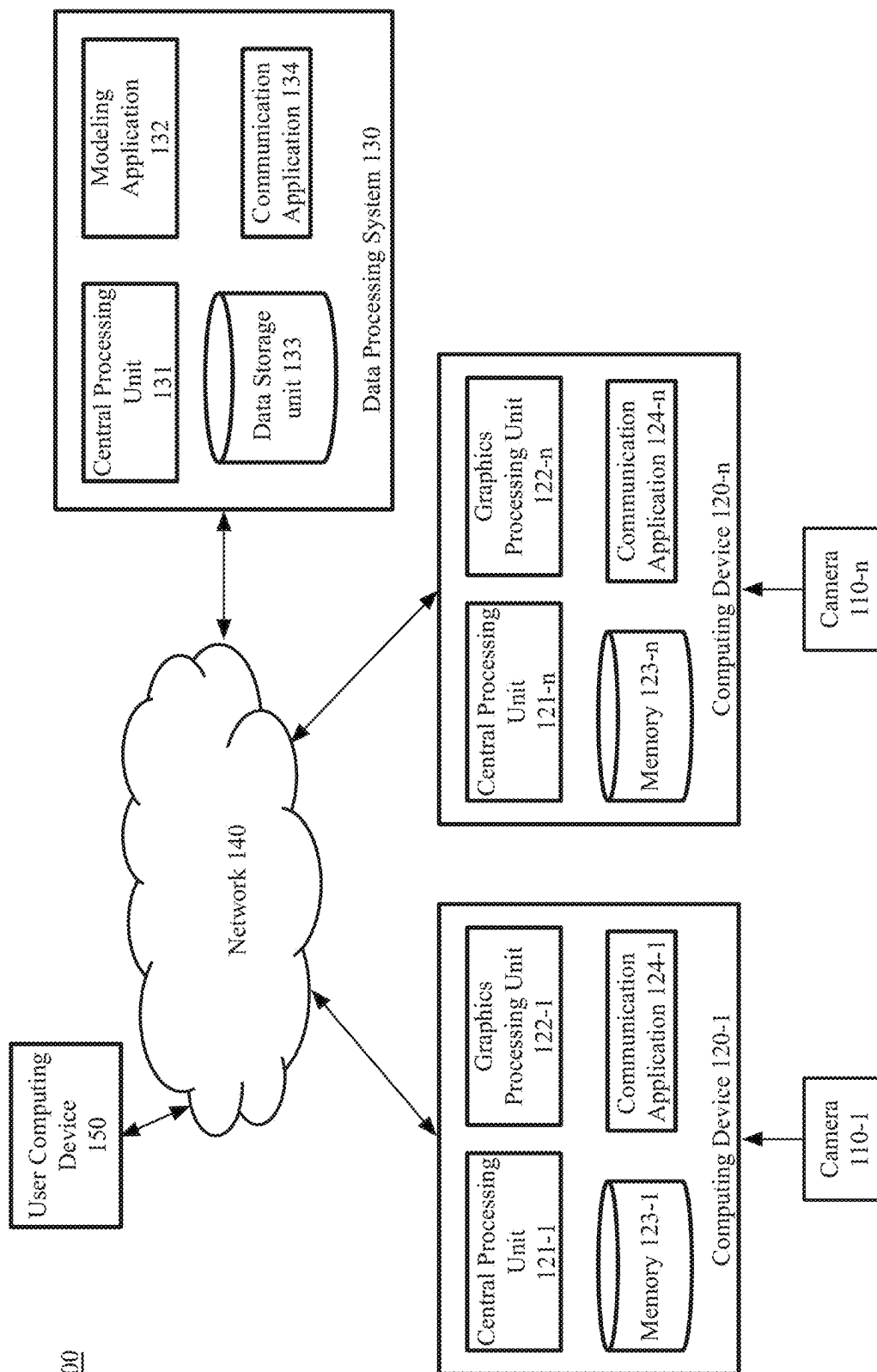


Figure 1

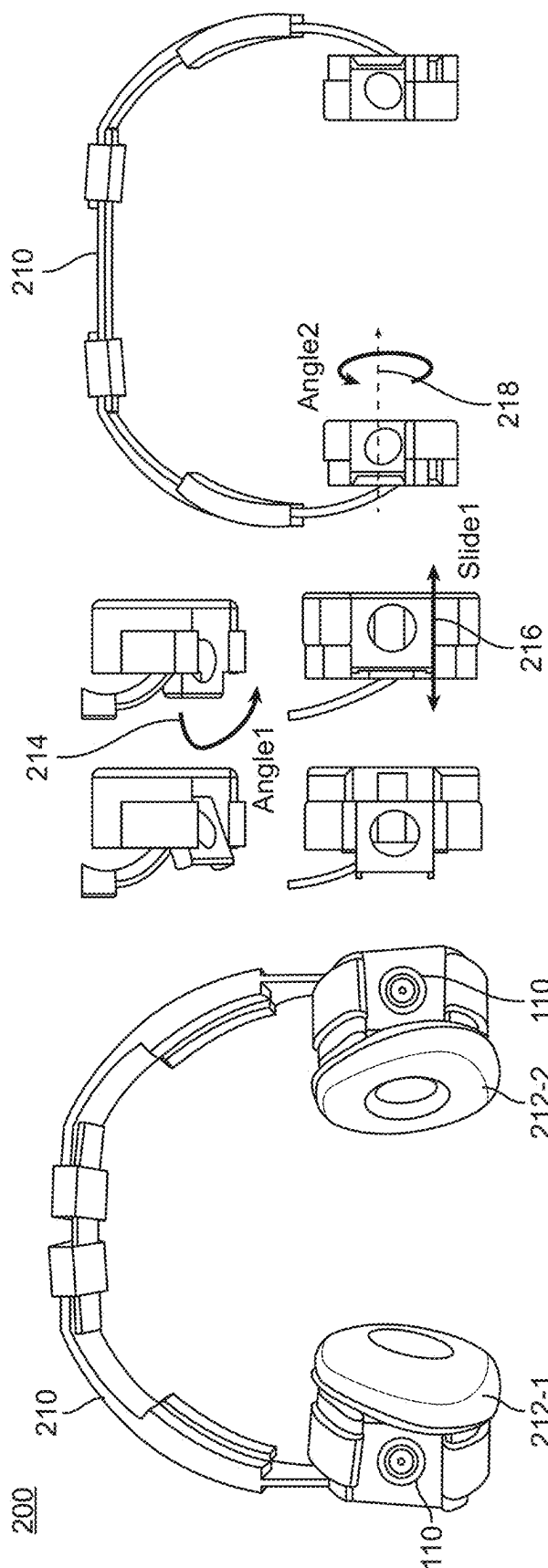


Figure 2A

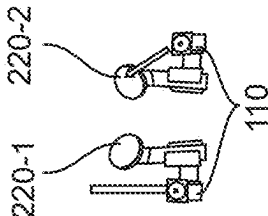


Figure 2B

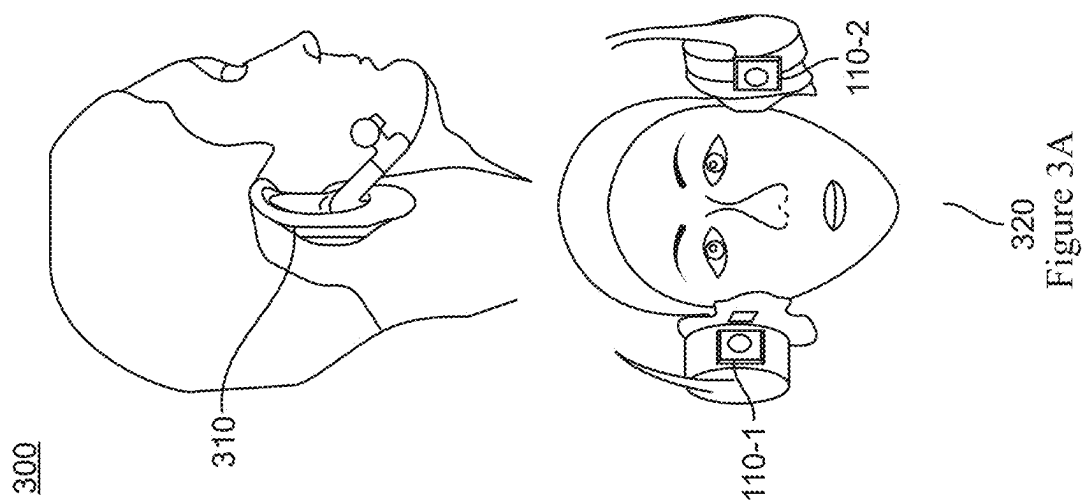


Figure 3B

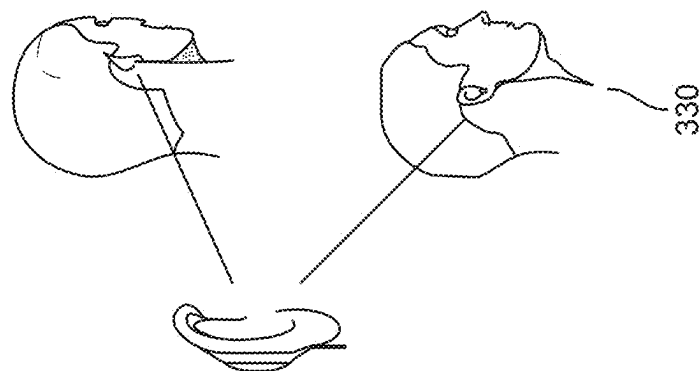
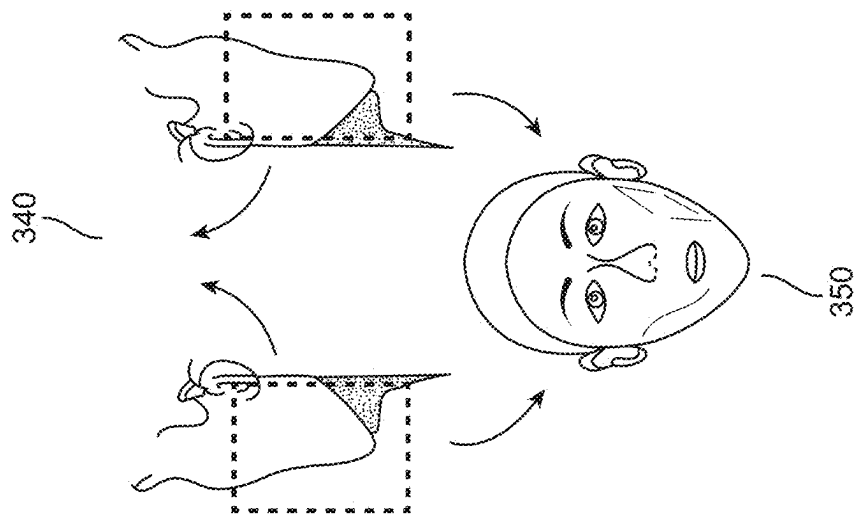


Figure 3C



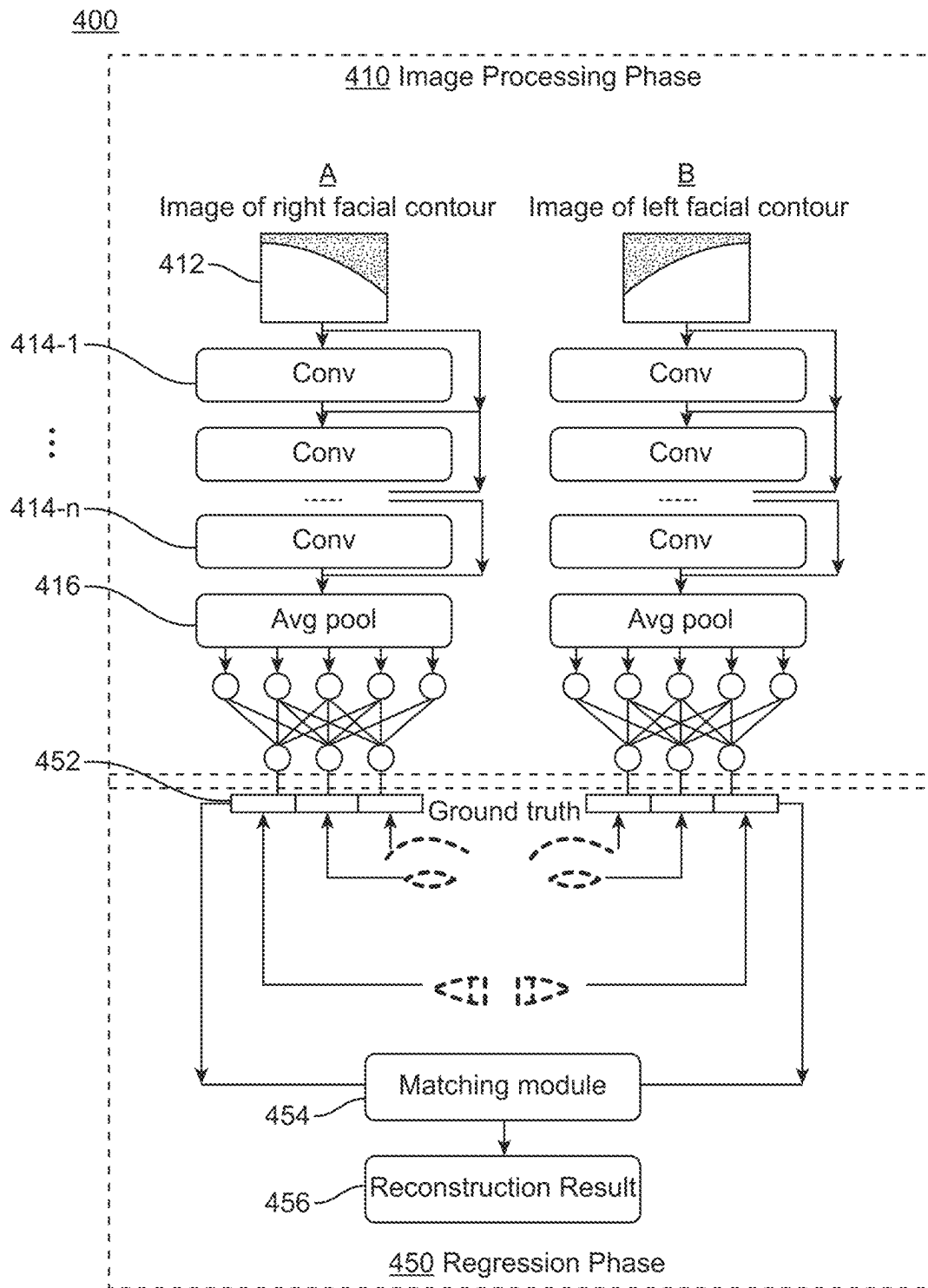


Figure 4

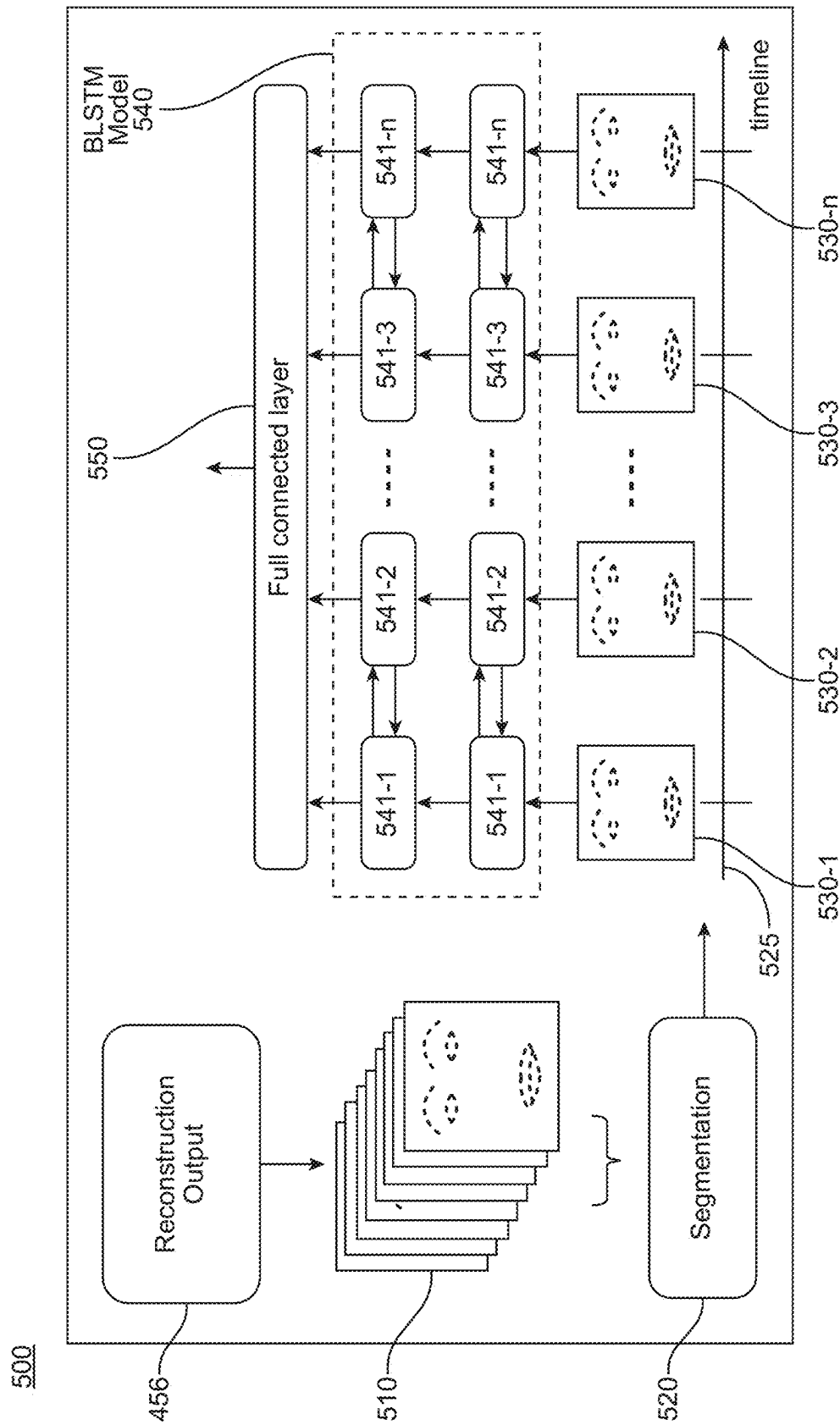


Figure 5

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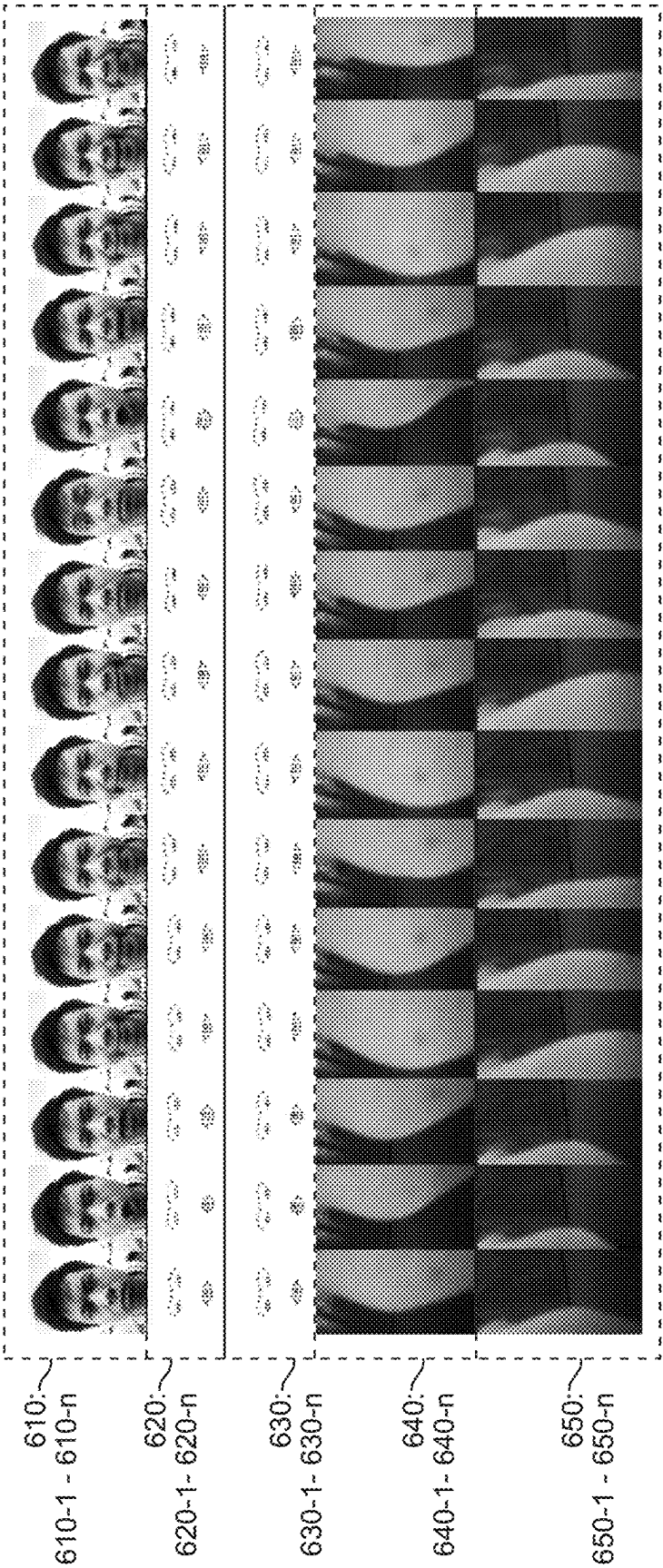


Figure 6

700

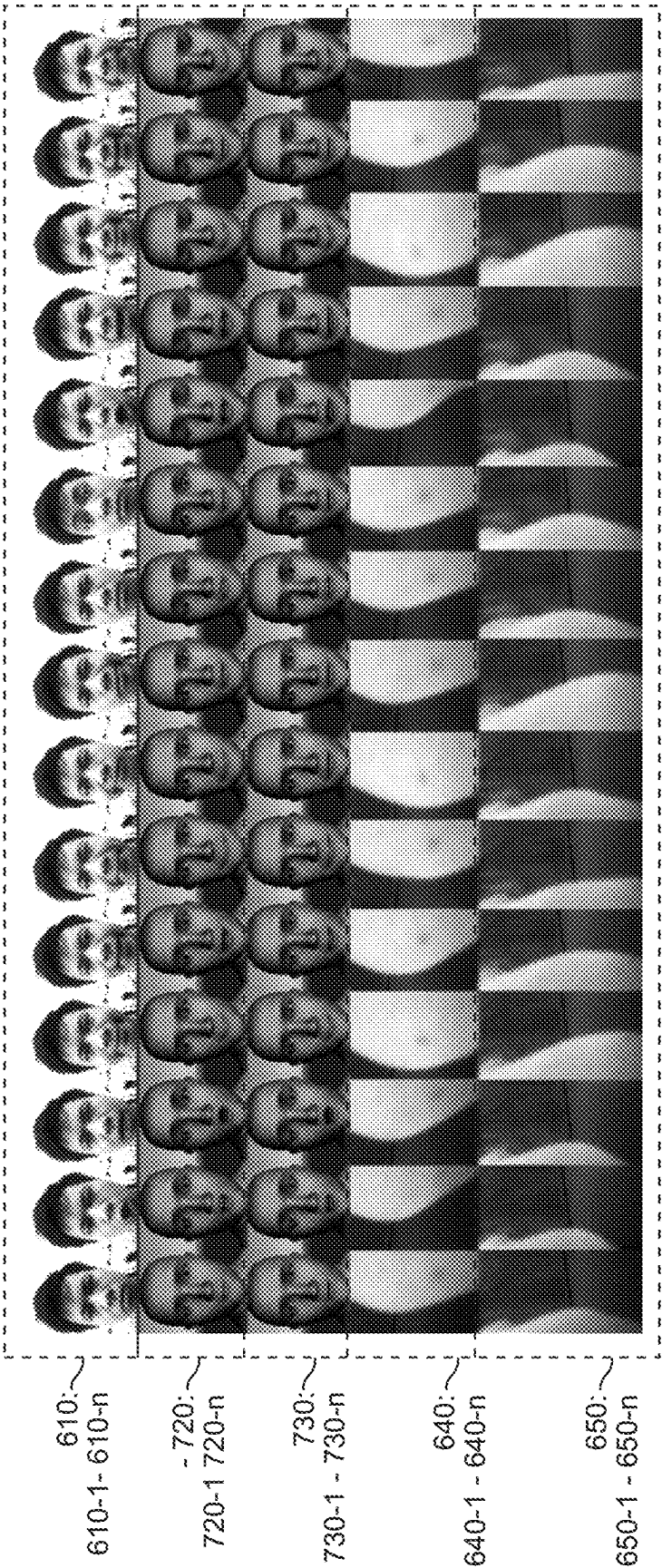


Figure 7

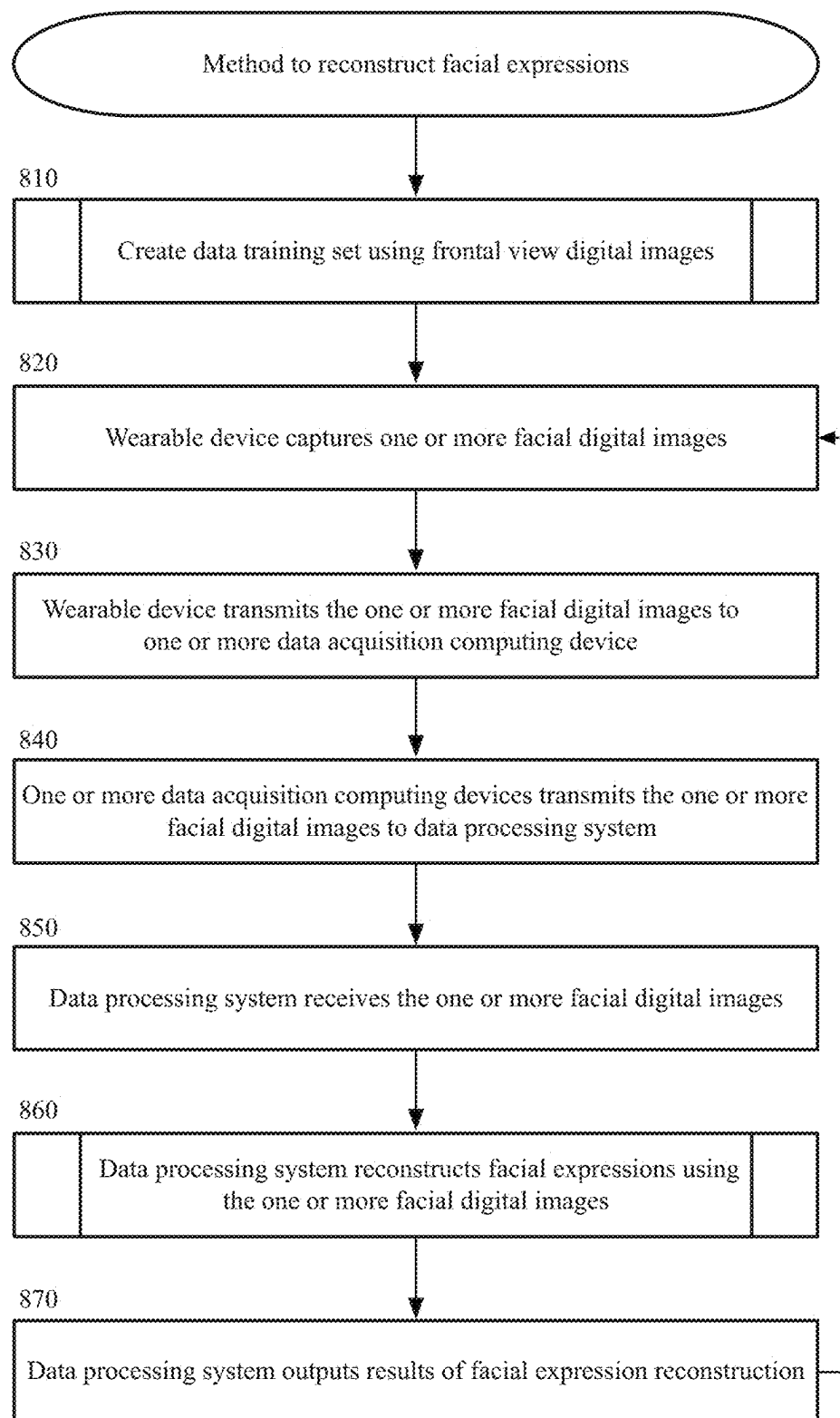
800

Figure 8

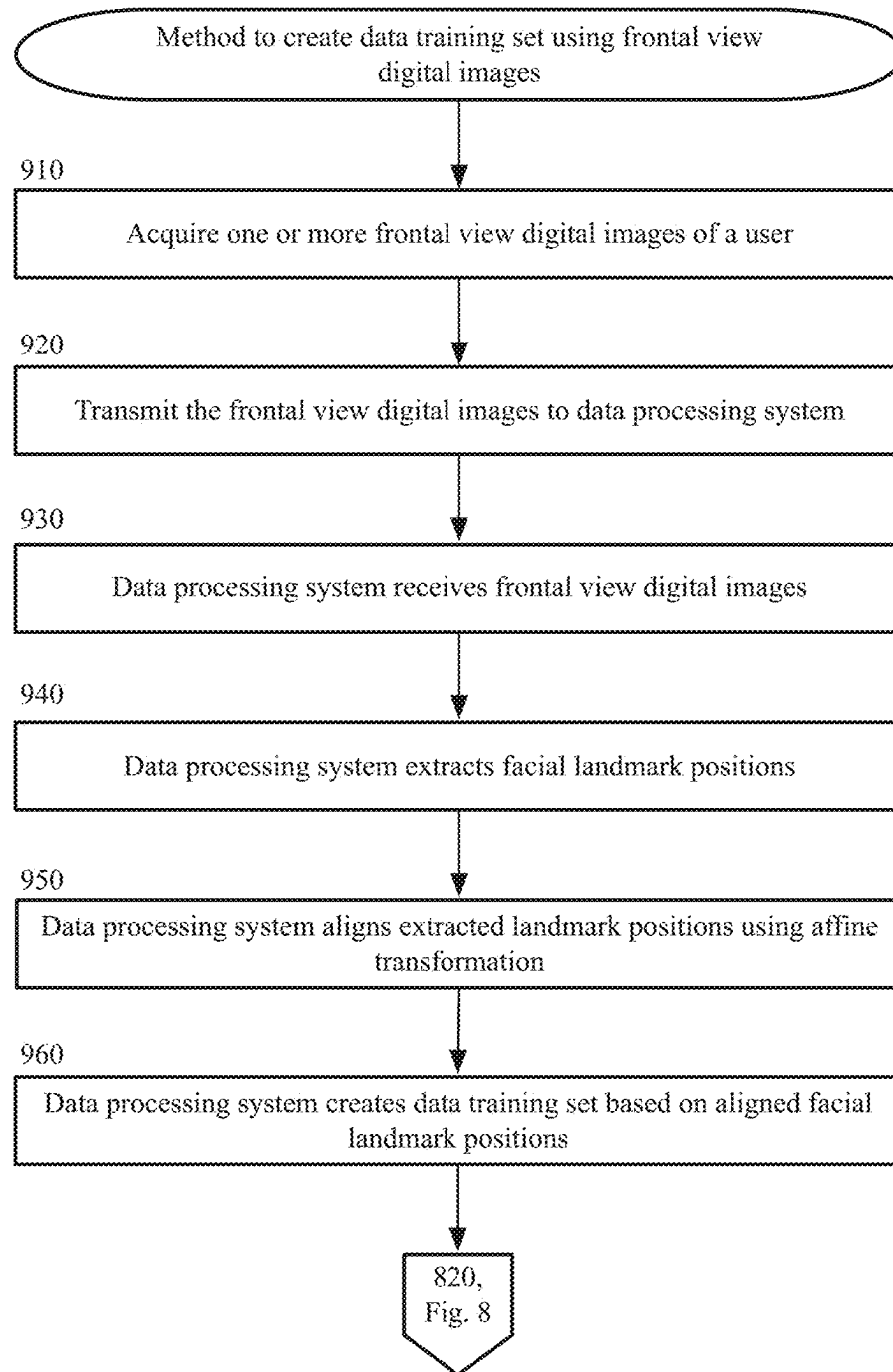
810

Figure 9

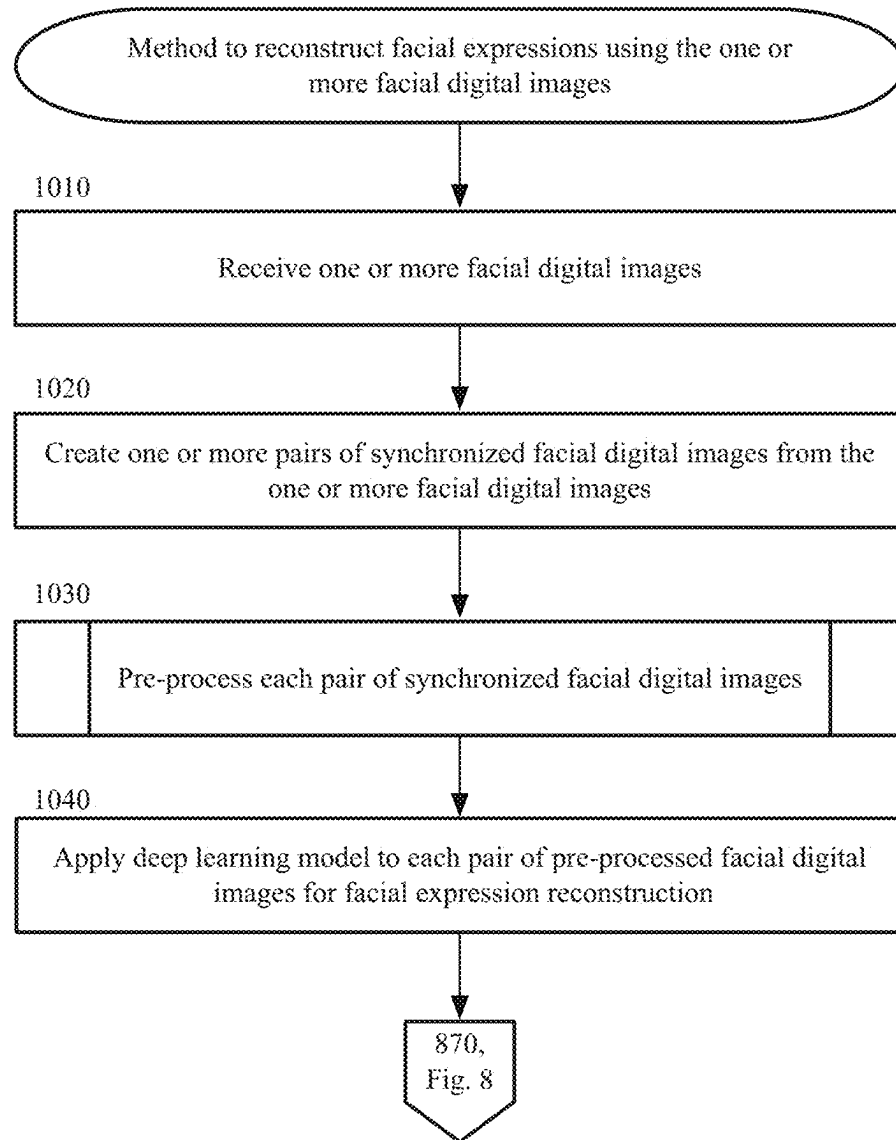
860

Figure 10

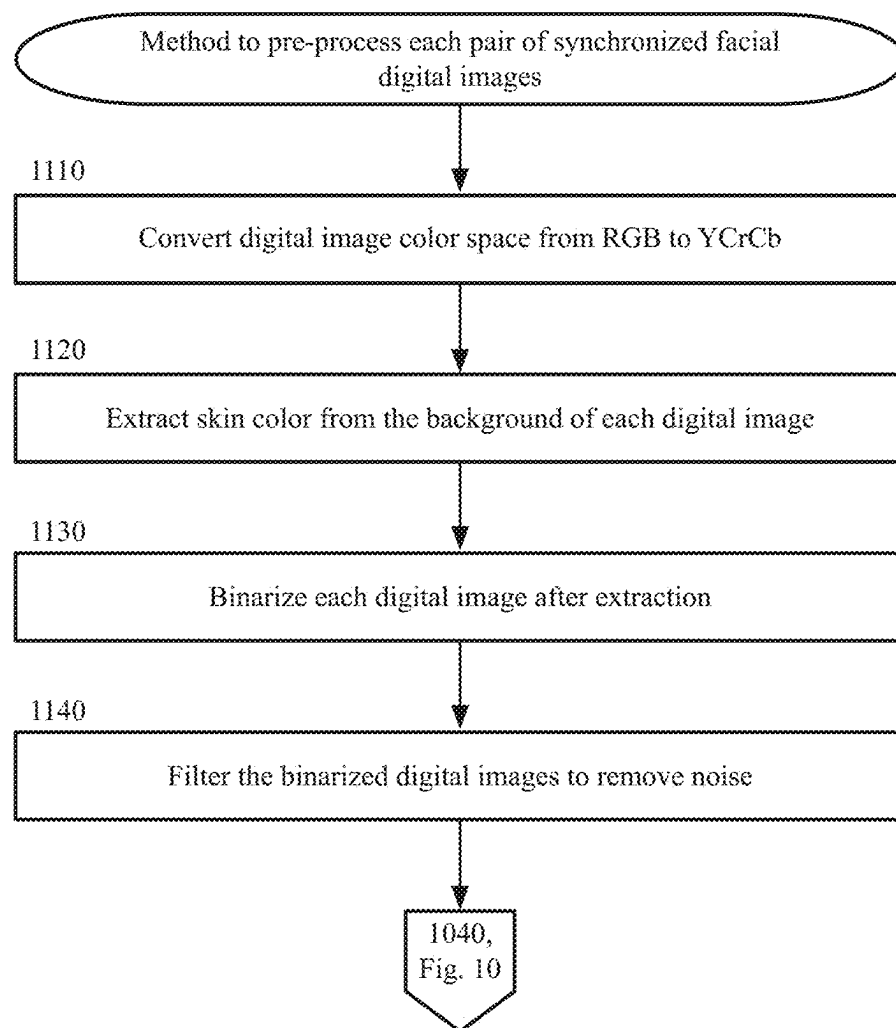
1030

Figure 11

1200

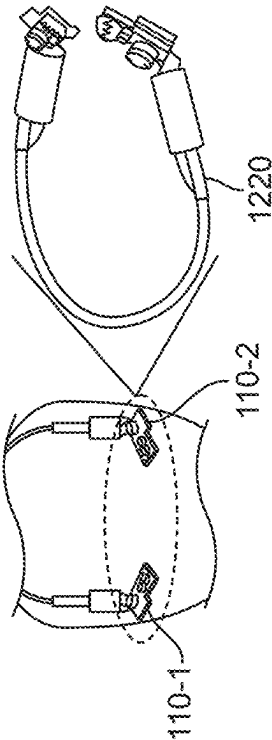


Figure 12B

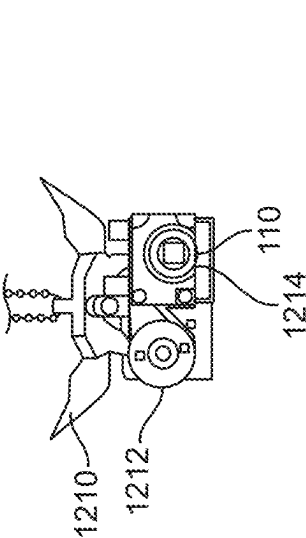


Figure 12A

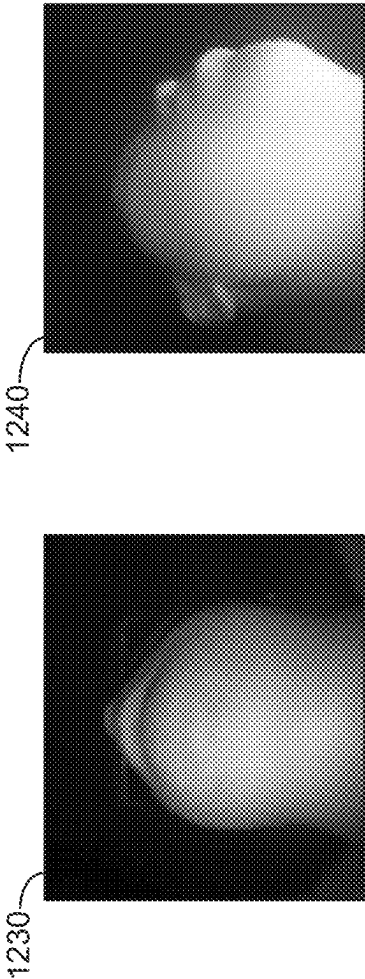


Figure 12C

1300

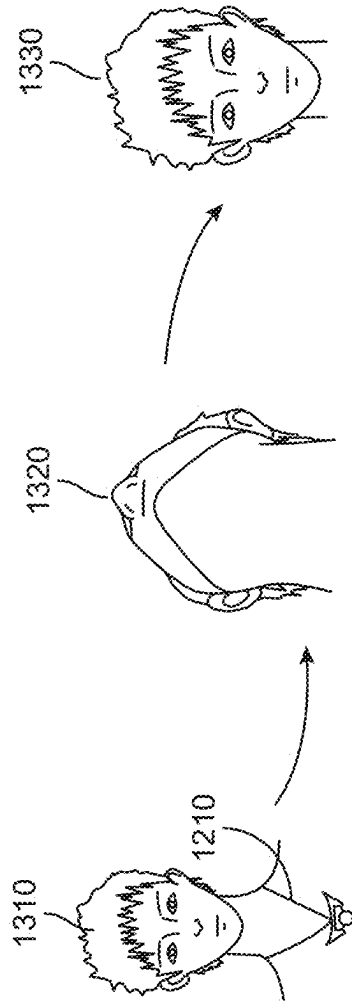


Figure 13A

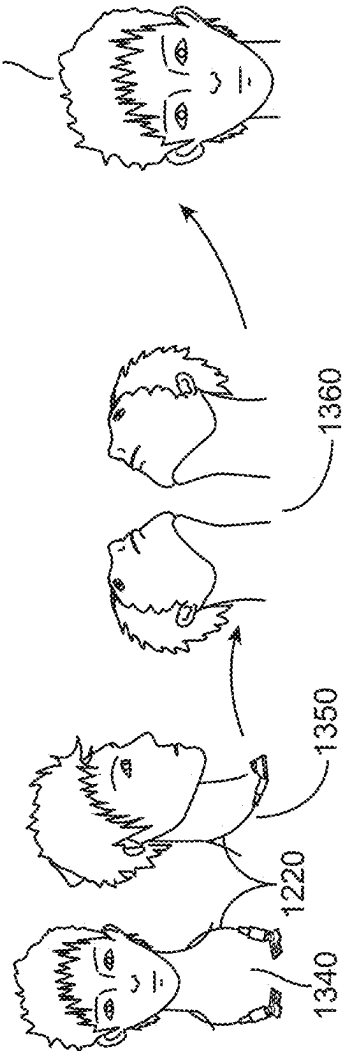


Figure 13B

1400

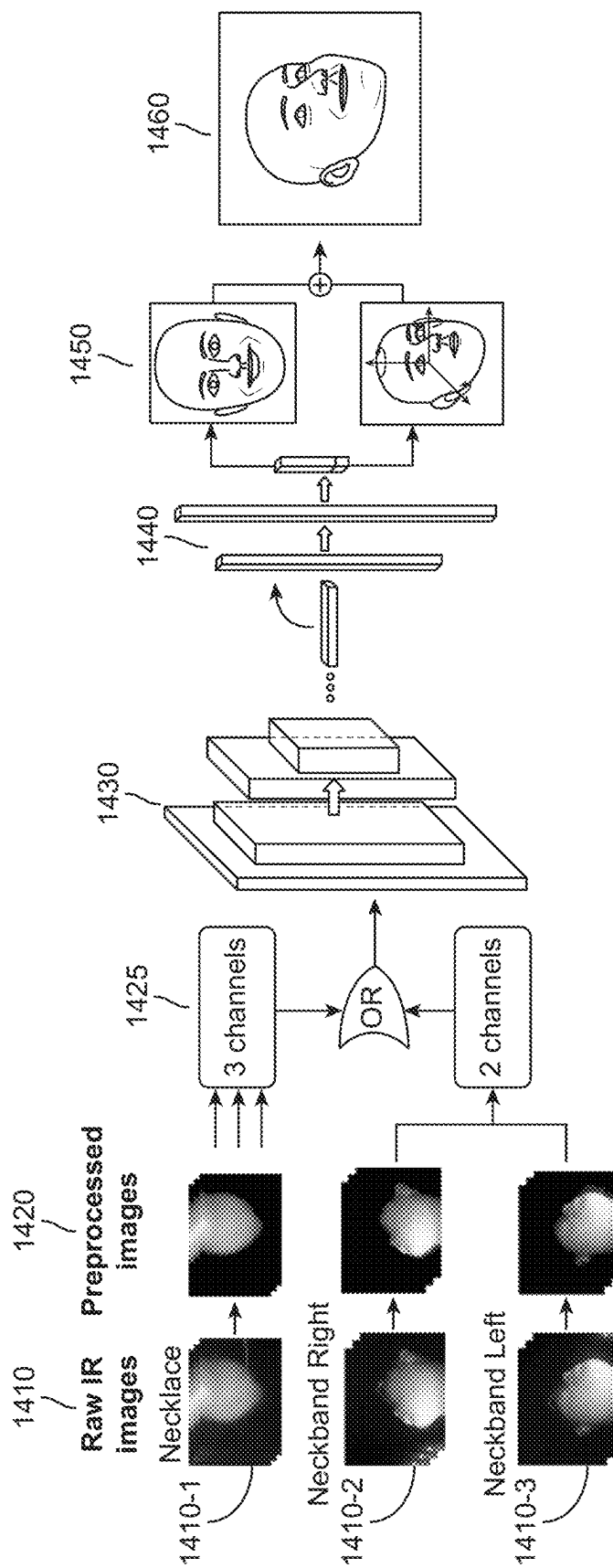


Figure 14

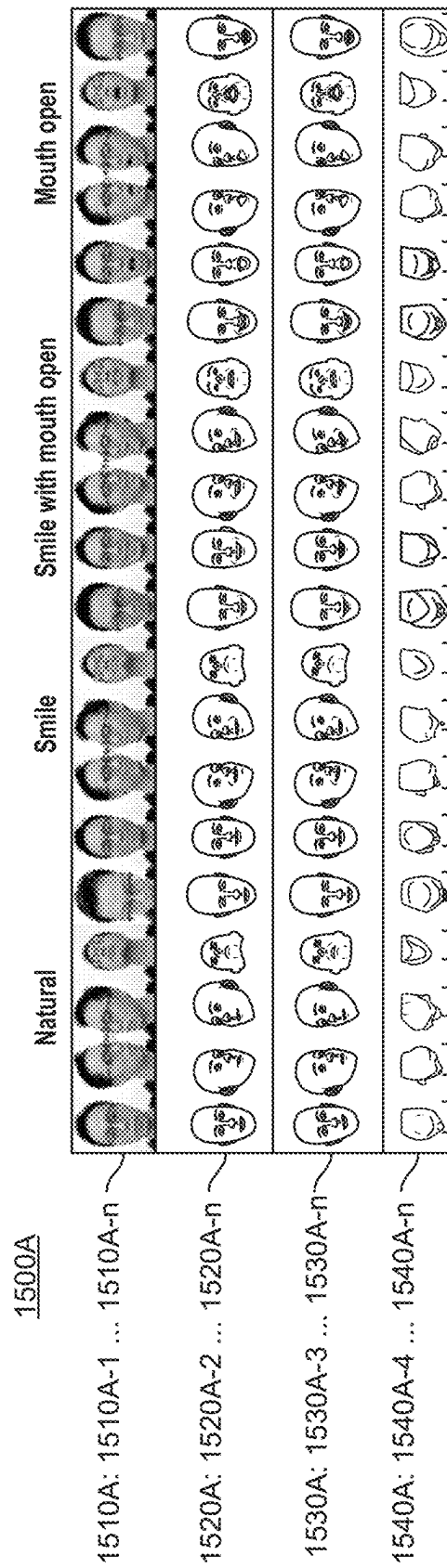


Figure 15A

1500B

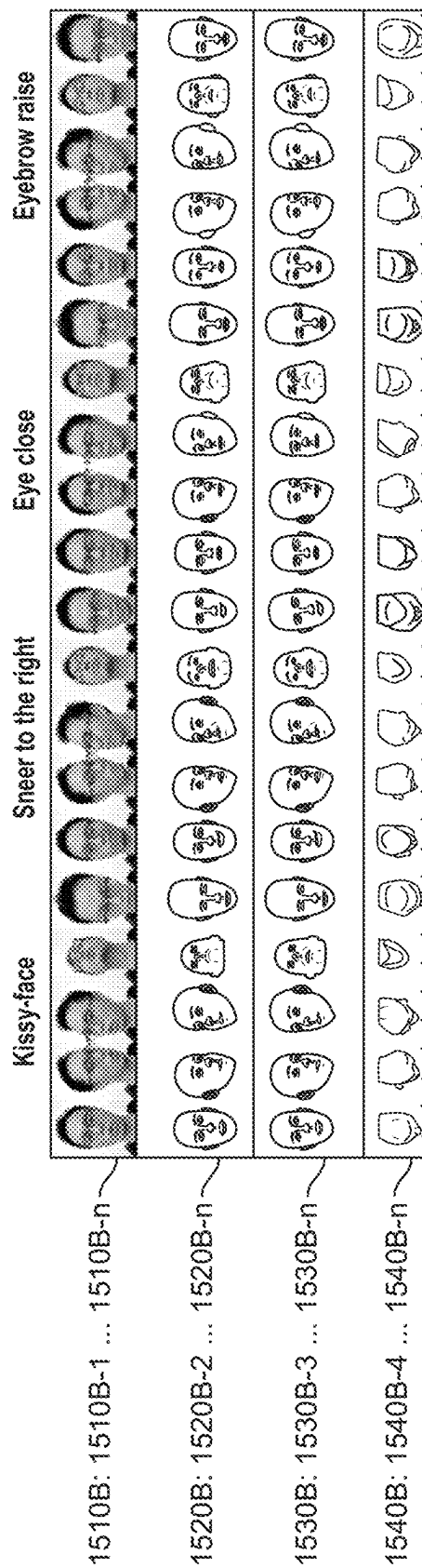


Figure 15B

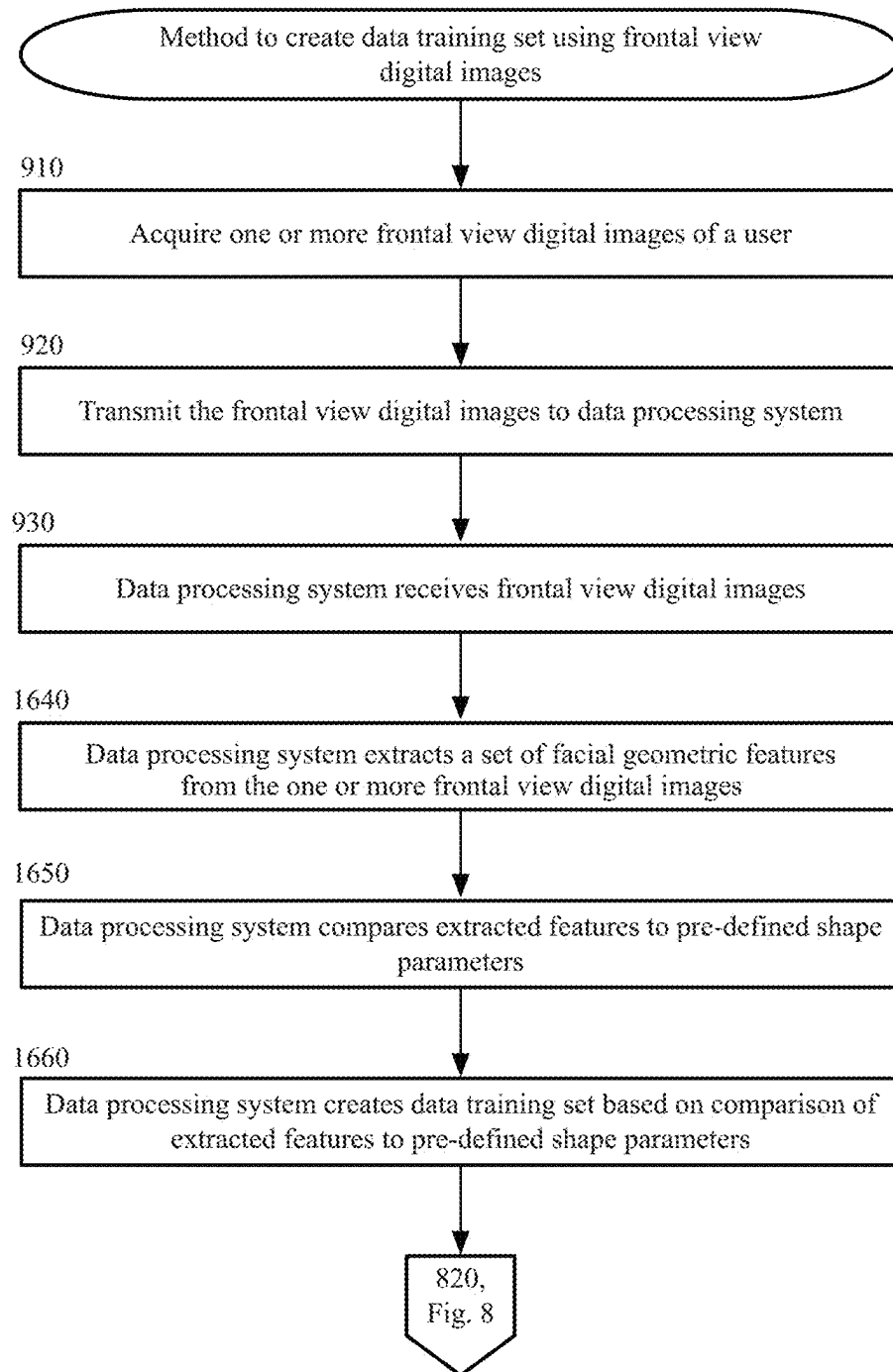
810'

Figure 16

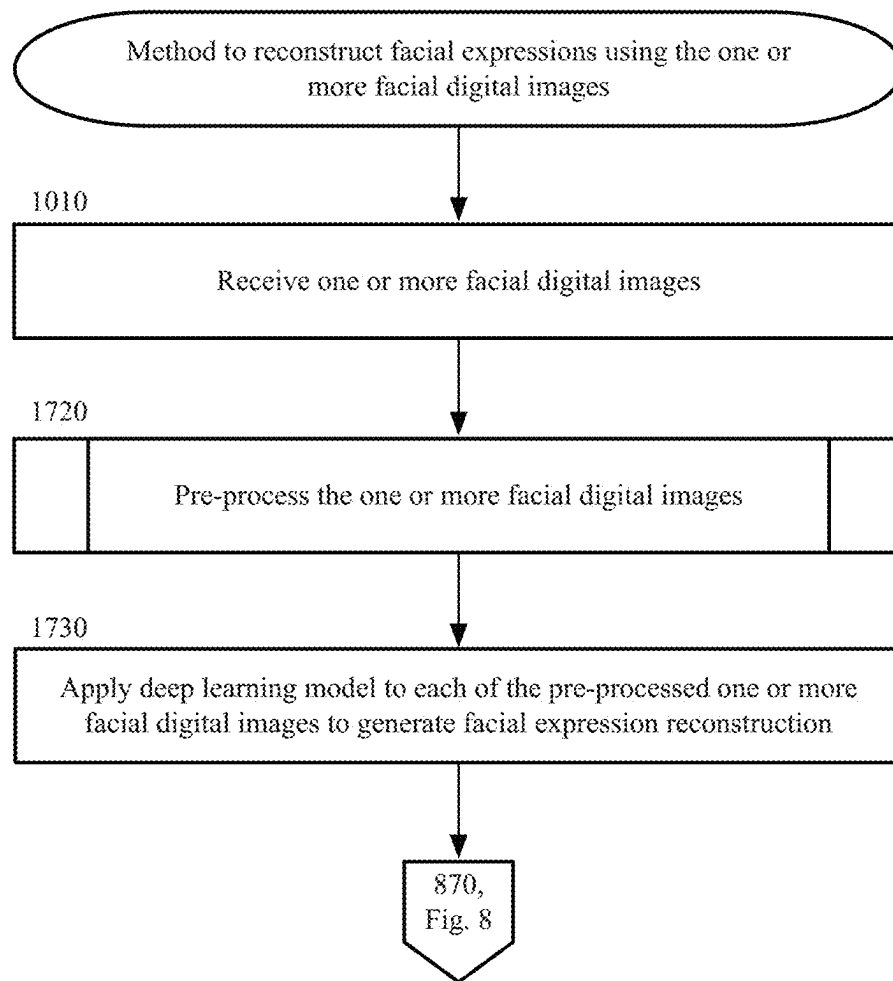
860'

Figure 17

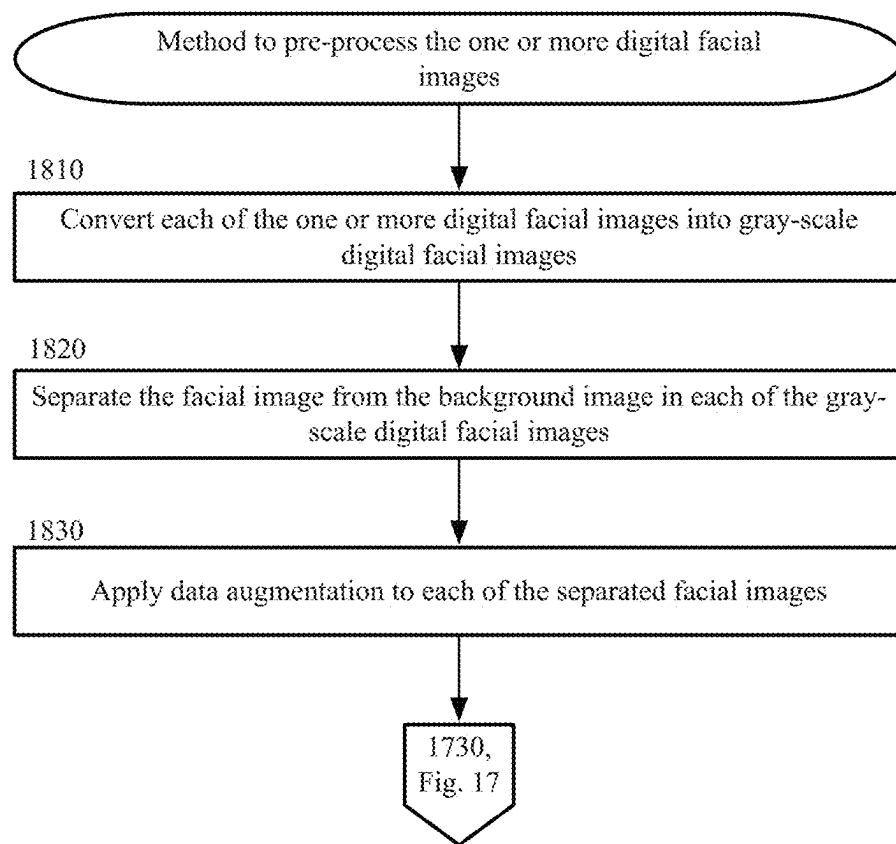
1720

Figure 18

1900

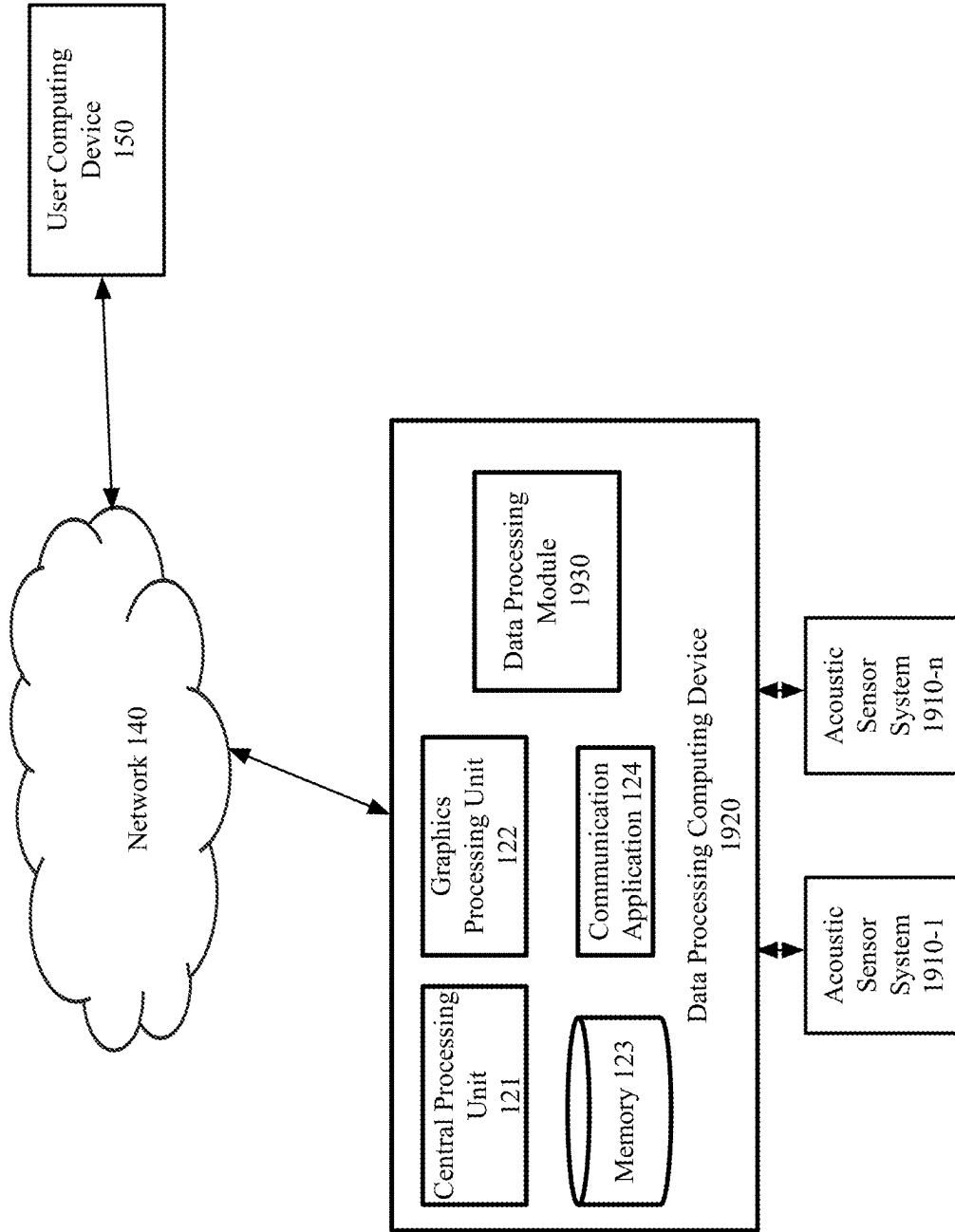


Figure 19

1910

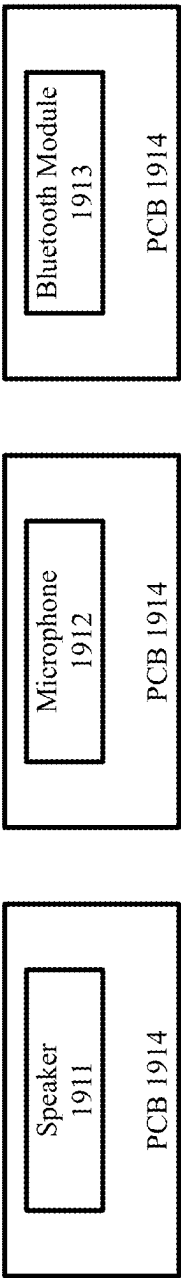


Figure 20A

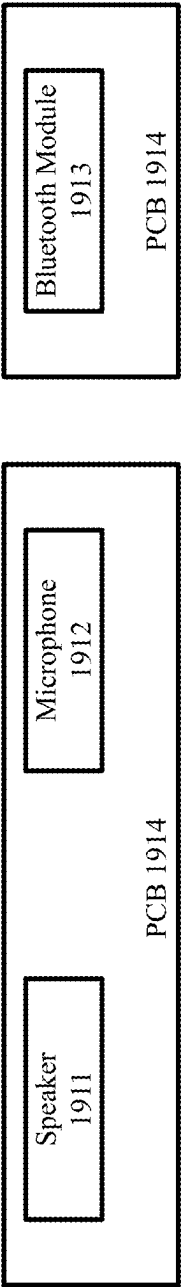


Figure 20B

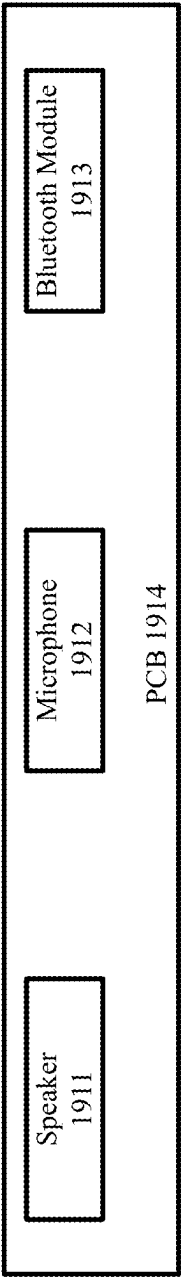


Figure 20C

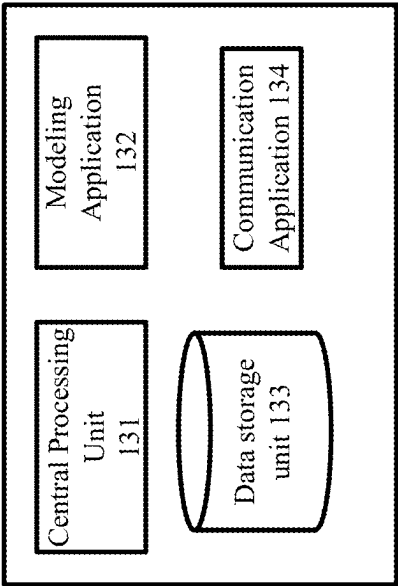


Figure 21

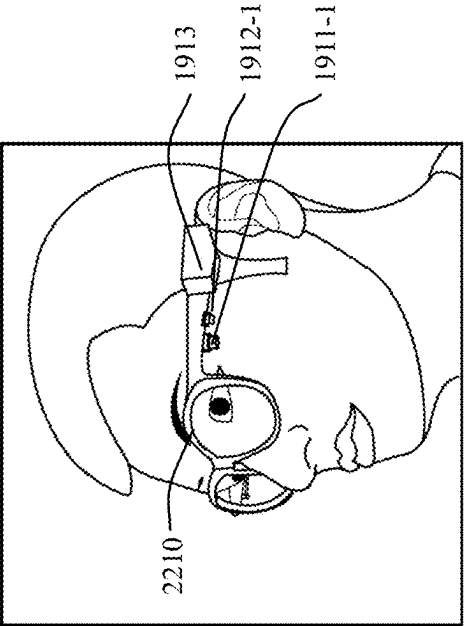


Figure 22B

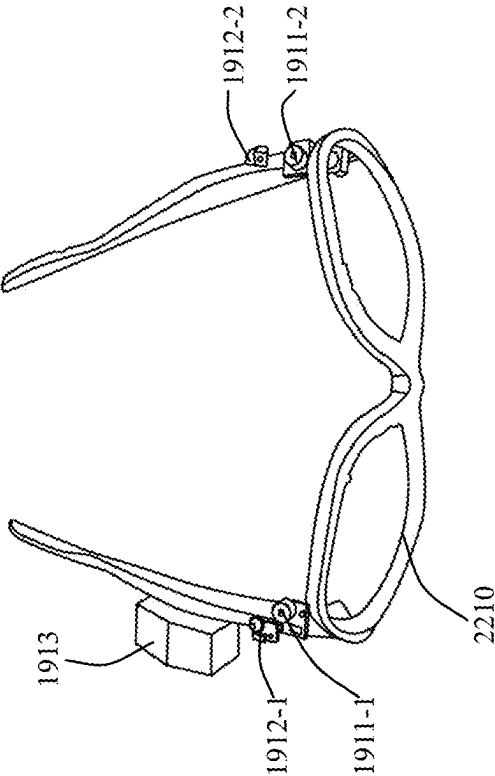


Figure 22A

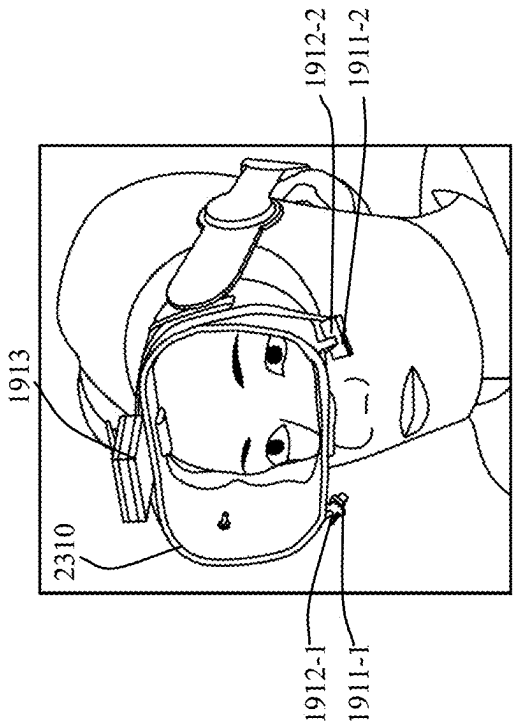


Figure 23B

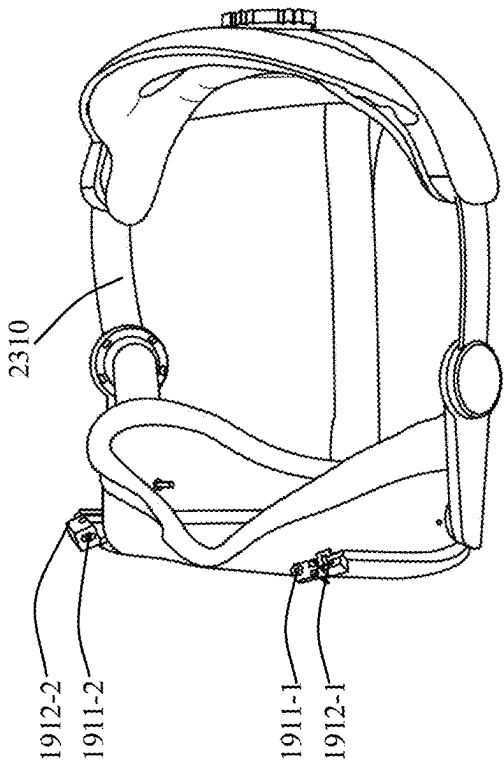


Figure 23A

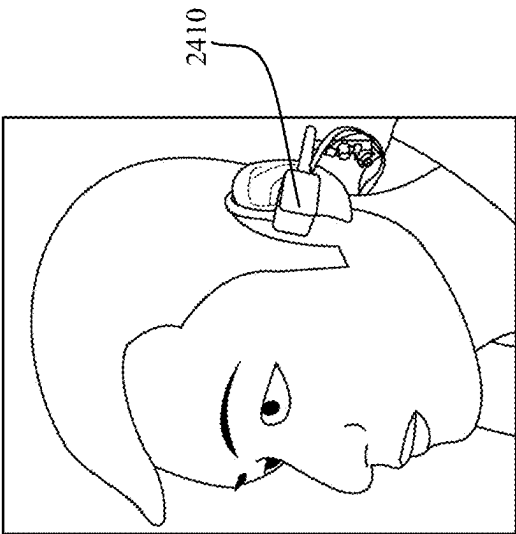


Figure 24B

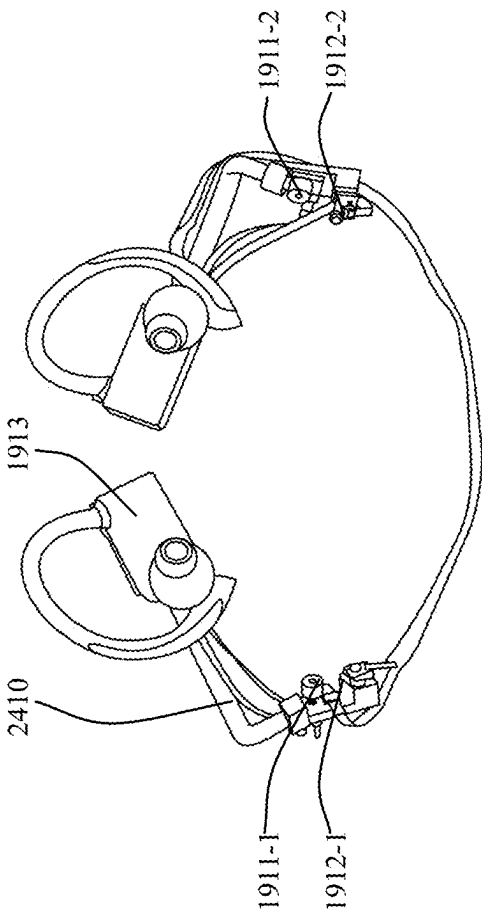


Figure 24A

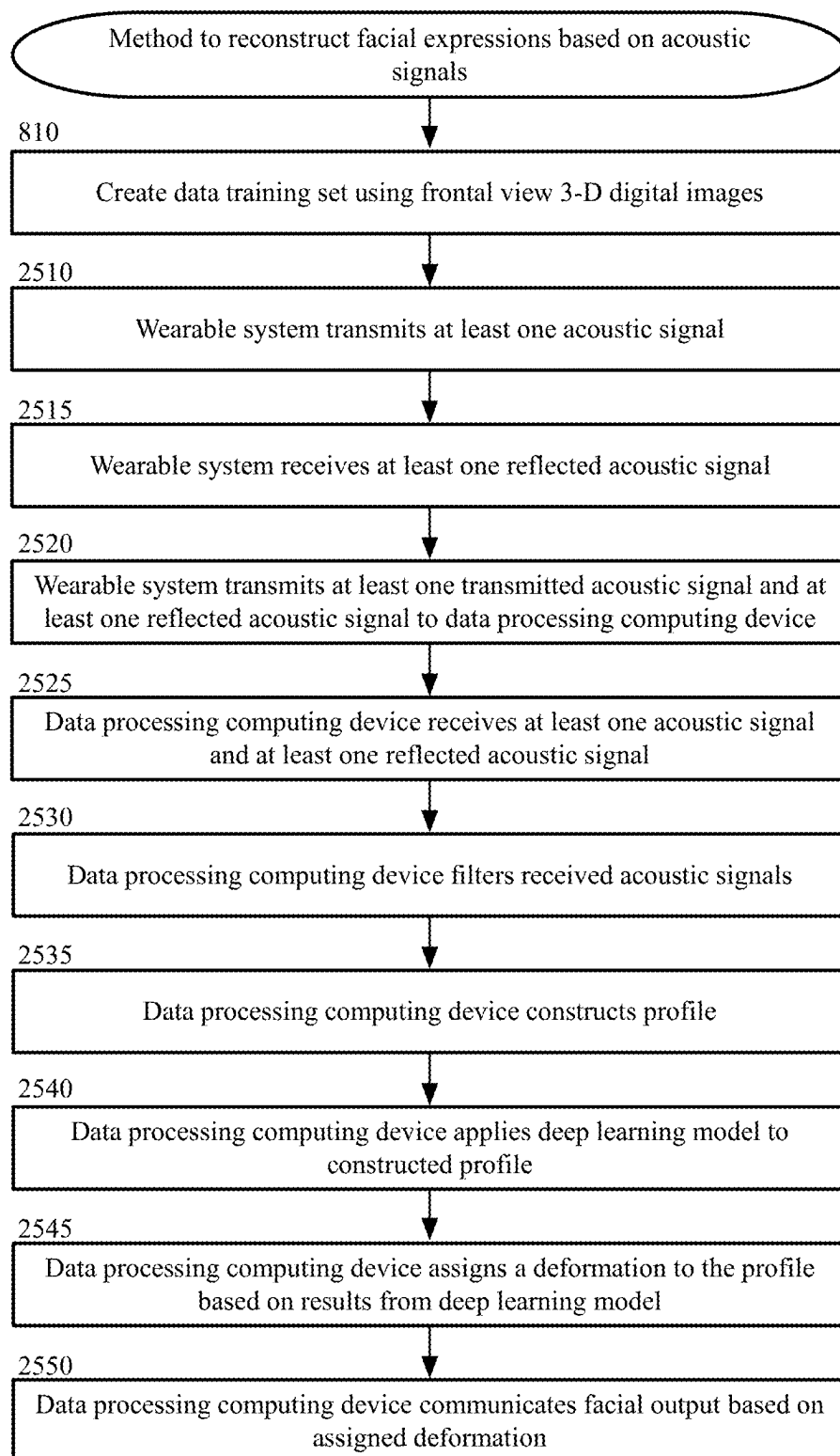
2500

Figure 25

2600

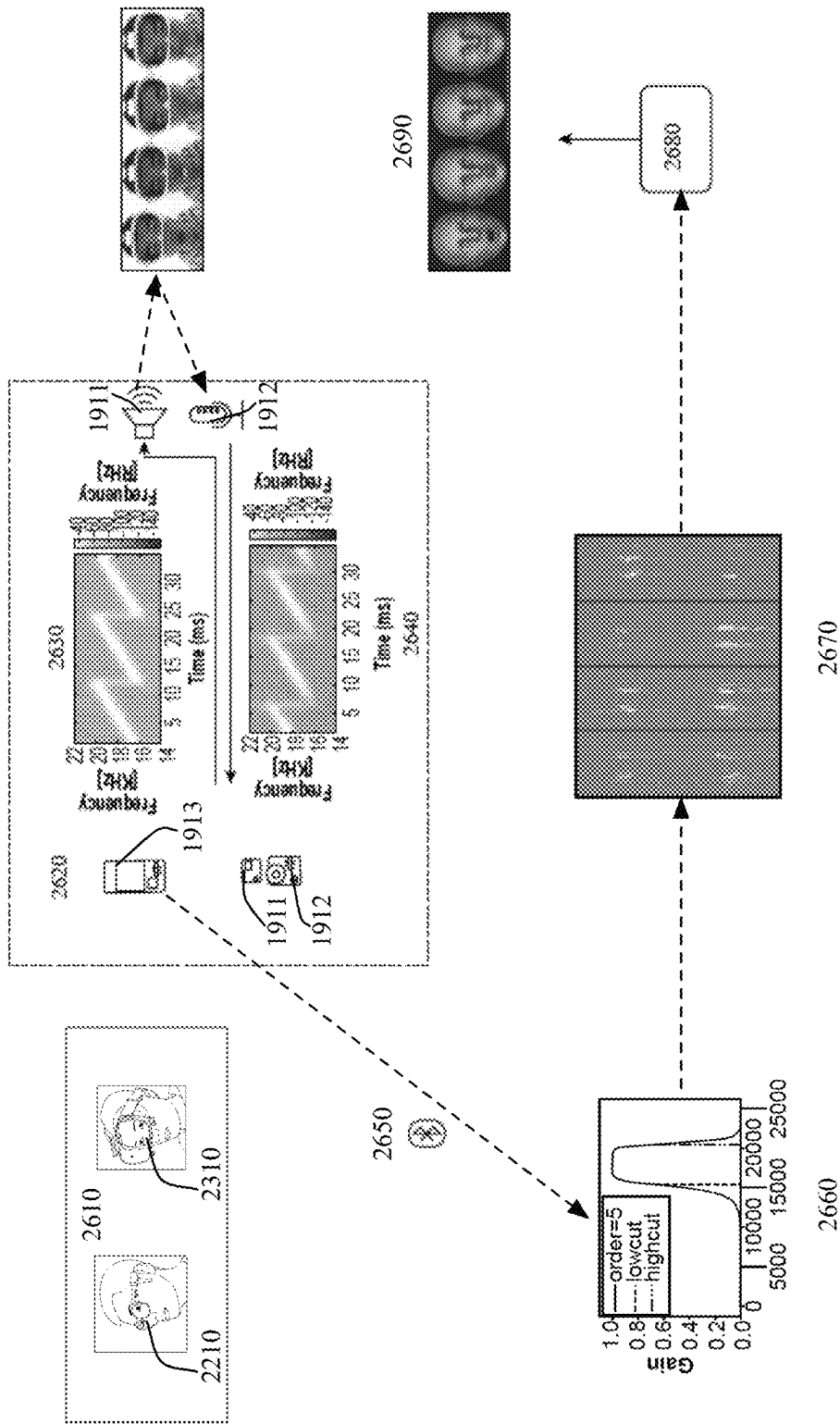


Figure 26

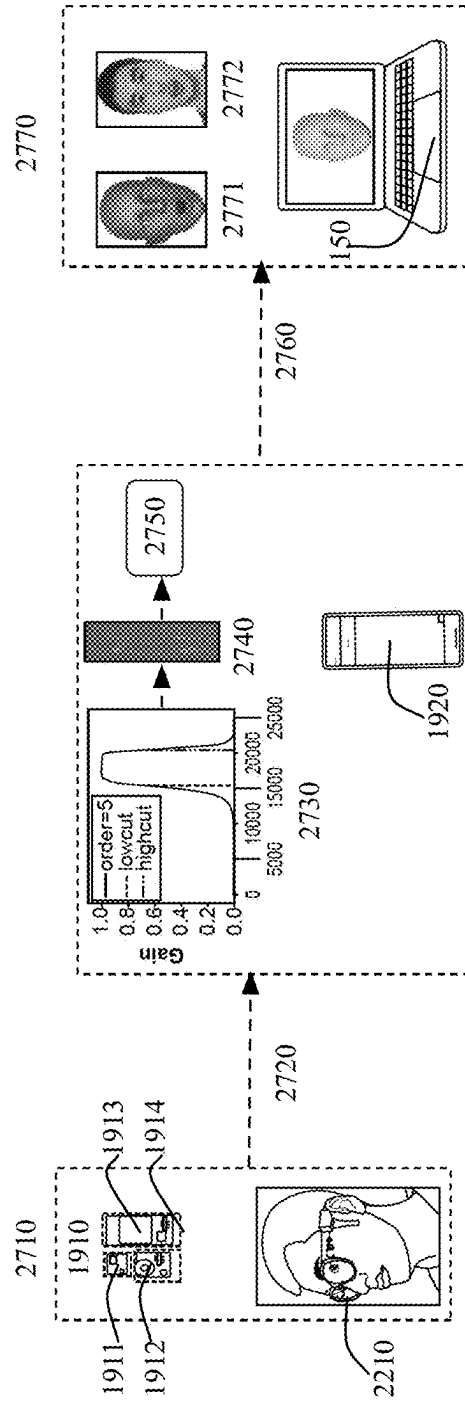


Figure 27

2800

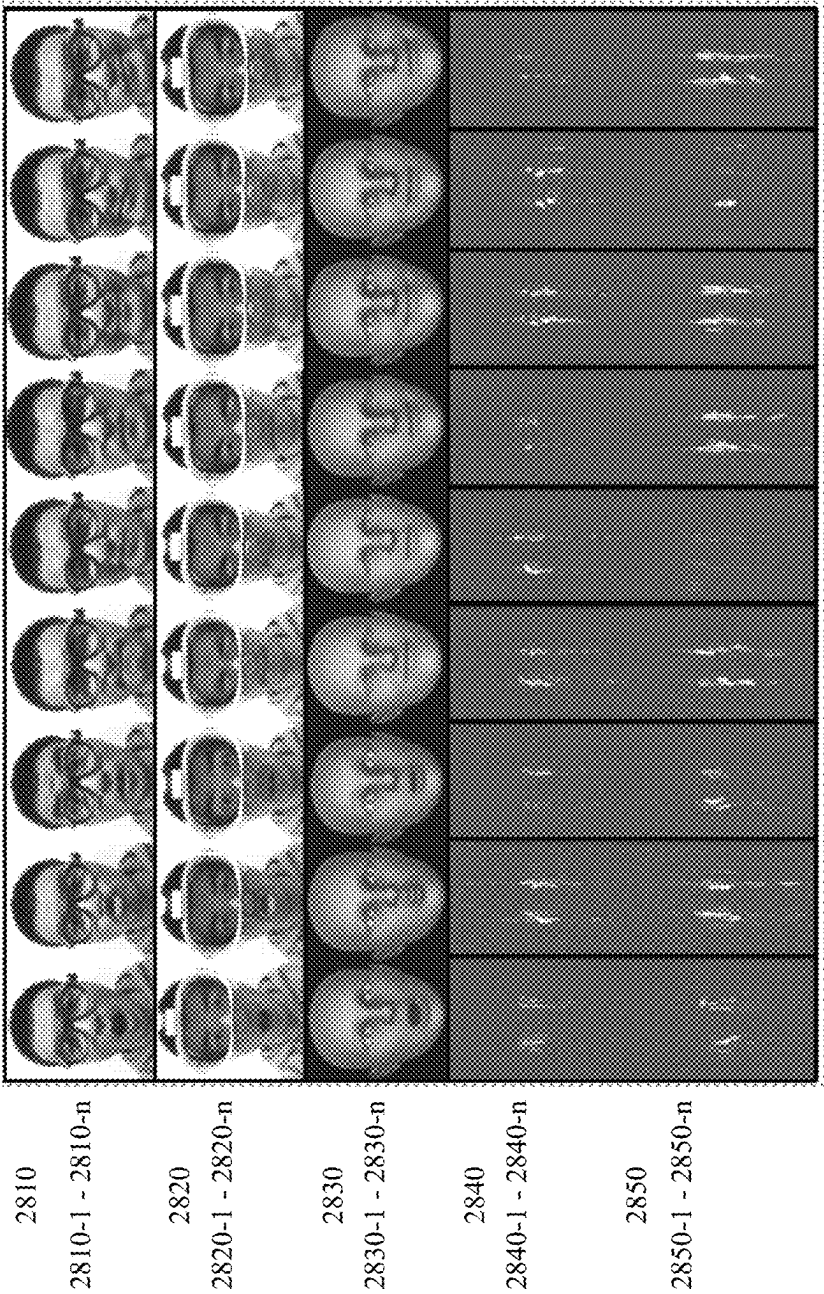


Figure 28

2900

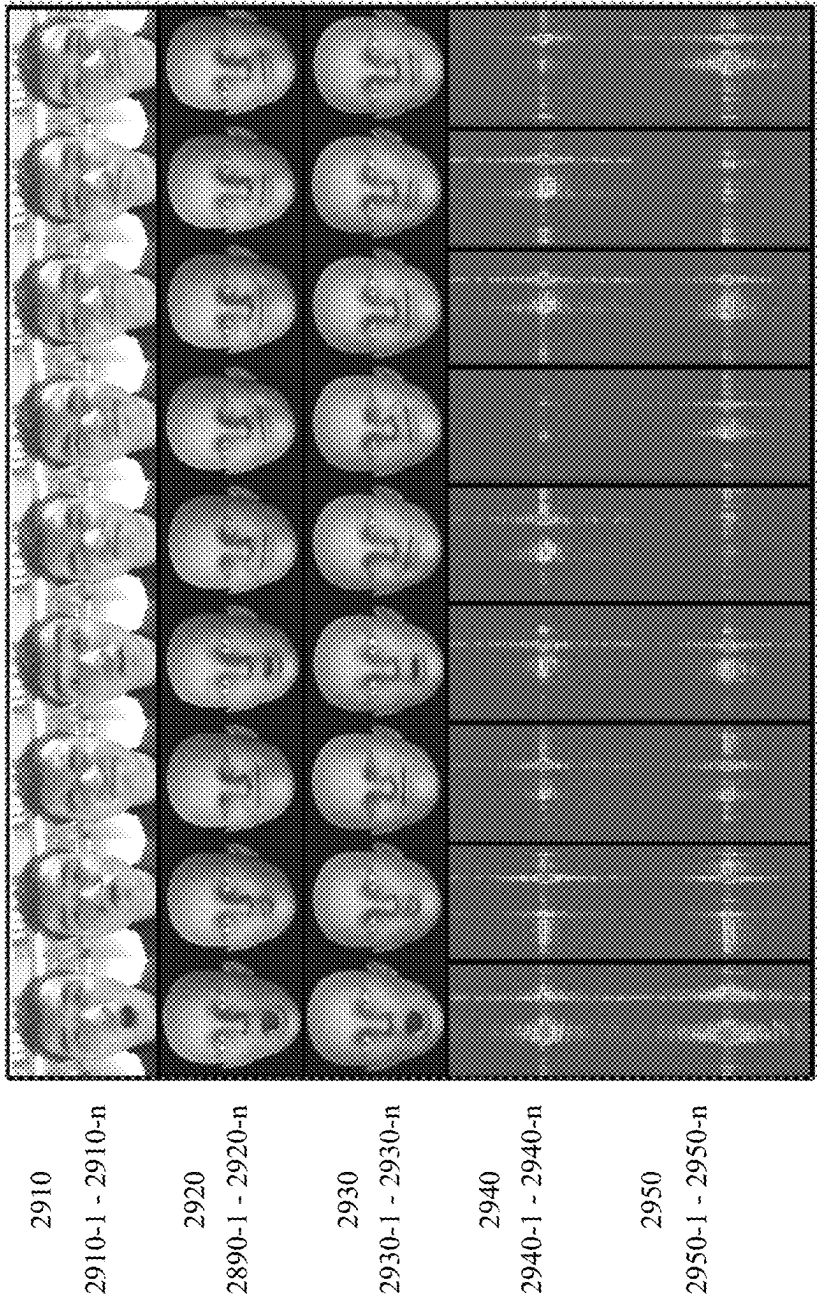


Figure 29

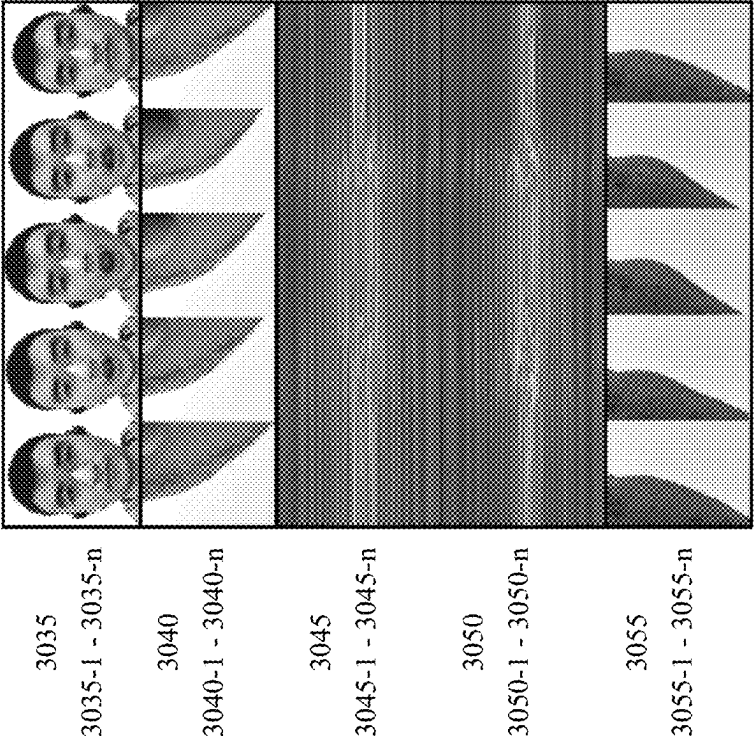


Figure 30A

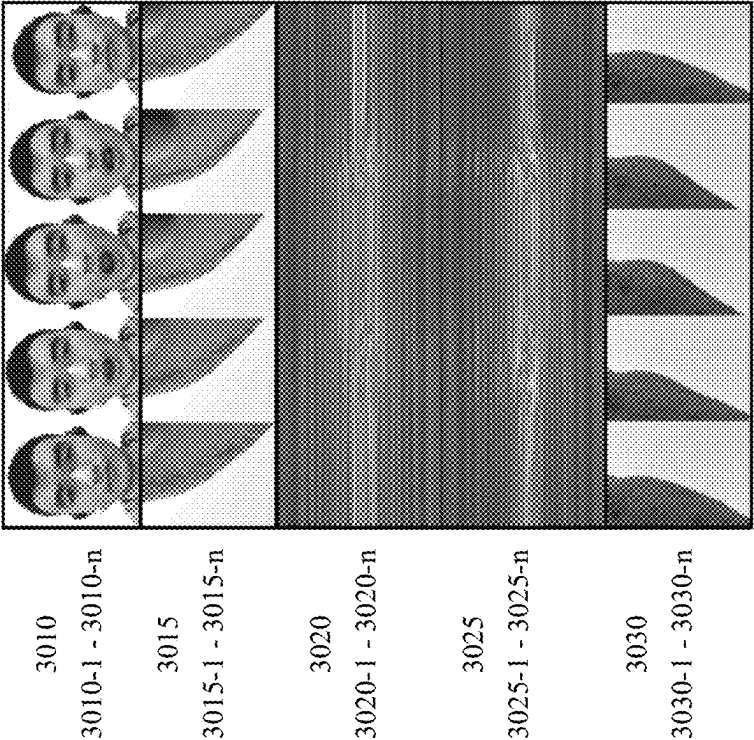


Figure 30B

4000

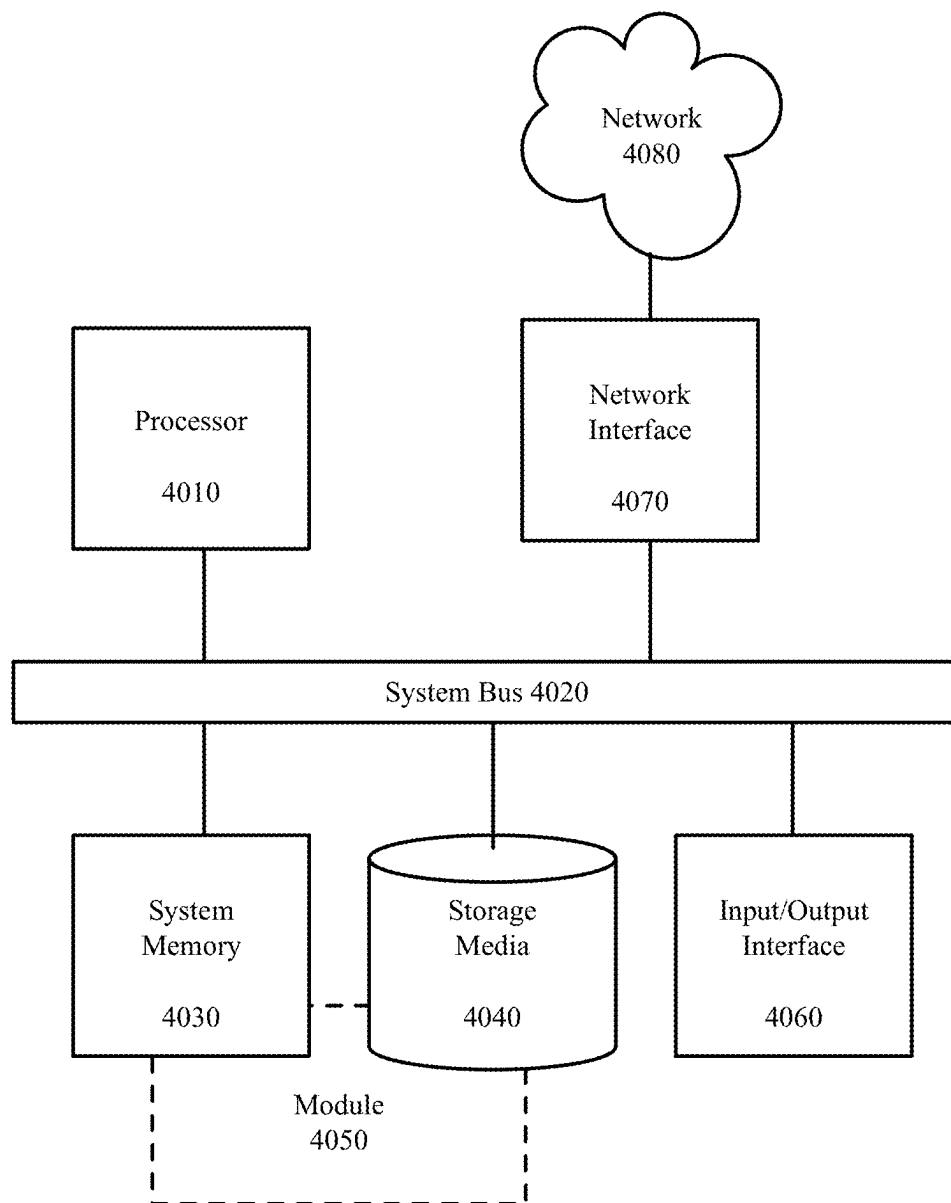


Figure 31

1

WEARABLE FACIAL MOVEMENT TRACKING DEVICES

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims the benefit of U.S. Provisional Patent Application No. 63/343,023, filed May 17, 2022 and entitled “Wearable Facial Movement Tracking Devices” and is a continuation-in-part of PCT Patent Application No. PCT/US2021/032511, filed May 14, 2021 and entitled “Wearable Devices For Facial Expression Recognition,” which claims the benefit of U.S. Provisional Patent Application No. 63/025,979, filed May 15, 2020 and entitled “C-Face: Continuously Reconstructing Facial Expressions By Deep Learning Contours Of The Face With Ear-Mounted Miniature Cameras.” The entire contents of the above-identified priority applications are hereby fully incorporated herein by reference.

TECHNICAL FIELD

This disclosure relates to systems and methods to reconstruct facial expressions and to track facial movements from head or neck-mounted wearable devices.

BACKGROUND

Humans use facial expressions as a natural mode of communication. The ability to continuously record and understand facial movements can improve interactions between humans and computers in a variety of applications.

Conventional facial reconstruction methods require a camera to be positioned in front of a user’s face with a specified position and angle relative to the user’s face. To achieve reliable facial reconstruction, the camera needs an entire view of the face without occlusions. Conventional facial reconstruction methods do not perform well if the user is in motion, the camera is not appropriately set up, the camera is not in front of the user, or the user’s face is partially occluded or not fully visible to the camera due to the camera’s position or angle relative to the user’s face.

As an alternative to frontal camera systems, wearable devices for facial expression reconstruction have been developed using sensing techniques, such as acoustic interference, pressure sensors, electrical impedance tomography, and electromyography. These wearable devices use instrumentation that is mounted directly on a user’s face. These conventional devices often cover the user’s face and only recognize discrete facial expressions. Examples of these conventional wearable devices include face masks with built-in ultrasonic transducers or electrodes secured to a human face with electromyography or capacitive sensing abilities. These wearable devices are attached directly to the user’s face or body and may block the field of vision and interfere with normal daily activities, such as eating or socializing.

Another alternative to frontal camera systems is smart eyewear including smart glasses, augmented reality glasses, and virtual reality headsets. However, these smart eyewear devices cannot track high-quality facial movements continuously. For example, virtual reality devices cannot depict 3D avatars in virtual worlds with facial expressions of the user.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a block diagram depicting a facial expression reconstruction system.

2

FIGS. 2A and 2B are perspective views of head-mounted wearable devices for facial expression reconstruction.

FIGS. 3A, 3B, and 3C are illustrations depicting examples of head-mounted wearable devices on a user.

FIG. 4 is a block diagram depicting a deep-learning model process.

FIG. 5 is a block diagram depicting a classifier process.

FIG. 6 is a diagram depicting the correlation between frontal view camera images, landmark training data, predicted facial expression landmarks, and right and left facial contour images.

FIG. 7 is a diagram depicting the correlation between frontal view camera images, two-dimensional training data, predicted two-dimensional facial expressions, and right and left facial contour images.

FIG. 8 is a block flow diagram depicting a method to reconstruct facial expressions.

FIG. 9 is a block flow diagram depicting a method to create a data training set using frontal view digital images.

FIG. 10 is a block flow diagram depicting a method to reconstruct facial expressions using one or more facial digital image(s).

FIG. 11 is a block flow diagram depicting a method to pre-process each pair of synchronized facial digital images.

FIGS. 12A, 12B, and 12C are perspective views of neck-mounted wearable devices for facial expression reconstruction.

FIGS. 13A and 13B are illustrations depicting examples of neck-mounted wearable devices on a user.

FIG. 14 is a block diagram depicting an alternate embodiment of a deep-learning model process.

FIG. 15A is a diagram depicting the correlation between frontal view camera images, three-dimensional training data, predicted three-dimensional face expressions, and infrared facial images.

FIG. 15B is a diagram depicting the correlation between frontal view camera images, three-dimensional training data, predicted three-dimensional face expressions, and infrared facial images.

FIG. 16 is a block flow diagram depicting an alternative method to create a data training set using frontal view images.

FIG. 17 is a block flow diagram depicting an alternative method to reconstruct facial expressions using one or more facial digital image(s).

FIG. 18 is a block flow diagram depicting an alternative method to pre-process the one or more digital facial image(s).

FIG. 19 is a block diagram depicting an acoustic sensor system based facial expression reconstruction system.

FIG. 20A is a block diagram depicting a first embodiment of an acoustic sensor system. FIG. 20B is a block diagram depicting a second embodiment of an acoustic sensor system. FIG. 20C is a block diagram depicting a third embodiment of an acoustic sensor system.

FIG. 21 is a block diagram depicting a data processing module.

FIG. 22A depicts a first embodiment of a head-mounted acoustic sensor system wearable device. FIG. 22B is an illustration depicting the first embodiment of the head-mounted acoustic sensor system wearable device on a user.

FIG. 23A depicts a second embodiment of a head-mounted acoustic sensor system wearable device. FIG. 23B is an illustration depicting the second embodiment of the head-mounted acoustic sensor system wearable device on a user.

3

FIG. 24A depicts a third embodiment of a head-mounted acoustic sensor system wearable device. FIG. 24B is an illustration depicting the third embodiment of the head-mounted acoustic sensor system wearable device on a user.

FIG. 25 is a block flow diagram depicting a method to reconstruct facial expressions based on acoustic signals.

FIG. 26 is a diagram depicting an acoustic sensor system based facial expression reconstruction process.

FIG. 27 is a diagram depicting an acoustic sensor system based facial expression reconstruction output process.

FIG. 28 is a diagram depicting the correlation between frontal view camera images, predicted facial expressions, and left and right differential echo profiles.

FIG. 29 is a diagram depicting the correlation between frontal view camera images, training data, predicted facial expressions, and left and right differential echo profiles.

FIG. 30A is a diagram depicting the correlation between frontal view camera images of an open mouth, a left view image, a left and right echo profile, and a right view image.

FIG. 30B is a diagram depicting the correlation between frontal view camera images of a right sneer, a left view image, a left and right echo profile, and a right view image.

FIG. 31 is a block diagram depicting a computing machine and a module.

DETAILED DESCRIPTION

Overview

The present technology allows reconstruction of facial expressions and tracking of facial movements using non-obtrusive, wearable devices that capture optical or acoustical images of facial contours, chin profiles, or skin deformation of a user's face. The wearable devices include head-mounted technology that continuously reconstruct full facial expressions by capturing the positions and shapes of the mouth, eyes, and eyebrows. Miniature cameras capture contours of the sides of the face, which are used to train a deep-learning model to predict facial expressions. An alternate embodiment of this technology includes a neck-mounted technology to continuously reconstruct facial expressions. Infrared cameras capture chin and face shapes underneath the neck, which are used to train a deep-learning model to predict facial expressions.

Additional embodiments include various camera types or acoustic imaging systems for the wearable devices. The acoustic imaging systems may comprise microphones and speakers. For example, the wearable devices may use the microphones and speakers to transmit and receive acoustical signals to determine skin deformation of a user. Full facial movements and expressions can be reconstructed from subtle skin deformations. The wearable devices may comprise ear mounted devices, eye mounted devices, or neck mounted devices.

The systems of this technology include wearable devices configured with miniature cameras or acoustical imaging systems in communication with computing devices to transmit images or acoustical signals from the cameras or acoustical imaging systems to a remote server, data acquisition system, or data processing device for facial expression reconstruction or to track facial movement. The wearable devices include headphone, earbud, necklace, neckband, glasses, virtual reality ("VR") headsets, augmented reality ("AR") headsets, and other form factors. Each of the form factors include miniature cameras or acoustical imaging systems and micro computing devices, such as Raspberry Pi™.

4

The exemplary headphone device includes two cameras and may include functionality with Bluetooth or Wi-Fi connectivity and speakers embedded within the earpieces of the headphone. The cameras are attached to the earpieces of the headphone device and are connected to a computing device to transmit acquired images to a remote server. The headphone device is configured to acquire images of the contours of a user's face by directing the cameras at adjustable angles and positions.

The exemplary earbud devices are constructed for wear as a left-side earbud and a right-side earbud. Both the left and right earbud devices include a camera to acquire images of the contours of a user's face. Each camera is connected to a computing device to transmit acquired images to a remote server.

The exemplary necklace device includes an infrared ("IR") camera with an IR LED and an IR bandpass filter. The necklace device is configured to acquire images of a user's chin profile by directing the camera at the profile of a user's chin. The IR LED projects IR light onto a user's chin to enhance the quality of the image captured by the camera. The IR bandpass filter filters visible light such that the camera captures infrared light reflected by a user's skin. The camera is connected a computing device to transmit acquired images to a remote server.

The exemplary neckband device includes two cameras fashioned to be positioned on the left and right sides of a user's neck. The neckband device includes IR LEDs and IR bandpass filters configured in proximity to each camera, similar to the necklace device. Each camera is connected a computing device to transmit acquired images to a remote server.

Using one of the wearable devices, facial expressions can be reconstructed from images acquired from the cameras within the devices. A data training set is created using frontal view images of a user in a machine learning algorithm. Multiple frontal view images of a user are acquired with a variety of facial expressions. The frontal view images are transmitted to a data processing system to create the data training set.

The wearable device captures one or more facial digital image(s) and transmits the digital images to data acquisition computing devices connected to each of the cameras of the wearable device. The data acquisition computing devices subsequently transmit the images to a remote server or data processing system. The data processing system reconstructs facial expressions using the images. The data processing system pre-processes the images by reducing the image size, removing noise from the image, extracting skin color from the background of the image, and binarizing each the image. In some embodiments, a wearable imaging system comprises one or more imaging sensor(s), wherein the one or more imaging sensor(s) are positioned to capture one or more image(s) with incomplete side contours of a face (for example, from ear(s) and/or from neck), a processor, and a non-transitory machine-readable storage medium comprising machine-readable instructions executed by the processor to extract a plurality of features (for example, landmarks or parameters) from the one or more image(s), compare the extracted features and/or changes of the extracted features with features from a ground truth, and output one or more recognition or prediction result(s), wherein the results comprise a word or a phrase spoken by a user, an emoji of a facial expression of a user, and/or a real-time avatar of a facial expression of a user.

The data processing system applies a deep-learning model to the pre-processed images for facial expression reconstruction.

tion. The reconstructed facial expressions may be used in applications such as silent speech and emoji recognition.

In an alternate embodiment, acoustic sensing technology can be used to reconstruct facial expressions using an array of microphones and speakers and deep-learning models. Acoustic sensing technology is small, lightweight and, therefore, suitable to mount to a variety of wearable devices. Facial movements may be tracked by detecting skin deformations observed from different positions on a user's head. When a user performs facial movements, the skin on the face deforms with unique patterns. Using acoustic sensing technology, skin deformations may be captured without capturing images of the entire face. The acoustic technology comprises a wearable device such as headphones, earbuds, necklaces, neckbands, and any other suitable form factors such that the microphones and speakers can be mounted or attached to the wearable device. The positions of the microphones and speakers may be adjustable to optimize the recording of reflected signals.

The acoustic wearable device actively sends signals from the speakers towards a user's face. The signals are reflected by the user's face and captured by the microphones on the acoustic wearable device. The signals are reflected differently back towards the microphones based on different facial expressions or movements. The acoustic wearable device sends the transmitted and reflected signals to a computing device for further processing. An echo profile is calculated based on the transmitted and reflected acoustic signals. The echo profile is calculated by calculating a cross-correlation between the received acoustic signals and the transmitted acoustic signals, which can display the deformations of the skin in temporal and spatial domains. The echo profile is input into a deep learning module to reconstruct the facial expressions of the user.

While acoustic sensing technology may be used to determine facial expressions by detecting skin deformations, acoustic sensing technology may also be used to directly determine other outputs associated with skin deformation without the need to first determine facial expressions. For example, the acoustic sensing technology may track blinking patterns and eyeball movements of a user. Blinking patterns and eyeball movements can be applied to the diagnosis and treatment processes of eye diseases. The acoustic sensing technology may detect movements of various parts of the face associated with speech, whether silent or voiced. While speaking, different words and/or phrases lead to subtle, yet distinct, skin deformations. The acoustic sensing technology can capture the skin deformation to recognize silent speech or to vocalize sound. The acoustic sensing technology may also be used to recognize and track physical activities. For example, while eating, a user opens the mouth, chews, and swallows. During each of those movements, the user's skin deforms in a certain way such that opening the mouth is distinguishable from chewing and from swallowing. The detected skin deformations may be used to determine the type and quantity of food consumed by a user. Similar to eating, a type and quantity of a consumed drink may be determined by skin deformation. The acoustic sensing technology may also be used to determine an emotional status of a user. An emotional status may be related to skin deformations. By detecting skin deformations, an emotional status can be determined and reported to corresponding applications.

These and other aspects, objects, features, and advantages of the disclosed technology will become apparent to those having ordinary skill in the art upon consideration of the following detailed description of illustrated examples.

Example System Architecture

Turning now to the drawings, in which like numerals indicate like (but not necessarily identical) elements throughout the figures, examples of the technology are described in detail.

FIG. 1 is a block diagram depicting a facial expression reconstruction system 100, in accordance with certain examples. As depicted in FIG. 1, the facial expression reconstruction system 100 comprises cameras 110-1 through 110-n, computing devices 120-1 through 120-n, data processing system 130, and user computing device 150. Computing devices 120, data processing system 130, and user computing device 150 are configured to communicate via a network 140.

In example embodiments, network 140 includes one or more wired or wireless telecommunications system(s) by which network devices may exchange data. For example, the network 140 may include one or more of a local area network (LAN), a wide area network (WAN), an intranet, an Internet, a storage area network (SAN), a personal area network (PAN), a metropolitan area network (MAN), a wireless local area network (WLAN), a virtual private network (VPN), a cellular or other mobile communication network, a BLUETOOTH® wireless technology connection, a near field communication (NFC) connection, any combination thereof, and any other appropriate architecture or system that facilitates the communication of signals, data, and/or messages.

Facial expression reconstruction system 100 comprises cameras 110 that may be any suitable sensors for capturing images. For example, cameras 110 may include depth cameras, red-green-blue ("RGB") cameras, infrared ("IR") sensors, acoustic cameras or sensors, speakers, microphones, thermal imaging sensors, charge-coupled devices ("CCDs"), complementary metal oxide semiconductor ("CMOS") devices, active-pixel sensors, radar imaging sensors, fiber optics, and any other suitable image device technology.

The image frame resolution of the camera 110 may be defined by the number of pixels in a frame. The image resolution of the camera 110 may comprise any suitable resolution, including of the following resolutions, without limitation: 32×24 pixels; 32×48 pixels; 48×64 pixels; 160×120 pixels, 249×250 pixels, 250×250 pixels, 320×240 pixels, 420×352 pixels, 480×320 pixels, 640×480 pixels, 720×480 pixels, 1280×720 pixels, 1440×1080 pixels, 1920×1080 pixels, 2048×1080 pixels, 3840×2160 pixels, 4096×2160 pixels, 7680×4320 pixels, or 15360×8640 pixels. The resolution of the camera 110 may comprise a resolution within a range defined by any two of the preceding pixel resolutions, for example, within a range from 32×24 pixels to 250×250 pixels (for example, 249×250 pixels). In some embodiments, at least one dimension (the height and/or the width) of the image resolution of the camera 110 can be any of the following, including but not limited to 8 pixels, 16 pixels, 24 pixels, 32 pixels, 48 pixels, 72 pixels, 96 pixels, 108 pixels, 128 pixels, 256 pixels, 360 pixels, 480 pixels, 720 pixels, 1080 pixels, 1280 pixels, 1536 pixels, or 2048 pixels. The camera 110 may have a pixel size smaller than 1 micron, 2 microns, 3 microns, 5 microns, 10 microns, 20 microns, and the like. The camera 110 may have a footprint (for example, a dimension in a plane parallel to a lens) on the order of 10 mm×10 mm, 8 mm×8 mm, 5 mm×5 mm, 4 mm×4 mm, 2 mm×2 mm, or 1 mm×1 mm, 0.8 mm×0.8 mm, or smaller.

Each camera 110 is in communication with a computing device 120. Each camera 110 is configured to transmit

images or data to a computing device 120. Each camera 110 may be communicatively coupled to a computing device 120. In an alternate example, each camera 110 may communicate wirelessly with a computing device 120, such as via near field communication (“NFC”) or other wireless communication technology, such as Bluetooth, Wi-Fi, infrared, or any other suitable technology.

Computing devices 120 comprise a central processing unit 121, a graphics processing unit 122, a memory 123, and a communication application 124. In an example, computing devices 120 may be small, single-board computing devices, such as a Raspberry Pi™ device. Computing devices 120 function to receive images or data from cameras 110 and to transmit the images via network 140 to a data processing system 130.

Computing device 120 comprises a central processing unit 121 configured to execute code or instructions to perform the operations and functionality described herein, manage request flow and address mappings, and perform calculations and generate commands. Central processing unit 121 may be configured to monitor and control the operation of the components in the computing device 120. Central processing unit 121 may be a general purpose processor, a processor core, a multiprocessor, a reconfigurable processor, a microcontroller, a digital signal processor (“DSP”), an application specific integrated circuit (“ASIC”), a graphics processing unit (“GPU”), a field programmable gate array (“FPGA”), a programmable logic device (“PLD”), a controller, a state machine, gated logic, discrete hardware components, any other processing unit, or any combination or multiplicity thereof. Central processing unit 121 may be a single processing unit, multiple processing units, a single processing core, multiple processing cores, special purpose processing cores, co-processors, or any combination thereof. Central processing unit 121 along with other components of the computing device 120 may be a virtualized computing machine executing within one or more other computing machine(s).

Computing device 120 comprises a graphics processing unit 122 that serves to accelerate rendering of graphics and images in two- and three-dimensional spaces. Graphics processing unit 122 can process multiple images, or data, simultaneously for use in machine learning and high-performance computing.

Computing device 120 comprises a memory 123. Memory 123 may include non-volatile memories, such as read-only memory (“ROM”), programmable read-only memory (“PROM”), erasable programmable read-only memory (“EPROM”), flash memory, or any other device capable of storing program instructions or data with or without applied power. Memory 123 may also include volatile memories, such as random access memory (“RAM”), static random access memory (“SRAM”), dynamic random access memory (“DRAM”), and synchronous dynamic random access memory (“SDRAM”). Other types of RAM also may be used to implement memory 123. Memory 123 may be implemented using a single memory module or multiple memory modules. While memory 123 is depicted as being part of the computing device 120, memory 123 may be separate from the computing device 120 without departing from the scope of the subject technology.

Computing device 120 comprises a communication application 124. Communication application 124 interacts with web servers or other computing devices or systems connected via network 140, including data processing system 130.

Facial expression reconstruction system 100 comprises a data processing system 130. Data processing system 130 serves to receive images or data from cameras 110 via computing devices 120 and network 140. Data processing system 130 comprises a central processing unit 131, a modeling application 132, a data storage unit 133, and a communication application 134.

Data processing system 130 comprises central processing unit 131 configured to execute code or instructions to perform the operations and functionality described herein, manage request flow and address mappings, and perform calculations and generate commands. Central processing unit 131 may be configured to monitor and control the operation of the components in the data processing system 130. Central processing unit 131 may be a general purpose processor, a processor core, a multiprocessor, a reconfigurable processor, a microcontroller, a digital signal processor (“DSP”), an application specific integrated circuit (“ASIC”), a graphics processing unit (“GPU”), a field programmable gate array (“FPGA”), a programmable logic device (“PLD”), a controller, a state machine, gated logic, discrete hardware components, any other processing unit, or any combination or multiplicity thereof. Central processing unit 131 may be a single processing unit, multiple processing units, a single processing core, multiple processing cores, special purpose processing cores, co-processors, or any combination thereof. Central processing unit 131 along with other components of the data processing system 130 may be a virtualized computing machine executing within one or more other computing machine(s).

Data processing system 130 comprises a modeling application 132. The modeling application 132 employs a variety of tools, applications, and devices for machine learning applications. The modeling application 132 may receive a continuous or periodic feed of images or data from one or more of the computing device(s) 120, the central processing unit 131, or the data storage unit 133. Collecting the data allows the modeling application 132 to leverage a rich dataset to use in the development of a training set of data or ground truth for further use in facial expression reconstructions. The modeling application 132 may use one or more machine learning algorithm(s) to develop facial expression reconstructions, such as a convolution neural network (“CNN”), Naïve Bayes Classifier, K Means Clustering, Support Vector Machine, Apriori, linear regression, logistic regression, decision trees, random forest, or any other suitable machine learning algorithm.

Data processing system 130 comprises a data storage unit 133. Data storage unit 133 may be accessible by the modeling application 132 and the communication application 134. The example data storage unit 133 can include one or more tangible computer-readable storage device(s). The data storage unit 133 can be within the data processing system 130 or can be logically coupled to the data processing system 130. For example, the data storage unit 133 can include on-board flash memory and/or one or more removable memory device(s) or removable flash memory. In certain embodiments, the data storage unit 133 may reside in a cloud-based computing system.

Data processing system 130 comprises a communication application 134. Communication application 134 interacts with web servers or other computing devices or systems connected via network 140, including the computing devices 120 and user computing device 150.

User computing device 150 is a computing device configured to receive and communicate results of facial expression reconstruction or facial movement. The results of the

facial expression reconstruction may be displayed as a graphical representation of a facial expression, such as **630** of FIG. **6**, a predicted facial expression with three-dimensional visualization, such as **730** of FIG. **7**, a three-dimensional facial expression, such as **1530A** of FIG. **15**, an emoji, text or audio representing silent speech recognition, or any other suitable display of facial expression reconstructions results. The user computing device **150** also may comprise a speaker to audibly communicate the facial expression reconstructions results directed to silent speech recognition. In an alternate embodiment by tracking facial movement, user computing device **150** may determine skin deformation or stretching and, subsequently, directly determine one or more of facial expression(s), eyeball movement, or eye-related disease. By tracking facial movement, user computing device **150** may recognize, or make audible, silent speech, and recognize and track human activities involving facial movements such as eating or drinking. By tracking facial movement, user computing device **150** may synthesize a user's voice or determine a user's emotional status.

FIGS. **2A** and **2B** are perspective views of head-mounted wearable devices **210**, **220**, respectively, for facial expression reconstruction, in accordance with certain examples.

FIG. **2A** depicts a headphone-style wearable device **210** (herein referred to as headphone device **210**) as a form factor to acquire images or data for use in facial expression reconstruction, in accordance with certain examples. Headphone device **210** comprises two cameras **110-1** and **110-2** to acquire images of a left and a right contour, respectively. Cameras **110** were previously described herein with reference to FIG. **1**. In an example, headphone device **210** is a device constructed of a band **211** suitable to wear across the top of a user's head with a left-side ear piece **212-1** and a right-side ear piece **212-2**. In an example, headphone device **210** may be over-the-ear headphones or outside-the-ear headphones. In an example, headphone device **210** may include functionality with Bluetooth or Wi-Fi connectivity and speakers embedded within the earpieces **212**. Camera **110-1** is connected to left-side ear piece **212-1**, and camera **110-2** is connected to right-side ear piece **212-2**. In an alternate example, two cameras **110** are connected to each of the left-side ear piece **212-1** and right-side ear piece **212-2**. Any suitable number of cameras may be connected to each of the ear pieces **212**. Each camera **110** is in communication with a computing device **120**, as previously described with respect to FIG. **1**. In an example, computing device **120** may be connected to each camera **110** and be embedded within the headphone device **210**. In an alternate example, each camera **110** may communicate with a computing device **120** via a wireless technology, such as Bluetooth or Wi-Fi.

Headphone device **210** is configured to acquire images of a left and a right contour of a user's face by directing cameras **110-1** and **110-2** at adjustable angles and positions. Each of the cameras **110** are independently adjustable by a first angle **214**, a slide **216**, and second angle **218**. First angle **214** adjusts a tilt of the position of the camera **110** relative to a plane parallel to an exterior surface of a user's ear. The first angle **214** may be adjusted so that the camera **110**/earpiece **212** assembly is tilted closer to a user's ear, or the first angle **214** may be adjusted so that the camera **110** via the earpiece **212** is tilted farther away or offset from a user's ear. In an example, first angle **214** may be 0° indicating that the camera **110**/earpiece **212** assembly is in a vertical position relative to a plane parallel to the exterior surface of a user's ear. First angle **214** may be adjusted by -10° , -20° , -30° , -40° , or any suitable angle measure relative to the

plane parallel to the exterior surface of a user's ear, such that each camera **110** may be aligned with the left and right contours of a user's face.

Slide **216** adjusts a position of the camera **110** relative to the earpiece **212** in a direction that is perpendicular to a plane parallel to the exterior surface of a user's ear, in other words, the position of the camera **110** may change along the slide **216** while the position of the earpiece **212** is fixed. The position of the camera **110** via the slide **216** may be adjusted such that the earpiece **212** is in close contact with an exterior surface of a user's ear. The position of the camera **110** via the slide **216** may be adjusted such that the camera **110** is extended a distance away from the plane parallel to the exterior surface of a user's ear. In an example, the extended distance may be 1 cm, 2 cm, 3 cm, or any suitable distance away from the plane parallel to the exterior surface of a user's ear. The slide **216** positions the cameras **110** in a manner similar to positioning the earpieces **212** of the headphone device **210**. In some embodiments, the position of an imaging sensor or a camera (for example, the optical center of the lens in the camera) is less than 5 cm, less than 4 cm, less than 3 cm, less than 2 cm, less than 1 cm, or less than 0.5 cm away from the surface plane of a user's ear. In some embodiments, the position of an imaging sensor or a camera projected to the closest skin surface is within the region of a user's ear or less than 5 cm, less than 4 cm, less than 3 cm, less than 2 cm, or less than 1 cm away from the nearest contour edge of a user's ear. In some embodiments, the system comprises at least one, at least two, or at least three imaging sensors (for example, cameras) located in different positions as described above. In some embodiments, an imaging sensor, such as a camera, is positioned below a chin of a user and less than 25 cm, less than 20 cm, less than 15 cm, less than 10 cm, or less than 5 cm away from the chin of a user, or 2-30 cm below and away from the chin of the user.

Second angle **218** adjusts a rotational position of the camera **110** along a horizontal axis through earpieces **212-1** and **212-2**. In an example, second angle **218** adjusts an angular position of the camera **110** relative to the horizontal axis while the position of the earpiece **212** remains unchanged. In an alternate example, second angle **218** adjusts an angular position of the camera **110**/earpiece **212** assembly. Relative to a left or right contour of a user's face, second angle **218** may be 0° indicating that the camera **110** is in a horizontal position. A second angle **218** of 10° indicates that the camera **110** is directed 10° upwards. A second angle **218** of -10° indicates that the camera **110** is directed 10° downwards. Any suitable measure of second angle **218** may be used to align the cameras **110** with the contour of a user's face.

Each of the cameras **110** are independently adjustable by the first angle **214** and the second angle **218**, by any suitable mechanism, for example, by mounting the cameras **110** to the headphone device **210** via rotational positioning devices allowing incremental changes of the direction of the cameras **110**.

FIG. **2B** depicts earbud-style wearable devices **220-1** and **220-2** (herein referred to as earbud devices **220**) as a form factor to acquire images or data for use in facial expression reconstruction, in accordance with certain examples. Earbud device **220-1** is constructed for wear as a left-side earbud, and earbud device **220-2** is constructed for wear as a right-side earbud. In an example, earbud device **220** may be ear pods, earphones, or in-the-ear ("ITE") headphones. Earbud device **220-1** comprises a camera **110-1**, and earbud device **220-2** comprises a camera **110-2** to acquire images of

11

a left and a right contour, respectively, of a user's face by directing cameras **110-1** and **110-2** at the left and right contours of a user's face. Cameras **110-1** and **110-2** were previously described in reference to FIG. 1. In an example, earbud devices **220** may include functionality with Bluetooth or Wi-Fi connectivity and speakers embedded within the earbuds **220**. Each camera **110** is in communication with a computing device **120**, as previously described with respect to FIG. 1. In an example, a computing device **120** may be connected to each camera **110** and be embedded within each earbud device **220**. In an alternate example, each camera **110** may communicate with a computing device **120** via a wireless technology, such as Bluetooth or Wi-Fi.

The camera **110** position for each earbud device **220** may be controlled by twisting and/or rotating the earbud device **220** in the user's ear. The earbud device **220** may be rotated such that the camera **110** is angled closer to the contour of a user's face. In an alternate example, the camera **110** may be attached to the earbud device **220** such that the camera **110** may be positioned independently of the earbud device **220**. The camera **110** may be attached to the earbud device **220** with a ball and socket joint or any other suitable attachment method such that the position of the camera **110** may be adjusted independently of the earbud device **220**.

FIGS. 3A, 3B, and 3C are illustrations depicting examples **310-350** of head-mounted, wearable devices **210**, **220** on a user, camera views **330**, captured facial contours **340**, and corresponding facial expressions **350**, in accordance with certain examples. In FIG. 3A, earbuds **220** are illustrated in a closeup view of the back of a user's ear. Camera **110** of the earbuds **220** is illustrated in this example as being positioned in a forward direction towards a right contour of a user's face. Although not depicted in FIG. 3A, a second earbud **210** is worn by the user in the opposite, left ear, having a camera **110** positioned in a forward direction towards a left contour of the user's face.

Also, in FIG. 3A, alternative headphone **210** is illustrated in a closeup view from a front perspective of the user. Cameras **110-1** and **110-2** of the headphone **210** are illustrated in this example as being positioned in a forward direction towards left and right contours of a user's face.

FIG. 3B depicts a back and side view of a user to illustrate the contour of the user's face from the point of view of a camera **110** in either the earbud device **220** or headphone device **210** depicted in FIG. 3A.

FIG. 3C depicts the left and right contours of a user's face as captured by cameras **110** through the use of the headphone **210** or the earbuds **220**. FIG. 3B also depicts the facial expression of a user reconstructed through the images captured by cameras **110** through the use of the headphone **210** or the earbuds **220**.

The computing devices **120**, **130**, and **150** and any other network computing devices other computing machines associated with the technology presented herein may be any type of computing machine, such as, but not limited to, those discussed in more detail with respect to FIG. 31. For example, each device can include any suitable processor-driven device.

Furthermore, any functions, applications, or components associated with any of these computing machines, such as those described herein or others (for example, scripts, web content, software, firmware, hardware, or modules) associated with the technology presented herein may be any of the components discussed in more detail with respect to FIG. 31.

The network connections illustrated are examples and other means of establishing a communications link between

12

the computers and devices can be used. The computing machines discussed herein may communicate with one another, as well as with other computing machines or communication systems over one or more network(s). Each network may include various types of data or communications networks, including any of the network technology discussed with respect to FIG. 31.

Additionally, those having ordinary skill in the art having the benefit of the present disclosure will appreciate that the devices illustrated in the figures may have any of several other suitable computer system configurations.

Example Processes

The methods illustrated in FIGS. 4 through 11 are described hereinafter with respect to the components of facial expression reconstruction system **100**, headphone device **210** of FIG. 2A, and earbud devices **220** of FIG. 2B. The methods of FIGS. 4 through 11 may also be performed with other systems and in other environments. The operations described with respect to FIGS. 4 through 11 can be implemented as executable code stored on a computer- or machine-readable, non-transitory, tangible storage medium (for example, floppy disk, hard disk, ROM, EEPROM, nonvolatile RAM, CD-ROM, etc.) that are completed based on execution of the code by a processor circuit implemented using one or more integrated circuit(s); the operations described herein also can be implemented as executable logic that is encoded in one or more non-transitory, tangible media for execution (for example, programmable logic arrays or devices, field programmable gate arrays, programmable array logic, application specific integrated circuits, etc.).

FIG. 4 is a block diagram depicting a deep-learning model process **400**, in accordance with certain examples. The deep-learning model process **400** may be performed by the modeling application **132** of the data processing system **130**, previously described herein in reference to FIG. 1. In an example where the machine learning algorithm is a convolutional neural network ("CNN"), the deep-learning model process **400** receives images as an input and assigns importance (learnable weights and biases) to various aspects/objects in the images. The CNN reduces the images into a form that is easier to process without losing the features of the images that are used to get a good prediction. In the current example, the prediction is a reconstructed facial expression based on the received right and left contour facial images.

The deep-learning model process **400** comprises an image processing phase **410** and a regression phase **450**. In the image processing phase **410**, image processing is divided into two parallel paths. Path A is directed to processing an image of a right facial contour, and path B is directed to processing an image of a left facial contour. The right facial contour images and left facial contour images are processed independently of each other and combined in block **454** of the deep-learning model process **400**.

In block **412**, the data processing system **130** receives an input of an image of a right facial contour and an input of an image of a left facial contour. Example right facial contour images are depicted in row **640** of FIGS. 6 and 7, and example left facial contour images are depicted in row **650** of FIGS. 6 and 7. The images are transmitted through blocks **414-1** through **414-n**. Blocks **414-1** through **414-n** comprise layers of the CNN that include multiple convolution filtering operations. In an example, the CNN may comprise between 32 and 512 layers to extract features from the images. The

13

first convolution layer in block **414-1** captures low-level features, such as edges, color, and gradient orientation. As the process progresses through the additional layers to layer **414-n**, higher-level features are captured such that the object of the image is captured. In the current example, the object of the image is a facial contour.

From block **414-n**, the process proceeds to block **416**. Block **416** is a pooling layer. The pooling layer may be a Max Pooling that returns a maximum value from the image. The pooling layer may be an Average Pooling that returns an average of all of the values from the image. The pooling layer of block **416** outputs a vector representation of each input image (in other words, a right image vector for the right facial contour and a left image vector for the left facial contour).

The right image vector and the left image vector are received as inputs to the regression phase **450**. Within the regression phase **450**, the right image vector and the left image vector are inputs into two fully connected layers **452** with a rectified linear unit (“ReLU”) between the fully connected layer **452**. The fully connected layer **452** learns facial landmarks of the right image vector and the left image vector based on facial landmarks in a training data set, or ground truth, set of facial expressions. The fully connected layer **452** compares features of the right image vector to right side facial landmarks in the training data set of facial expressions to match the features of the right image vector to a right side facial expression in the training data set. Similarly, the fully connected layer **452** compares features of the left image vector to left side facial landmarks in the training data set of facial expressions to match the features of the left image vector to a left side facial expression in the training data set. The fully connected layer **452** outputs landmarks of both the right and left side of a user’s face. The output landmarks are inputs to a matching module **454**. The matching module **454** concatenates the landmarks from the right and left sides by aligning key landmarks that are present in both the right and left sides using translation and scaling. The matching module **454** outputs the final facial expression reconstruction, or predicted facial expression, as a set of facial landmarks as the reconstruction result in block **456**. Examples of predicted facial expressions as a set of facial landmarks are depicted in row **630** of FIG. **6**. In an alternate example, the facial expression reconstruction is further modeled to create a three-dimensional visualization of the predicted facial expression as depicted in row **730** of FIG. **7**. The process **400** continues to repeat as additional right facial contour images and left facial contour images are received.

FIG. **5** is a block diagram depicting a classifier process **500**, in accordance with certain examples. The classifier process **500** classifies the facial expression reconstruction output **456** of the deep-learning model process **400** for use in applications such as silent speech recognition, emoji input, real-time avatar facial expressions, or other suitable applications. In block **510**, the reconstruction output **456** from the deep-learning model process **400** (FIG. **4**) is received as input to the classifier process **500**. In block **510**, a stream of landmark positions from the reconstruction output **456** are received as frames. In block **520**, the landmark positions are segmented. Segmentation extracts an event from a stream of data and labels its start and end time. In the example of silent speech recognition, segmentation would identify the initial facial expression associated with the pronunciation of a word and the final facial expression associated with the pronunciation of the same word. For each frame, two indicators are calculated to measure the

14

degree of transformation for the mouth and eyes, known as the Mouth Transformation Degree (“MTD”) and Eye Transformation Degree (“ETD”). The MTD and ETD are calculated by summing the differences between the landmarks outlining the mouth or eye/eyebrow at the time and the landmarks of the natural facial expression. Then, a lowpass filter is applied to filter out high-frequency noise. When an event happens, the MTD or ETD raises to a single peak or a primary peak with several secondary peaks. A peaking seeking algorithm is applied to find the peaks in the stream. Thresholds are set for peak height, peak prominence, and peak width to filter peaks that may be associated with noise. After finding peak positions, the first-order difference of the ETD and MTD are calculated. In the first-order difference sequence, the left nearest peak to the event is labeled as the beginning, and the right nearest valley to the event is labeled as the end. The frames associated with the event are placed sequentially on a timeline **525** as frames **530-1** through frames **530-n**.

After the sequential placement of frames **530-1** through **530-n**, the process **500** proceeds to block **540** where the frames are inputs into a Bidirectional Long Short-Term Memory (“BLSTM”) model for classification. Block **540** depicts a two-layer BLSTM model. However, any suitable number of layers may be used. The blocks **541-1** through **541-n** depicted within the BLSTM model **540** are recurrently connected memory blocks. Each of the blocks **541** have feedback connections such that sequences of data can be processed as opposed to single images or data points. As depicted in block **540**, blocks **541-1** through **541-n** comprise bidirectional connections within each of the blocks **541** in each layer of block **540**. Processing sequences of data allows for speech recognition, emoji recognition, real-time avatar facial expressions, or other suitable applications. The output from block **540** is a vector representation of the input frames.

In block **550**, the vector representation is received by a fully connected layer comprising a SoftMax function. The SoftMax function transforms the vector representation into a probability distribution to predict a facial event. The output from block **550** is an encoding of a facial event class. The facial event classification may be a facial expression. In the example of silent speech recognition, the facial event classification may be the pronunciation of a specific word or phrase, such as “hello” or “how are you.” In the example of emoji recognition, the facial event classification may be an emoji indicating a smiling face, a frowning face, a crying face, or any other suitable emoji associated with a facial expression. In the example of real-time avatar facial expressions, the facial event classification may be a three-dimensional visualization of a facial expression.

FIG. **6** is a diagram **600** depicting the correlation between frontal view camera images **610**, landmark training data **620**, predicted facial expressions **630**, and captured right and left facial contour images **640**, **650**, in accordance with certain examples. Row **610** illustrates frontal view camera images of a user. To construct a training data set or “ground truth,” frontal view camera images of a user are acquired, via an imaging capturing device, with the user making various different facial expressions as depicted in images **610-1** through **610-n**. The method to create the training data set from the frontal view camera images is described herein in greater detail with reference to method **810** of FIG. **9**. Row **620** illustrates a training data set in the form of facial landmarks. The landmark training data images depicted in images **620-1** through **620-n** correlate with images **610-1**

15

through 610-*n*. The landmark training data images 620 are the output of method 810 of FIG. 9.

Row 640 depicts right facial contour images 640-1 through 640-*n* captured by a head-mounted, wearable device, such as headphone device 210 or earbud device 220 described herein with reference to FIG. 2. Row 650 depicts left facial contour images 650-1 through 650-*n* captured by a head-mounted, wearable device, such as headphone device 210 or earbud device 220 described herein with reference to FIG. 2. The right and left facial contour images 640, 650 are synchronized such that the images are representing the same facial expression. The synchronization of the images is described in greater detail herein with reference to block 1020 of FIG. 10. The synchronized images are used as inputs to the deep-learning process 400, previously described herein with reference to FIG. 4.

The right facial contour images 640, the left facial contour images 650, and the landmark training data images 620 are used in the deep-learning model process 400 to construct the predicted facial expressions in row 630. The predicted facial expressions 630 are depicted as landmark images in images 630-1 through 630-*n*. The predicted facial expressions 630 are illustrated in FIG. 6 beneath the landmark training data images 620 to illustrate the accuracy of predicting facial expressions using right facial contour images 640 and left facial contour images 650 acquired from headphone device 210 or earbud device 220.

FIG. 7 is a diagram 700 depicting the correlation between frontal view camera images 610, training data with three-dimensional visualization 720, predicted facial expressions with three-dimensional visualization 730, and right and left facial contour images 640, 650, in accordance with certain examples. FIG. 7 is an alternate embodiment of FIG. 6, wherein training data images with three-dimensional visualization are depicted in row 720 and predicted facial expressions with three-dimensional visualization are depicted in row 730. The training data images 720-1 through 720-*n* are correlated with predicted facial expression images 730-1 through 730-*n*. The three-dimensional visualizations depicted in blocks 720 and 730 are created using the facial landmarks described in rows 620 and 630 in reference to FIG. 6. The facial landmarks are input into a three-dimensional modeling application to create a three-dimensional visualization of the training data and the predicted facial expressions.

FIG. 6 and FIG. 7 present two-dimensional and three-dimensional correlations between training data and predicted facial expressions. Any suitable correlation can be made based on the type of camera used and the type of data collected. For example, correlations can be made based on imaging data such as thermal gradients or acoustic reflections.

FIG. 8 is a block flow diagram depicting a method 800 to reconstruct facial expressions, in accordance with certain examples. In block 810, a data training set is created using frontal view digital images. Block 810 is described in greater detail in method 810 of FIG. 9.

FIG. 9 is a block flow diagram depicting a method 810 to create a data training set using frontal view digital images, in accordance with certain examples. In block 910, one or more frontal view digital image(s) are acquired of a user. The one or more frontal view digital image(s) depict the user making a variety of facial expressions. For example, the user may smile, frown, squint, close one or both eyes, make an angry facial expression, raise one or both eyebrows, or any other suitable expressions such that the one or more frontal view digital image(s) represent a wide range of potential

16

facial expressions. Example frontal view digital images are depicted in row 610 of FIG. 6. The user may take the one or more frontal view digital image(s) of himself, another person may take the one or more frontal view digital image(s), or any other suitable method to acquire the one or more frontal view digital image(s) may be used. The one or more frontal view digital image(s) may be taken by any device or sensor capable of capturing a digital image including, but not limited to, depth cameras, red-green-blue (“RGB”) cameras, infrared (“IR”) sensors, thermal imaging sensors, charge-coupled devices (“CCDs”), complementary metal oxide semiconductor (“CMOS”) devices, active-pixel sensors, radar imaging sensors, fiber optics, and any other suitable image device technology.

In block 920, the frontal view digital images are transmitting to data processing system 130. The frontal view digital images may be transmitted to the data processing system 130 by a user’s computing device, such as user computer device 150, or any other suitable device to transmit data.

In block 930, the data processing system 130 receives the frontal view digital images.

In block 940, the data processing system 130 extracts facial landmark positions. To extract the facial landmark positions, the data processing system 130 uses a computer vision library as a ground truth acquisition method. In an example, the computer vision library is a Dlib library. The computer vision library is used to extract key feature points or landmarks from the frontal view digital images. The computer vision library may extract 42, 68, or any suitable number of key feature points.

In block 950, the data processing system 130 aligns the extracted landmark positions using an affine transformation. Aligning the extracted landmark positions accounts for variations in a user’s head position in the frontal view digital images. When a user acquires the frontal view digital images as described in block 910, the same facial expressions may vary if the user slightly changes his face orientation. To align the extracted landmark positions, a set of landmarks are selected whose relative positions change very little when making facial expressions. For example, the selected landmarks may be one or more of a right canthus, left canthus, or apex of the nose. The selected landmarks are used to calculate an affine matrix for each frontal view digital image. The landmark positions are aligned to the same range to reduce the influence of head position change when the frontal view digital images are acquired.

In block 960, the data processing system 130 creates the data training set based on the aligned facial landmark positions. In an example, after the extracted landmark positions are aligned, the data processing system 130 selects the most informative feature points from the extracted landmark positions. In an example, the Dlib library may extract 68 facial landmarks from each of the frontal view digital images. When making facial expressions, changes mainly occur in the areas around the mouth, eyes, and eyebrows. The less informative features points may be removed, leaving a smaller set of facial landmarks, such as 42 facial landmarks. Any suitable number of facial landmarks may be used. The data training set is the set of the most informative landmark positions for each facial expression from each of the frontal view digital images. Example data training set images are depicted in row 620 of FIG. 6. In an alternate example, the facial expression reconstruction is further modeled to create a three-dimensional visualization of the training data images as depicted in row 720 of FIG. 7. In an example, the data processing system saves the data training

17

set in the data storage unit **133**. In some embodiments, the system may extract information of at least 10, at least 20, at least 30, at least 40, at least 50, at least 60, at least 70, at least 80, at least 90, or at least 100 parameters or features from one or more image(s) captured by one or more imaging sensor(s) for a facial expression, wherein the parameters or features comprise facial landmark positions, shape parameters (for example, blendshape parameters), orientations (for example, head orientations), or any combination thereof.

FIG. 6 depicts frontal view digital images of a user at row **610**, which are used to create the corresponding data training set depicted in row **620**. In an alternate example, training data with three-dimensional visualization, as depicted in row **720** of FIG. 7, may be utilized. Any suitable method or visualization may be used to create the data training set.

From block **960**, the method **810** returns to block **820** of FIG. 8.

Referring back to FIG. 8, in block **820**, a wearable device captures one or more facial digital image(s). In an example, the wearable device may be a head-mounted device, such as headphone device **210** or earbud device **220**, previously described herein with reference to FIG. 2. The head-mounted device is configured to capture right facial contour images and left facial contour images of a user's face directed towards the right and left cheekbones using two or more cameras **110**. In the current example, the right and left facial contour images are two-dimensional images of the contour of a user's face. Example right and left facial contour images are depicted in rows **640** and **650**, respectively, of FIGS. 6 and 7.

In an alternate example, the wearable device may be a neck-mounted device such as necklace device **1210** or neckband device **1220**, described hereinafter with reference to FIG. 12. The neck-mounted device is configured to take digital images of the contour of a user's face directed towards the user's chin using one or more camera(s) **110**.

In block **830**, the wearable device transmits the one or more facial digital image(s) to one or more data acquisition computing device(s) **120**. As depicted and described in reference to FIG. 1, each camera **110** is in communication with a data acquisition computing device **120**.

In block **840**, the one or more data acquisition computing device(s) **120** transmit the one or more facial digital image(s) to the data processing system **130**.

In block **850**, the data processing system **130** receives the one or more facial digital image(s).

In block **860**, the data processing system **130** reconstructs facial expressions using the one or more facial digital image(s). Block **860** is described in greater detail herein with reference to method **860** of FIG. 10.

FIG. 10 is a block flow diagram depicting a method **860** to reconstruct facial expressions using one or more facial digital image(s), in accordance with certain examples. The examples of FIG. 10 are directed to the method to reconstruct facial expressions from one or more facial digital image(s) received from a head-mounted wearable device. An alternate method to reconstruct facial expressions from one or more facial digital image(s) received from a neck-mounted wearable device is described herein with reference to method **860'** of FIG. 17.

In block **1010**, the data processing system **130** receives the one or more facial digital image(s).

In block **1020**, the data processing system **130** creates one or more pair(s) of synchronized facial digital images from the one or more facial digital image(s). In the example where the one or more facial digital image(s) are acquired from a head-mounted wearable device, the wearable device cap-

18

tures right facial contour images and left facial contour images. To accurately reconstruct a facial expression of a user, the data processing system **130** synchronizes the images from the left camera **110-1** and the right camera **110-2** such that each pair of right and left facial contour images represents a particular facial expression.

In block **1030**, the data processing system **130** pre-processes each pair of synchronized facial digital images. Block **1030** is described in greater detail herein with reference to method **1030** of FIG. 11.

FIG. 11 is a block flow diagram depicting a method **1030** to pre-process each pair of synchronized facial digital images, in accordance with certain examples. In block **1110**, the data processing system **130** converts a digital image color space of each pair of synchronized facial digital images from a red-green-blue ("RGB") color space to a luminance, chrominance (red-yellow), chrominance (blue-yellow) ("YCrCb") color space. The YCrCb color space is used to take advantage of a lower resolution with respect to luminosity amount of data stored with respect to each pair of synchronized facial digital images. The conversion from the RGB color space to the YCrCb color space is accomplished using a standard process.

In block **1120**, the data processing system **130** extracts skin color from the background of each converted pair of facial digital images. In an example, the skin color is extracted using Otsu's thresholding method. Otsu's thresholding method determines whether pixels in an image fall into a foreground or a background. In the current example, the foreground represents a facial contour of each facial digital image, and the background represents an area of the image outside of the facial contour.

In block **1130**, the data processing system **130** binarizes each facial digital image after the extraction of the skin color from the background. Image binarization is the process of taking the image in YCrCb color space and converting it to a black and white image. The binarization of the image allows for an object to be extracted from an image, which in this example is a facial contour.

In block **1140**, the data processing system **130** filters the binarized digital images to remove noise from the images. Filtering the binarized digital images produces a smoother image to assist in more accurate facial expression reconstructions.

From block **1140**, the method **1030** returns to block **1040** of FIG. 10. Referring back to FIG. 10, in block **1040**, the data processing system **130** applies a deep-learning model to each pair of pre-processed facial digital images for facial expression reconstruction. The deep-learning model for facial expression reconstruction was described in detail herein in the deep-learning model process **400** with reference to FIG. 4 and in the classifier process **500** with reference to FIG. 5. The results of the facial expression reconstruction from block **1040** may include the reconstruction output from block **456** of FIG. 4 and/or the encoding of a facial event class from block **550** of FIG. 5. For example, the data processing system **130** may output from block **1040** the landmark images from the process **400** of FIG. 4. The data processing system **130** also, or alternatively, may further process the facial expression reconstruction such that the output from block **1040** comprises a three-dimensional visualization of the facial expression, silent speech recognition, an emoji associated with the facial expression reconstruction, and/or real-time avatar facial expressions associated with the facial expression reconstruction, such as described by the classifier process **500** of FIG. 5.

19

From block 1040, the method 860 returns to block 870 of FIG. 8. Referring back to FIG. 8, in block 870, the data processing system 130 outputs the results of the facial expression reconstruction. In an example, the results may be sent to user computing device 150 to be displayed visually or audibly to a user. In an example, the results of the facial expression reconstruction may be displayed as a graphical representation of a facial expression, such as row 630 of FIG. 6; a predicted facial expression with three-dimensional visualization, such as row 730 of FIG. 7; a three-dimensional facial expression, such as row 1530A of FIG. 15; an emoji; text representing silent speech recognition; or any other suitable display of facial expression reconstruction results. User computing device 150 may comprise a speaker to audibly communicate the facial expression reconstruction results directed to silent speech recognition.

From block 870 of FIG. 8, the method 800 returns to 820 where the method 800 repeats the process of reconstructing facial expressions by capturing additional facial digital images.

Other Example Architectures

FIGS. 12A, 12B, and 12C are perspective views of neck-mounted wearable devices 1210, 1220 for facial expression reconstruction with example IR images 1230 and 1240, in accordance with certain examples.

FIG. 12A depicts a necklace style neck-mounted wearable device 1210 (herein referred to as necklace device 1210) as a form factor to acquire images or data for use in facial expression reconstruction. Necklace device 1210 comprises a camera 110, previously described herein in reference to FIG. 1. In the current example, camera 110 is an infrared ("IR") camera. Necklace device 1210 also comprises an IR LED 1212 and an IR bandpass filter 1214. Necklace device 1210 is configured to acquire images of a user's chin profile by directing the camera 110 at the profile of a user's chin. The IR LED 1212 projects IR light onto a user's chin to enhance the quality of the image captured by camera 110. The IR bandpass filter 1214 filters visible light such that camera 110 captures infrared light reflected by the skin of a user.

Necklace device 1210 comprises a chain 1218 or other suitable device for securing the necklace device 1210 about the neck of a user. In an alternate example, necklace device 1210 may attach to a user's clothing instead of being secured about the neck of a user. In the current example, necklace device 1210 may be attached to a user's clothing underneath the user's chin. In an alternate example, multiple necklace devices 1210 may be attached to a user's clothing to capture camera 110 images from multiple viewpoints. In the current example, a necklace device 1210 may be attached on a user's clothing close to each shoulder of the user. The necklace device 1210 may comprise a clip, a pin, a clasp, or any other suitable device to attach necklace device 1210 to a user's clothing.

The camera 110 is in communication with a computing device 120, as previously described with respect to FIG. 1. In an example, a computing device 120 may be connected to the camera 110 and be embedded within the necklace device 1210. In an alternate example, the camera 110 may communicate with a computing device 120 via a wireless technology, such as Bluetooth or Wi-Fi.

FIG. 12B depicts a neckband style neck-mounted wearable device 1220 (herein referred to as neckband device 1220) as a form factor to acquire images or data for use in facial expression reconstruction. Neckband device 1220

20

comprises two cameras 110-1 and 110-2 fashioned to be positioned on the left and right sides of a user's neck. Neckband device 1220 also comprises IR LEDs and IR bandpass filters configured in proximity to each camera 110-1 and 110-2 (not depicted in FIG. 12). In an example, neckband device 1220 may include functionality with Bluetooth or Wi-Fi connectivity and speakers embedded within the neckband 1220. Each camera 110 is in communication with a computing device 120, as previously described with respect to FIG. 1. In an example, a computing device 120 may be connected to each camera 110 and be embedded within the neckband device 1220. In an alternate example, each camera 110 may communicate with a computing device 120 via a wireless technology, such as Bluetooth or Wi-Fi.

FIG. 12C depicts two example IR images 1230 and 1240. Example 1230 is an IR image acquired from necklace device 1210. Example 1240 is a right camera 110 IR image acquired from neckband device 1220. Additional examples of IR images acquired by either necklace device 1210 or neckband device 1220 are depicted in row 1540A of FIG. 15A and row 1540B of FIG. 15B.

FIGS. 13A and 13B are illustrations 1300 depicting examples 1310-1360 of neck-mounted wearable devices 1210, 1220 on a user, camera views 1320 and 1360, and corresponding facial expressions 1330, in accordance with certain examples. In FIG. 13A, example 1310 illustrates a necklace device 1210 as being worn by a user in a front view of the user. Example 1320 illustrates a view of the user as captured by camera 110 of the necklace device 1210. The corresponding example 1330 illustrates a reconstructed facial expression based on IR images captured by the necklace device 1210.

In FIG. 13B, example 1340 illustrates a neckband device 1220 as being worn by a user in a front view of the user. Example 1350 illustrates a neckband device 1220 as being worn by a user in a right-side view of the user. Example 1360 illustrates two views of the user as captured by cameras 110-1 and 110-2 of the neckband device 1220. The corresponding example 1330 illustrates a reconstructed facial expression based on IR images captured by the necklace device 1220.

Other Example Processes

The methods illustrated in FIGS. 14 through 18 are described hereinafter with respect to the components of facial expression reconstruction system 100, necklace device 1210 of FIG. 12A, and neckband device 1220 of FIG. 12B. The methods of FIGS. 14 through 18 may also be performed with other systems and in other environments. The operations described with respect to FIGS. 14 through 18 can be implemented as executable code stored on a computer- or machine-readable, non-transitory, tangible storage medium (for example, floppy disk, hard disk, ROM, EEPROM, nonvolatile RAM, CD-ROM, etc.) that are completed based on execution of the code by a processor circuit implemented using one or more integrated circuit(s); the operations described herein also can be implemented as executable logic that is encoded in one or more non-transitory, tangible media for execution (for example, programmable logic arrays or devices, field programmable gate arrays, programmable array logic, application specific integrated circuits, etc.).

FIG. 14 is a block diagram depicting an alternate embodiment of a deep-learning model process 1400, in accordance with certain examples. The deep-learning model process 1400 may be performed by the modeling application 132 of

21

the data processing system 130, previously described herein with reference to FIG. 1. In an example where the machine learning algorithm is a convolution neural network (“CNN”), the deep-learning model process 1400 receives images as an input and assigns importance (learnable weights and biases) to various aspects/objects in the images. The CNN reduces the images into a form that is easier to process without losing the features of the images that are critical for getting a good prediction. In the current example, the prediction is a reconstructed facial expression based on received IR images of a user’s chin profile.

In block 1410, the data processing system 130 receives IR images from either the necklace device 1210 or the neckband device 1220. Example necklace 1210 IR images are depicted at 1410-1. Example neckband 1220 IR images are depicted at 1410-2 and 1410-3 as the neckband acquires both a right and left side IR image of a user’s chin profile. Other example IR images are depicted in row 1540A of FIG. 15A and row 1540B of FIG. 15B.

In block 1420, the data processing system 130 pre-processes the IR images. Pre-processing the IR images is described in greater detail in reference to method 1720 of FIG. 18.

In block 1425, the data processing system 130 duplicates the pre-processed IR images of the necklace device 1210 into three channels to improve the expressiveness and the ability to extract features of the model. As the neckband device 1220 already comprises two pre-processed images, the images are not duplicated into additional channels.

The pre-processed IR images are input into an image processing phase of the deep-learning model process 1400 depicted at block 1430. Block 1430 comprises convolution layers, normalization layers, and an averaging pooling layer. The processing of block 1430 is described in greater detail herein with reference to blocks 414 and 416 of FIG. 4. The output from block 1430 is a vector representation of each of the pre-processed IR images.

The vector representations of each of the pre-processed IR images are input into a regression phase 1440 of the deep-learning model process 1400. The architecture of the regression phase 1440 is similar to the architecture of regression phase 450, previously described herein with reference to FIG. 4. Within the regression phase 1440, the data processing system 130 extracts details from the vector representations of the pre-processed IR images regarding facial components including the cheek, mouth, eyes, eyebrows, chin, and nose and three-dimensional head rotation angles. The data processing system 130 represents each facial expression with blend shapes that are a three-dimensional representation of the vector representations. The parameters of the blend shapes and the three-dimensional head rotation angles, or Euler angles, are obtained by learning from the IR images captured from the neck. The data processing system 130 compares the blend shapes and Euler angles from the vector representations to the blend shapes and Euler angles of the data training set to match the blend shapes/Euler angles to a facial expression from the data training set.

In block 1450, the data processing system 130 combines the blend shapes with three-dimensional angles of rotation of the user’s head. In an example, the three-dimensional angles of rotation are represented by Euler’s angles of roll, yaw, and pitch. In block 1460, the final facial expression reconstruction is output as a three-dimensional image. Example three-dimensional facial expression reconstructions are depicted in row 1530A of FIG. 15A and row 1530B of FIG. 15B. In an example, the three-dimensional facial expression reconstructions may be further processed by a

22

classifier process, such as classifier process 500 for applications such as silent speech recognition, emoji input, real-time avatar facial expressions, or any other suitable applications.

FIG. 15A is a diagram 1500A depicting the correlation between frontal view camera images 1510A, three-dimensional training data 1520A, predicted three-dimensional face expressions 1530A, and infrared facial images 1540A, in accordance with certain examples.

Row 1510A illustrates frontal view camera images of a user. To construct a training data set or ground truth, frontal camera images of a user are acquired with the user making various different facial expressions, as depicted in images 1510A-1 through 1510-n. The method to create the training data set from the frontal view camera images is described herein in greater detail with reference to method 810' of FIG. 16. Row 1520A illustrates three-dimensional training data in three-dimensional blend shapes and head rotation angles. The three-dimensional training data images depicted in images 1520A-1 through 1520A-n correlate with images 1510A-1 through 1510-n. The three-dimensional training data images 1520A are the output of method 810' of FIG. 16.

Row 1540A depicts IR images 1540A-1 through 1540A-n captured by a neck-mounted wearable device, such as necklace device 1210 or neckband device 1220 described herein in greater detail in reference to FIG. 12. The IR images 1540A, and the three-dimensional training data images 1520A are used in the deep-learning model process 1400 to construct the predicted facial expressions 1530A illustrated in row 1530A of FIG. 15A. The predicted facial expressions 1530A are depicted as three-dimensional images in images 1530A-1 through 1530A-n. The predicted facial expressions 1530A are illustrated in FIG. 15A beneath the three-dimensional training data images 1520A to illustrate the accuracy of predicting facial expressions using IR images 1540A acquired from necklace device 1210 or neckband device 1220.

FIG. 15B is a diagram 1500B depicting the correlation between additional frontal view camera images 1510B, three-dimensional training data 1520B, predicted three-dimensional face expressions 1530B, and infrared facial images 1540B, in accordance with certain examples. Each of the frontal view camera images 1510B, three-dimensional training data 1520B, predicted three-dimensional face expressions 1530B, and infrared facial images 1540B correspond to the frontal view camera images 1510A, three-dimensional training data 1520A, predicted three-dimensional face expressions 1530A, and infrared facial images 1540A described herein with reference to FIG. 15A, except these items correlate to different facial expressions.

In this example, the images captured by the cameras of the neck-mounted devices can be processed for facial reconstruction similarly to the methods discussed previously with reference to FIGS. 8-11. Various alternatives to the methods of FIGS. 8-11 will be discussed.

FIG. 16 is a block flow diagram depicting an alternative method 810' to create a data training set using frontal view images, in accordance with certain examples.

Blocks 910, 920, and 930 of FIG. 16 were previously described herein with reference to FIG. 9.

In block 1640, the data processing system 130 extracts a set of facial geometric features from the one or more frontal view digital image(s). In an example, the one or more frontal view digital image(s) are three-dimensional digital images captured from a camera that provides depth data in real time along with visual information. Example frontal view digital images are depicted in rows 1510A and 1510B, respectively,

of FIGS. 15A and 15B. In an example, the data processing system 130 extracts full facial expressions using an augmented reality ("AR") application. The AR application extracts high-quality and fine-grained details of facial expressions/movements of the eyes, eyebrows, cheeks, mouth, nose, and chin. The AR application can extract facial features such as facial geometry and rotation from a depth map associated with the three-dimensional digital images. In an example, the rotation is represented by Euler's angles of roll, yaw, and pitch.

In block 1650, the data processing system 130 compares the extracted features to pre-defined shape parameters. In an example, the AR application comprises pre-defined blend shapes as templates for complex facial animations. In the example, the AR application comprises blend shapes with features for left and right eyes, mouth and jaw movement, eyebrows, cheeks, nose, tongue, and any other suitable facial features.

In block 1660, the data processing system 130 creates a data training set based on the comparison of the extracted features to the pre-defined shape parameters. The data training set comprises a blend shape with a Euler angle of head rotation as depicted in block 1520A of FIG. 15A and block 1520B of FIG. 15B. In an example, the data processing system 130 stores the data training set in the data storage unit 133.

From block 1660, the method 810' returns to block 820 of FIG. 8.

FIG. 17 is a block flow diagram depicting an alternative method 860' to reconstruct facial expressions using one or more facial digital image(s), in accordance with certain examples.

Block 1010 of FIG. 17 was previously described herein with reference to FIG. 10.

In block 1720, the data processing system 130 pre-processes the one or more facial digital image(s). Example facial digital images are depicted in rows 1540A and 1540B of FIGS. 15A and 15B, respectively. Block 1720 is described in greater detail herein with reference to method 1720 of FIG. 18.

FIG. 18 is a block flow diagram depicting an alternative method 1720 to pre-process the one or more digital facial image(s), in accordance with certain examples.

In block 1810, the data processing system 130 converts each of the one or more digital facial image(s) into gray-scale digital facial images. The one or more digital facial image(s) are converted to gray-scale to remove any potential color variance. As the IR bandpass filter 1214 only allows monochrome light into the camera 110, any color present in the one or more digital facial image(s) does not represent details related to the facial expression of the user.

In block 1820, the data processing system 130 separates the facial image from the background image in each of the gray-scale digital facial images. Using the IR technology previously discussed in reference to FIG. 12, the user's skin in the images appears brighter than the surrounding background. However, some light sources in the background may contain IR light that would introduce noise in the background. To remove the noise, the gray-scale digital facial images are binarized based on a set brightness threshold. The data processing system 130 separates the facial image as the brighter component of the gray-scale digital facial image.

In block 1830, the data processing system 130 applies data augmentation to each of the separated facial images. As a user wears a necklace device 1210 or a neckband device 1220 and performs activities, the position and angles of the

cameras 110 of the necklace device 1210 and neckband device 1220 may not be constant. To mitigate the issue, a probability of 60% to conduct three types of image transformations is set, which can be caused by camera shifting causing translation, rotation, and scaling on the images. In an alternate example, any suitable probability may be used. Three Gaussian Models are deployed to generate the parameters for the translation ($\mu=0$, $\sigma^2=30$), rotation ($\mu=0$, $\sigma^2=10$), scaling ($\mu=1$, $\sigma^2=0.2$) on the synthesized training data. The data augmentation is performed on all the images in the training dataset during each training epoch before feeding the images into deep-learning model process 1400. Data augmentation improves the deep-learning model's ability to confront camera shifting and avoid over-fitting during model training.

From block 1830, the method 1720 returns to block 1730 of FIG. 17.

In block 1730, the data processing system 130 applies the deep-learning model to each of the pre-processed one or more facial digital image(s) to generate facial expression reconstruction. The deep-learning model for facial expression reconstruction was described in the deep-learning model process 1400 herein with reference to FIG. 14 and in the classifier process 500 with reference to FIG. 5. The results of the facial expression reconstruction from block 1730 may include the reconstruction output from block 1460 of FIG. 14 and/or the encoding of a facial event class from block 550 of FIG. 5. For example, the data processing system 130 may output from block 1730 the images from the process 1400 of FIG. 14. The data processing system 130 also, or alternatively, may further process the facial expression reconstruction such that the output from block 1730 comprises a three-dimensional visualization of the facial expression, silent speech recognition, an emoji associated with the facial expression reconstruction, and/or real-time avatar facial expressions associated with the facial expression reconstruction, such as described by the classifier process 500 of FIG. 5.

From block 1730, the method 860' returns to block 870 of FIG. 8 to output results of the facial reconstruction.

Cameras 110 were described herein with reference to FIG. 1 as any suitable sensors for capturing images. For example, cameras 110 may include depth cameras, red-green-blue ("RGB") cameras, infrared ("IR") sensors, acoustic cameras or sensors, microphones and speakers, thermal imaging sensors, charge-coupled devices ("CCDs"), complementary metal oxide semiconductor ("CMOS") devices, active-pixel sensors, radar imaging sensors, fiber optics, and any other suitable image device technology. While the examples herein described specific cameras for the headband device 210, earbud devices 220, necklace device 1210, and neckband device 1220, it should be appreciated that any type of camera can be used with any of the wearable devices. The methods to process the images and recreate the facial expressions are adjusted based on camera/image type and are not dependent on the type of device used to acquire the images.

The embodiments herein describe wearable devices in the form of head-mounted devices headband device 210 and earbud devices 220, and neck-mounted devices necklace device 1210 and neckband device 1220. Any suitable wearable device may be used such that the one or more camera(s) may be directed towards a user's face including, but not limited to, glasses, smart glasses, a visor, a hat, a helmet, headgear, a virtual reality ("VR") headset.

The embodiments herein describe head-mounted and neck-mounted wearable devices comprises cameras 110

25

with adjustable positions and angles. Each camera **110** may be positioned such that at least one of a buccal region, a zygomatic region, and/or a temporal region of a user's face are included in the field of view of the camera **110**.

In an alternate embodiment, the wearable devices previously described herein may be configured for use in a hand-face touching detection system to recognize and predict a time and position that a user's hand touches the user's face. An important step in reducing the risk of infection is avoiding touching the face because a virus, such as COVID-19, may enter the mucous membranes of the eyes, nose, and/or mouth. Touching different areas of the face carries different health related risks. For example, contacting a mucous membrane may introduce a higher risk of transmitting a virus than touching non-mucous areas such as the chin and cheek. In addition, the frequency of touching the face may be an indicator regarding the stress level of a person. Understanding how people touch their face may alleviate multiple health challenges. Accurately recognizing where the hand touches the face is an important step towards alleviating health risks introduced by hand-face touching behaviors. In order to implement behavior intervention technologies, the hand-face touching detection system predicts the behavior in advance rather than simply detecting the touching behavior.

A data training set is created similar to the data training sets described herein with reference to FIGS. **9** and **16**. The difference is that instead of frontal view images of a user capturing just the user's face, the data training set uses frontal view images of a user with different hand-face touching positions. For example, frontal view images of the user may be acquired with the user touching his eye, nose, mouth, cheek, or any other suitable facial feature. In addition to touching faces directly, people move their hands closer to their faces when performing daily activities such as eating and drinking. The daily activities are classified as separate behaviors to avoid false-positive errors of a person touching their face. Additional frontal view images are acquired for the data training set depicting example daily activities. For example, the frontal view images may comprise a user eating with his hand, a fork, a spoon, or chopsticks. The user may be drinking with a straw, a bottle, or a cup. The user may be placing an earbud in or out, putting on or taking off eyeglasses, adjusting eyeglasses, applying lip balm, or adjusting his hair. Any suitable daily activities may be used in the frontal view images to create the data training set.

The frontal view images of the user are sent to a server, such as data processing system **130**, to create the data training set, as previously described herein with reference to FIGS. **9** and **16**. However, instead of extracting facial landmark positions as described in block **940** of FIG. **9** or facial geometric features as described in block **1640** of FIG. **16**, the data processing system extracts hand/facial landmark positions or hand/geometric features to create the data training set.

In an example, necklace device **1210** may be positioned on a user to acquire images of the user's facial area, as previously described herein, to also include a user's hand if positioned or in motion near the user's face. Any suitable wearable device may be used to acquire the images of the user's face. To predict the time and location of a hand-face touch, camera images are monitored over a period of time. Camera images are sent to a server, such as data processing system **130**, for processing. The data processing system receives the hand/facial images and reconstructs the position of the user's hand relative to the user's face using the data training set and a deep-learning model, such as the models

26

previously described herein with reference to FIGS. **4** and **14**. The data processing system monitors the hand-face position over time to predict where and when a user may touch his face. Alternatively, the data processing system may determine that the user is participating in a daily activity and that the user is not in the process of a hand-face touch. In each case, the system outputs a reconstructed image (or other similar information as described herein) depicting the user's facial expression and a representation of the hand-touch activity. The system also can log each hand-touch to the facial area and output notifications of the activity.

In an alternate embodiment, acoustic sensing technology can be used to reconstruct facial expressions using an array of microphones and speakers and deep-learning models. The acoustic technology comprises a wearable device such as headphones, earbuds, necklaces, neckbands, and any other suitable form factors such that the microphones and speakers can be mounted or attached to the wearable device. In an example, the microphones are Micro-Electro-Mechanical System ("MEMS") microphones that are placed on a printed circuit board ("PCB"). In an example, each microphone/speaker assembly may comprise four microphones and one speaker. Any suitable number of microphones and speakers may be included. The positions of the microphones and speakers may be adjustable to optimize the recording of reflected signals.

The acoustic wearable device actively sends signals from the speakers towards a user's face. In an example, the speakers transmit inaudible acoustic signals within a frequency range of 16 kHz to 24 kHz towards the user's face. Any suitable frequency range may be used. The signals are reflected by the user's face and captured by the microphones on the acoustic wearable device. The signals are reflected differently back towards the microphones based on different facial expressions or movements. A Channel Impulse Response ("CIR") is calculated based on the acoustic signals received at each microphone.

Each microphone is in communication with a computing device, such as computing device **120**, such that the CIR images/data can be transmitted for processing to a server, such as data processing system **130**.

The data processing system creates a data training set using frontal view images of a user in a machine learning algorithm, previously described herein with reference to FIGS. **9** and **16**. The data processing system receives the CIR images/data and reconstructs the user's facial expression using the data training set and a deep-learning model, such as the models previously described herein with reference to FIGS. **4** and **14**. The data processing system outputs the resultant facial expression reconstruction, such as previously described herein with reference to block **870** of FIG. **8**.

The wearable devices and systems described herein may be used in applications such as silent speech recognition, emoji input, and real-time avatar facial expressions. Silent speech recognition is a method to recognize speech when vocalization is inappropriate, background noise is excessive, or vocalizing speech is challenging due to a disability. The data training set for silent speech recognition comprises a set of frontal view facial images directed to the utterance of a word or phrase. To recognize silent speech, the wearable device, such as necklace **1220**, captures a series of facial movements from underneath the chin of a user while the user silently utters words or commands and transfers the series of digital facial images to the data processing system **130** for facial expression reconstruction. The results of the facial expression reconstruction, previously described herein with

reference to block 456 or block 1460, are used as inputs to the classifier process 500, previously described herein with reference to FIG. 5. Using the series of facial expression reconstruction images, the classifier process 500 detects the start time and end time of an utterance and selects the frames from the series of facial expression reconstruction images associate with the utterance. A vector representation of the utterance is created and input into a probability distribution to predict a word or phrase. In some embodiments for applications, such as in a testing process after the training, or using a trained system, a front view of a user is not needed. In some embodiments for speech recognition, one or more imaging sensor(s), such as camera(s) in a necklace and/or a wearable device worn in or on ear(s) of a user, capture a series of images for speaking a syllable, word, or phrase with or without audio (for example, at least two images of a syllable, a word, or a phrase), wherein each image comprises only a portion of a full face. In some embodiments, one image used in the application only captures less than 50%, less than 40%, less than 30%, less than 25%, less than 20%, or less than 15% of a full face. In some embodiments, the one or more imaging sensor(s) used in the application do not capture direct visual information for eyes, nose, and/or mouth. In some embodiments, the one or more image(s) and the combination thereof used in the application capture incomplete visual information for eyes, nose, and/or mouth. In some embodiments, the system and method are configured to recognize or predict speech, and/or other facial expression(s) at a high accuracy, wherein the accuracy is at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, or at least 98% for one or multiple words, phrases, or facial expressions on average. In some embodiments, the application is a speech recognition based on images/videos captured by the system. In other embodiments, the image-based or video-based speech recognition may be combined with an audio-based speech recognition to improve the accuracy of speech recognition. In some embodiments, the image-based or video-based speech recognition may be used to validate a result of an audio-based speech recognition.

Alternate Embodiment

Alternate Embodiment Example System Architecture

FIG. 19 is a block diagram depicting an acoustic sensor system based facial expression reconstruction system 1900, in accordance with certain examples. As depicted in FIG. 19, acoustic sensor system based facial expression reconstruction system 1900 comprises user computing device 150, acoustic sensor systems 1910-1 through 1910-n, and data processing computing device 1920. User computing device 150 and data processing computing device 1920 are configured to communicate via network 140. Network 140 and user computing device 150 were previously described herein with reference to FIG. 1. User computing device 150 may be configured to receive display a facial output, such as a predicted facial expression with three-dimensional visualization with examples depicted at 2830 of FIGS. 28 and 2930 of FIG. 29. Acoustic sensor systems 1910 are described in greater detail herein with reference to FIGS. 20A, 20B, and 20C.

FIG. 20A is a block diagram depicting a first embodiment of an acoustic sensor system 1910 in accordance with certain examples. Acoustic sensor system 1910 comprises speaker 1911, microphone 1912, Bluetooth module 1913, and printed circuit boards ("PCB") 1914. Speaker 1911 is a small

form factor speaker. In an example, speaker 1911 is a speaker configurable to transmit signals in an inaudible frequency range. For example, speaker 1911 may be configured to transmit signals in a frequency range of 16-24 kHz. Any suitable frequency range may be used. In an example, speaker 1911 transmits signals that are frequency modulation continuous wave ("FMCW") transmissions, Channel Impulse Response ("CIR") transmissions, or any other suitable type of acoustic signal. In an example, the speakers transmit inaudible acoustic signals within the frequency range of 16 kHz to 24 kHz towards a user's face. As depicted in FIG. 20A, speaker 1911 is affixed to a PCB 1914.

Microphone 1912 is a small form factor microphone. In an example, microphone 1912 is a Micro-Electro-Mechanical System ("MEMS") microphone. Microphone 1912 may be a microphone chip, a silicon microphone, or any other suitable microphone. In an example, microphone 1912 is configured to receive reflected signals transmitted by speaker 1911. As depicted in FIG. 20A, microphone 1912 is affixed to a PCB 1914.

Bluetooth module 1913 may be a Bluetooth low energy ("BLE") module. Bluetooth module 1913 may be a System-on-Chip ("SoC") module. In an example, Bluetooth module 1913 may comprise a microcontroller unit ("MCU") with memory. In an example, the memory may be an on-board SanDisk ("SD") card. The SD card may function to save acoustic data on the MCU. Bluetooth module 1913 may be configured to control a battery (not depicted) to provide power to the acoustic sensing system 1910. In an example, the battery may be a lithium polymer ("LiPo") battery. Bluetooth module 1913 may be configured to power and control speaker 1911 transmissions, microphone 1912 signal receptions, and data transmissions external to a wearable device, for example transmissions to computing device 1920. As depicted in FIG. 20A, Bluetooth module 1913 is affixed to a PCB 1914.

FIG. 20B is a block diagram depicting a second embodiment of an acoustic sensor system 1910, in accordance with certain examples. FIG. 20B depicts speaker 1911 and microphone 1912 affixed to a common PCB 1914 with Bluetooth module 1913 affixed to a separate PCB 1914.

FIG. 20C is a block diagram depicting a third embodiment of an acoustic sensor system 1910, in accordance with certain examples. FIG. 20C depicts speaker 1911, microphone 1912, and Bluetooth module 1913 affixed to a common PCB 1914. Any suitable configuration of speaker 1911, microphone 1912, Bluetooth module 1913 and PCBs 1914 may be used in addition to the configurations depicted in FIGS. 20A, 20B, and 20C.

In an example, a wearable device system may comprise one or more speaker(s) 1911 and one or more microphone(s) 1912, with a pair of a speaker 1911 and a microphone 1912 affixed on a PCB 1914. Any suitable number of speakers 1911 and microphones 1912 may be included on a particular wearable device system. The positions and/or orientations of the speakers 1911 and microphones 1912 may be adjustable to optimize the recording of reflected signals.

Returning to FIG. 19, acoustic sensor system based facial expression reconstruction system 1900 comprises data processing computing device 1920. Data processing computing device 1920 functions to receive data from Bluetooth module 1913 to construct a facial output. The facial output may comprise one of a facial movement, an avatar movement associated with the facial movement, a speech recognition, text associated with the speech recognition, a physical activity, an emotional status, a two-dimensional visualization of a facial expression, a three-dimensional visualization

29

of a facial expression, an emoji associated with a facial expression, or an avatar image associated with a facial expression. Data processing computing device 1920 may be a smart phone, a mobile device, or any type of computing device described herein with reference to FIG. 31. Data processing computing device 1920 comprises central processing unit 121, graphics processing unit 122, memory 123, communication application 124, and data processing module 1930. Central processing unit 121, graphics processing unit 122, memory 123, and communication application 124 were previously described herein with reference to FIG. 1. Data processing module 1930 is described in greater detail herein with reference to FIG. 21.

FIG. 21 is a block diagram depicting a data processing module 1930, in accordance with certain examples. Data processing module 1930 comprises central processing unit 131, modeling application 132, data storage unit 133, and communication application 134. Central processing unit 131, modeling application 132, data storage unit 133, and communication application 134 were previously described herein with reference to FIG. 1.

FIG. 22A depicts a first embodiment of a head-mounted acoustic sensor system wearable device 2210. FIG. 22A depicts a glasses style wearable device 2210 (herein referred to as glasses device 2210). Glasses device 2210 comprises two speakers 1911-1 and 1911-2, two microphones 1912-1 and 1912-2, and a Bluetooth module 1913. While two speakers 1911 and two microphones 1912 are depicted, any suitable quantity of speakers 1911 and microphones 1912 may be used.

FIG. 22B is an illustration depicting the first embodiment of the head-mounted acoustic sensor system wearable device 2210 on a user. As illustrated in FIG. 22B, glasses device 2210 may be positioned on a user such that a pair of speaker 1911-1 and microphone 1912-1 are located on the left side of the user's face and a pair of speaker 1911-2 and microphone 1912-2 are located on the right side of the user's face. The disposition of each pair on the left and right side of the user allows data associated with skin deformations to be collected for the left and right contours of the user's face.

FIG. 23A depicts a second embodiment of a head-mounted acoustic sensor system wearable device 2310. FIG. 23A depicts a headset style wearable device 2310 (herein referred to as headset device 2310). In an example, headset device 2310 may be a virtual reality headset. Headset device 2310 comprises two speakers 1911-1 and 1911-2, two microphones 1912-1 and 1912-2, and a Bluetooth module 1913 (depicted in FIG. 23B). While two speakers 1911 and two microphones 1912 are depicted, any suitable quantity of speakers 1911 and microphones 1912 may be used.

FIG. 23B is an illustration depicting the second embodiment of the head-mounted acoustic sensor system wearable device 2310 on a user. As illustrated in FIG. 23B, headset device 2310 may be positioned on a user such that a pair of speaker 1911-1 and microphone 1912-1 are located on the left side of the user's face and a pair of speaker 1911-2 and microphone 1912-2 are located on the right side of the user's face. The disposition of each pair on the left and right side of the user allows data associated with skin deformations to be collected for the left and right contours of the user's face.

FIG. 24A depicts a third embodiment of a head-mounted acoustic sensor system wearable device 2410. FIG. 24A depicts a headphone style wearable device 2410 (herein referred to as headphone device 2410). In an example, headphone device 2410 may be a headband style headphone, ear pods, earphones, ITE headphones, over the ear headphones, or any other headphone type device. Headphone

30

device 2410 comprises two speakers 1911-1 and 1911-2, two microphones 1912-1 and 1912-2, and a Bluetooth module 1913. While two speakers 1911 and two microphones 1912 are depicted, any suitable quantity of speakers 1911 and microphones 1912 may be used.

FIG. 24B is an illustration depicting the third embodiment of the head-mounted acoustic sensor system wearable device 2410 on a user. As illustrated in FIG. 24B, headphone device 2410 may be positioned on a user such that a pair of speaker 1911-1 and microphone 1912-1 are located on the left side of the user's face and a pair of speaker 1911-2 and microphone 1912-2 are located on the right side of the user's face. The disposition of each pair on the left and right side of the user allows data associated with skin deformations to be collected for the left and right contours of the user's face.

As described herein, FIGS. 22A, 23A, and 24A depict example embodiments of a head-mounted acoustic sensor system wearable device. Other example embodiments may include ear buds, ear pods, in-the-ear ("ITE") headphones, over-the-ear headphones, outside-the-ear ("OTE") headphones, glasses, smart glasses, a visor, a hat, a helmet, headgear, a virtual reality headset, another head-borne device, a necklace, a neckband, a garment-attachable system, or any other type of device or system to which at least one acoustic sensor system 1910 may be affixed.

Alternate Embodiment Example Processes

FIG. 25 is a block flow diagram depicting a method 2500 to reconstruct facial expressions based on acoustic signals, in accordance with certain examples. In block 810, a data training set is created using frontal view digital images. Block 810 was previously described herein with reference to method 810 of FIG. 9.

In block 2510, a wearable system transmits at least one acoustic signal. The wearable system may be a system as described herein with reference to glasses device 2210, headset device 2310, headphone device 2410. In an example, the wearable system may include other embodiments such as ear buds, ear pods, ITE headphones, over-the-ear headphones, OTE headphones, glasses, smart glasses, a visor, a hat, a helmet, headgear, a virtual reality headset, another head-borne device, a necklace, a neckband, a garment-attachable system, or any other type of device or system to which at least one acoustic sensor system 1910 may be affixed. In an example, the wearable system is positioned on a wearer such that the distance from the wearable system to the wearer's face is less than 10 cm, 11 cm, 12 cm, 13 cm, 14 cm, 15 cm, or any suitable distance. The at least one acoustic signal is transmitted by a speaker 1911 of at least one acoustic sensor system 1910. Acoustic sensor system 1910 may be configured to transmit signals in a frequency range of 16-24 kHz. Any suitable frequency range may be used. Acoustic sensor system 1910 may transmit signals that are FMCW transmissions, CIR transmissions, or any other suitable type of acoustic signal. In an example, the transmitted signals have an associated sample length. For example, the sample length may be set to 600 corresponding to a sample length of 0.012 seconds. In this example, approximately 83.3 frames may be collected per second. Any suitable sample length may be used. In an example, acoustic sensor system 1910 transmits inaudible acoustic signals within the frequency range of 16 kHz to 24 kHz towards a wearer's face. In an example, acoustic sensor system 1910 may be positioned on the wearable system to transmit acoustic signals towards a first facial feature on a first side of a sagittal plane of a wearer and to transmit

31

acoustic signals towards a second facial feature on a second side of the sagittal plane of a wearer. In an example, acoustic sensor system 1910 may be positioned on the wearable system to transmit acoustic signals towards an underside contour of a chin of a wearer.

In block 2515, the wearable system receives at least one reflected acoustic signal. Subsequent to transmitting the at least one acoustic signal toward a wearer's face as described herein with reference to block 2510, the signal is reflected by the wearer's face. The at least one reflected acoustic signal is received by microphone 1912 of the at least one acoustic sensor system 1910. The at least one transmitted signal is reflected differently to microphone 1912 based on skin deformations associated with different facial movements.

In block 2520, the wearable system transmits the at least one transmitted acoustic signal and the at least one reflected acoustic signal to data processing computing device 1920. In an example, the at least one transmitted acoustic signal and the at least one reflected acoustic signal are transmitted in frames. The at least one transmitted acoustic signal and the at least one reflected acoustic signal are transmitted by the Bluetooth module 1913 of the at least one acoustic sensor system 1910.

In block 2525, data processing computing device 1920 receives the at least one transmitted acoustic signal and the at least one reflected acoustic signal.

In block 2530, data processing computing device 1920 filters the received acoustic signals. In an example, the at least one transmitted acoustic signal and the at least one reflected acoustic signal are filtered to remove noise outside of a target frequency range. In an example, the target frequency range is 15.5 kHz to 20.5 kHz. Any suitable target frequency range may be used. In an example, the at least one transmitted acoustic signal and the at least one reflected acoustic signal are filtered using a Butterworth band-pass filter. Any suitable filter may be used.

In block 2535, data processing computing device 1920 constructs a profile using the filtered acoustic signals. In an example, the profile is an echo profile. The echo profile is determined by calculating a cross-correlation between the filtered at least one transmitted acoustic signal and the filtered at least one reflected acoustic signal. The echo profile depicts deformations of the skin of the wearer's face in temporal and spatial domains. In an example, a differential echo profile may be calculated. A differential echo profile is calculated by subtracting echo profiles between two adjacent echo frames. The differential echo profile removes static objects that may be present in the echo profile. Further, as the wearable system was positioned to be less than 10-15 cm away from a wearer's face, echo profiles comprising distance greater than that range may also be removed. For example, echo profiles with a distance greater than ± 16 cm, ± 17 cm, ± 18 cm, or any other suitable distance may be removed. In an example, the transmitted acoustic signal is a FMCW signal from which the echo profile is constructed. In alternate examples, the transmitted acoustic signal may be a CIR signal with global system for mobiles ("GSM"), characteristic impedance ("ZC"), or Barker sequence encoding; angle of arrival ("AoA"); doppler effect; or phase change detection.

In block 2540, data processing computing device 1920 applies a deep learning model to the constructed profile. The deep-learning model for facial expression reconstruction was previous described herein with reference to the processes and methods of FIG. 4, FIG. 5, and FIG. 14. In an example with reference to acoustic data, a sliding window of one second may be applied to the differential echo profiles

32

to provide 84 frames of differential echo profiles in each window. In an example, each differential echo profile may have approximately 200 data points providing an input vector of size 200×84 . The input vector is used in the deep learning model along with the training data to estimate facial expressions.

In block 2545, data processing computing device 1920 assigns a deformation to the constructed profile based on the results of the deep learning model. In an example, the assigned deformation has a predetermined degree of correspondence to a selected one of a plurality of deformations in the deep learning model.

In block 2550, the data processing computing device 1920 communicates a facial output based on the assigned deformation. In an example, the facial output comprises one of a facial movement, an avatar movement associated with the facial movement, a speech recognition, text associated with the speech recognition, a physical activity, an emotional status, a two-dimensional visualization of a facial expression, a three-dimensional visualization of a facial expression, an emoji associated with a facial expression, or an avatar image associated with a facial expression. In an example, the facial output is communicated to user computing device 150. In an example, the facial output is communicated in real time. In an example, the facial output is continuously updated.

FIG. 26 is a diagram 2600 depicting an acoustic sensor system based facial expression reconstruction process. FIG. 26 is a visual depiction of the method 2500 described herein with reference to FIG. 25. At 2610, a wearer is depicted with glasses device 2210 and headset device 2310. In an example, a wearer may select a wearable device from the various embodiments previously described herein. At 2620, speaker 1911, microphone 1912, and Bluetooth module 1913 of an acoustic sensor system 1910 of the wearable device are illustrated. At 2630, an acoustic signal is depicted as being transmitted by speaker 1911 and directed to the wearer's face. At 2630, the acoustic signal is deflected by the wearer's face and received by microphone 1912 at 2640. At 2650, the transmitted acoustic signal and the deflected acoustic signal are transmitted to data processing computing device 1920 (not depicted) by Bluetooth module 1913. At 2660, the transmitted acoustic signal and the deflected acoustic signal are filtered. At 2670, differential echo profiles are depicted. At 2680, the differential echo profiles are input into a deep learning model. At 2690, predicted facial expressions are depicted.

FIG. 27 is a diagram 2700 depicting an acoustic sensor system based facial expression reconstruction output process. FIG. 27 is a second visual depiction of the method 2500 described herein with reference to FIG. 25. At 2710, a wearer is depicted with glasses device 2210 with an acoustic sensor system 1910 comprising speaker 1911, microphone 1912, Bluetooth module 1913, and PCB 1914. At 2720, acoustic data, i.e., transmitted acoustic signal and deflected acoustic signal, are transmitted by acoustic sensor system 1910 to data processing computing device 1920. At 2730, the transmitted acoustic signal and the deflected acoustic signal are filtered. At 2740, differential echo profiles are depicted. At 2750, the differential echo profiles are input into a deep learning model. At 2760, data processing computing device 1920 transmits the output of the deep learning model to user computing device 150. At 2770, example facial outputs are depicted as received by user computing device 150. Example facial outputs may be a 3D avatar, such as depicted at 2771, or a personalized avatar, such as depicted at 2772.

33

FIG. 28 is a diagram 2800 depicting the correlation between frontal view camera images 2810 and 2820, predicted facial expressions 2830, and left and right differential echo profiles, 2840 and 2850 respectively, in accordance with certain examples. Rows 2810 and 2820 illustrate frontal view camera images of a user. To construct a training data set or “ground truth,” frontal view camera images of a user are acquired, via an imaging capturing device, with the user making various different facial expressions as depicted in images 2810-1 through 2810-*n* and images 2820-1 through 2820-*n*. The method to create the training data set from the frontal view camera images is described herein in greater detail with reference to method 810 of FIG. 9. Row 2840 depicts left differential echo profiles 2840-1 through 2840-*n* captured by a head-mounted, wearable device, such as glasses device 2210 or headset device 2310 described herein with reference to FIG. 22 and FIG. 23. Row 2850 depicts right differential echo profiles 2850-1 through 2850-*n* captured by a head-mounted, wearable device, such as glasses device 2210 or headset device 2310 described herein with reference to FIG. 22 and FIG. 23. Left differential echo profiles 2840 and right differential echo profiles 2850 are input into a deep learning model, as previously described herein, to create the predicted facial expressions depicted at 2830. Predicted facial expressions 2830-1 through 2830-*n* are illustrated in FIG. 28 beneath the frontal view camera images in rows 2810-1 through 2810-*n* and 2820-1 through 2820-*n* to illustrate the accuracy of the predicted facial expressions 2830.

FIG. 29 is a diagram 2900 depicting the correlation between frontal view camera images 2910, training data 2920, predicted facial expressions 2930, and left and right differential echo profiles, 2940 and 2950 respectively. Row 2910 illustrates frontal view camera images of a user. To construct a training data set or “ground truth,” frontal view camera images of a user are acquired, via an imaging capturing device, with the user making various different facial expressions as depicted in images 2910-1 through 2910-*n*. The method to create the training data set from the frontal view camera images is described herein in greater detail with reference to method 810 of FIG. 9. Row 2920 depicts a training data image or ground truth for the frontal view camera images in row 2910. Training data images 2920-1 through 2920-*n* correspond with frontal view camera images 2910-1 through 2910-*n*. Row 2940 depicts left differential echo profiles 2940-1 through 2940-*n* captured by a head-mounted, wearable device. Row 2950 depicts right differential echo profiles 2950-1 through 2950-*n* captured by a head-mounted, wearable device. Left differential echo profiles 2940 and right differential echo profiles 2950 are input into a deep learning model, as previously described herein, to create the predicted facial expressions depicted at 2930. Predicted facial expressions 2930-1 through 2930-*n* are illustrated in FIG. 29 beneath the ground truth images 2920-1 through 2920-*n* to illustrate the accuracy of the predicted facial expressions 2930.

FIG. 30A is a diagram depicting the correlation between frontal view camera images of an open mouth 3010, a left view image 3015, a left and right echo profile, 3020 and 3025 respectively, and a right view image 3030. The skin deformations, depicted in left view images 3015-1 through 3015-*n* and right view images 3030-1 through 3030-*n*, created by the various facial expressions 3010-1 through 3010-*n* result in distinctive patterns changes depicted in left echo profiles 3020-1 through 3020-*n* and right echo profiles 3025-1 through 3025-*n*. FIG. 30B is a diagram depicting the correlation between frontal view camera images of a right

34

sneer 3035, a left view image 3040, a left and right echo profile, 3045 and 3050 respectively, and a right view image 3055. The skin deformations, depicted in left view images 3040-1 through 3040-*n* and right view images 3055-1 through 3055-*n*, created by the various facial expressions 3035-1 through 3035-*n* result in distinctive patterns changes depicted in left echo profiles 3045-1 through 3045-*n* and right echo profiles 3050-1 through 3050-*n*.

While acoustic sensing technology may be used to determine facial expressions by detecting skin deformations, acoustic sensing technology may be used to directly determine other outputs associated with skin deformation without the need to first determine facial expressions. For example, the acoustic sensing technology may track blinking patterns and eyeball movements of a user. Blinking patterns and eyeball movements can be applied to the diagnosis and treatment processes of eye diseases. The acoustic sensing technology may detect movements of various parts of the face associated with speech, whether silent or voiced. While speaking, different words and/or phrases lead to subtle, yet distinct, skin deformations. The acoustic sensing technology can capture the skin deformation to recognize silent speech or to vocalize sound. By tracking subtle skin deformations, the acoustic sensing technology can be used to synthesize voice. The acoustic sensing technology may also be used to recognize and track physical activities. For example, while eating, a user opens the mouth, chews, and swallows. During each of those movements, the user's skin deforms in a certain way such that opening the mouth is distinguishable from chewing and from swallowing. The detected skin deformations may be used to determine the type and quantity of food consumed by a user. Similar to eating, a type and quantity of a consumed drink may be determined by skin deformation. The acoustic sensing technology may also be used to determine an emotional status of a user. An emotional status may be related to skin deformations. By detecting skin deformations, an emotional status can be determined and reported to corresponding applications. In an example, acoustic sensor system based facial expression reconstruction system 1900, described herein with reference to FIG. 19, may be configured to directly determine the other outputs associated with skin deformation without the need to first determine facial expressions, including, but not limited to, facial movements including eyeball movement, speech recognition, physical activities, and emotional status. Acoustic sensor system based facial expression reconstruction system 1900 may be configured to directly determine the other outputs associated with skin deformation using any type acoustic sensor system 1910 as described herein with reference to FIG. 19.

Other Examples

FIG. 31 depicts a computing machine 4000 and a module 4050 in accordance with certain examples. The computing machine 4000 may correspond to any of the various computers, servers, mobile devices, embedded systems, or computing systems presented herein. The module 4050 may comprise one or more hardware or software element(s) configured to facilitate the computing machine 4000 in performing the various methods and processing functions presented herein. The computing machine 4000 may include various internal or attached components such as a processor 4010, system bus 4020, system memory 4030, storage media 4040, input/output interface 4060, and a network interface 4070 for communicating with a network 4080.

35

The computing machine **4000** may be implemented as a conventional computer system, an embedded controller, a laptop, a server, a mobile device, a smartphone, a set-top box, a kiosk, a router or other network node, a vehicular information system, one or more processor(s) associated with a television, a customized machine, any other hardware platform, or any combination or multiplicity thereof. The computing machine **4000** may be a distributed system configured to function using multiple computing machines interconnected via a data network or bus system.

The processor **4010** may be configured to execute code or instructions to perform the operations and functionality described herein, manage request flow and address mappings, and to perform calculations and generate commands. The processor **4010** may be configured to monitor and control the operation of the components in the computing machine **4000**. The processor **4010** may be a general purpose processor, a processor core, a multiprocessor, a reconfigurable processor, a microcontroller, a digital signal processor ("DSP"), an application specific integrated circuit ("ASIC"), a graphics processing unit ("GPU"), a field programmable gate array ("FPGA"), a programmable logic device ("PLD"), a controller, a state machine, gated logic, discrete hardware components, any other processing unit, or any combination or multiplicity thereof. The processor **4010** may be a single processing unit, multiple processing units, a single processing core, multiple processing cores, special purpose processing cores, co-processors, or any combination thereof. The processor **4010** along with other components of the computing machine **4000** may be a virtualized computing machine executing within one or more other computing machine(s).

The system memory **4030** may include non-volatile memories such as read-only memory ("ROM"), programmable read-only memory ("PROM"), erasable programmable read-only memory ("EPROM"), flash memory, or any other device capable of storing program instructions or data with or without applied power. The system memory **4030** may also include volatile memories such as random access memory ("RAM"), static random access memory ("SRAM"), dynamic random access memory ("DRAM"), and synchronous dynamic random access memory ("SDRAM"). Other types of RAM also may be used to implement the system memory **4030**. The system memory **4030** may be implemented using a single memory module or multiple memory modules. While the system memory **4030** is depicted as being part of the computing machine **4000**, one skilled in the art will recognize that the system memory **4030** may be separate from the computing machine **4000** without departing from the scope of the subject technology. It should also be appreciated that the system memory **4030** may include, or operate in conjunction with, a non-volatile storage device such as the storage media **4040**.

The storage media **4040** may include a hard disk, a floppy disk, a compact disc read only memory ("CD-ROM"), a digital versatile disc ("DVD"), a Blu-ray disc, a magnetic tape, a flash memory, other non-volatile memory device, a solid state drive ("SSD"), any magnetic storage device, any optical storage device, any electrical storage device, any semiconductor storage device, any physical-based storage device, any other data storage device, or any combination or multiplicity thereof. The storage media **4040** may store one or more operating system(s), application programs and program modules such as module **4050**, data, or any other information. The storage media **4040** may be part of, or connected to, the computing machine **4000**. The storage media **4040** may also be part of one or more other computing

36

machine(s) that are in communication with the computing machine **4000** such as servers, database servers, cloud storage, network attached storage, and so forth.

The module **4050** may comprise one or more hardware or software element(s) configured to facilitate the computing machine **4000** with performing the various methods and processing functions presented herein. The module **4050** may include one or more sequence(s) of instructions stored as software or firmware in association with the system memory **4030**, the storage media **4040**, or both. The storage media **4040** may therefore represent machine or computer readable media on which instructions or code may be stored for execution by the processor **4010**. Machine or computer readable media may generally refer to any medium or media used to provide instructions to the processor **4010**. Such machine or computer readable media associated with the module **4050** may comprise a computer software product. It should be appreciated that a computer software product comprising the module **4050** may also be associated with one or more process(es) or method(s) for delivering the module **4050** to the computing machine **4000** via the network **4080**, any signal-bearing medium, or any other communication or delivery technology. The module **4050** may also comprise hardware circuits or information for configuring hardware circuits such as microcode or configuration information for an FPGA or other PLD.

The input/output ("I/O") interface **4060** may be configured to couple to one or more external device(s), to receive data from the one or more external device(s), and to send data to the one or more external device(s). Such external devices along with the various internal devices may also be known as peripheral devices. The I/O interface **4060** may include both electrical and physical connections for operably coupling the various peripheral devices to the computing machine **4000** or the processor **4010**. The I/O interface **4060** may be configured to communicate data, addresses, and control signals between the peripheral devices, the computing machine **4000**, or the processor **4010**. The I/O interface **4060** may be configured to implement any standard interface, such as small computer system interface ("SCSI"), serial-attached SCSI ("SAS"), fiber channel, peripheral component interconnect ("PCP"), PCI express (PCIe), serial bus, parallel bus, advanced technology attached ("ATA"), serial ATA ("SATA"), universal serial bus ("USB"), Thunderbolt, FireWire, various video buses, and the like. The I/O interface **4060** may be configured to implement only one interface or bus technology. Alternatively, the I/O interface **4060** may be configured to implement multiple interfaces or bus technologies. The I/O interface **4060** may be configured as part of, all of, or to operate in conjunction with, the system bus **4020**. The I/O interface **4060** may include one or more buffer(s) for buffering transmissions between one or more external device(s), internal device(s), the computing machine **4000**, or the processor **4010**.

The I/O interface **4060** may couple the computing machine **4000** to various input devices including mice, touch-screens, scanners, electronic digitizers, sensors, receivers, touchpads, trackballs, cameras, microphones, keyboards, any other pointing devices, or any combinations thereof. The I/O interface **4060** may couple the computing machine **4000** to various output devices including video displays, speakers, printers, projectors, tactile feedback devices, automation control, robotic components, actuators, motors, fans, solenoids, valves, pumps, transmitters, signal emitters, lights, and so forth.

The computing machine **4000** may operate in a networked environment using logical connections through the network

interface 4070 to one or more other system(s) or computing machines across the network 4080. The network 4080 may include WANs, LANs, intranets, the Internet, wireless access networks, wired networks, mobile networks, telephone networks, optical networks, or combinations thereof. The network 4080 may be packet switched, circuit switched, of any topology, and may use any communication protocol. Communication links within the network 4080 may involve various digital or an analog communication media such as fiber optic cables, free-space optics, waveguides, electrical conductors, wireless links, antennas, radio-frequency communications, and so forth.

The processor 4010 may be connected to the other elements of the computing machine 4000 or the various peripherals discussed herein through the system bus 4020. It should be appreciated that the system bus 4020 may be within the processor 4010, outside the processor 4010, or both. Any of the processor 4010, the other elements of the computing machine 4000, or the various peripherals discussed herein may be integrated into a single device such as a system on chip ("SOC"), system on package ("SOP"), or ASIC device.

Examples may comprise a computer program that embodies the functions described and illustrated herein, wherein the computer program is implemented in a computer system that comprises instructions stored in a machine-readable medium and a processor that executes the instructions. However, it should be apparent that there could be many different ways of implementing examples in computer programming, and the examples should not be construed as limited to any one set of computer program instructions. Further, a skilled programmer would be able to write such a computer program to implement an example of the disclosed examples based on the appended flow charts and associated description in the application text. Therefore, disclosure of a particular set of program code instructions is not considered necessary for an adequate understanding of how to make and use examples. Further, those skilled in the art will appreciate that one or more aspect(s) of examples described herein may be performed by hardware, software, or a combination thereof, as may be embodied in one or more computing system(s). Moreover, any reference to an act being performed by a computer should not be construed as being performed by a single computer as more than one computer may perform the act.

The examples described herein can be used with computer hardware and software that perform the methods and processing functions described herein. The systems, methods, and procedures described herein can be embodied in a programmable computer, computer-executable software, or digital circuitry. The software can be stored on computer-readable media. Computer-readable media can include a floppy disk, RAM, ROM, hard disk, removable media, flash memory, memory stick, optical media, magneto-optical media, CD-ROM, etc. Digital circuitry can include integrated circuits, gate arrays, building block logic, field programmable gate arrays ("FPGA"), etc.

The systems, methods, and acts described in the examples presented previously are illustrative, and, alternatively, certain acts can be performed in a different order, in parallel with one another, omitted entirely, and/or combined between different examples, and/or certain additional acts can be performed, without departing from the scope and spirit of various examples. Accordingly, such alternative examples are included in the scope of the following claims, which are to be accorded the broadest interpretation so as to encompass such alternate examples.

Although specific examples have been described above in detail, the description is merely for purposes of illustration. It should be appreciated, therefore, that many aspects described above are not intended as essential elements unless explicitly stated otherwise. Modifications of, and equivalent components or acts corresponding to, the disclosed aspects of the examples, in addition to those described above, can be made by a person of ordinary skill in the art, having the benefit of the present disclosure, without departing from the spirit and scope of examples defined in the following claims, the scope of which is to be accorded the broadest interpretation so as to encompass such modifications and equivalent structures.

Various embodiments are described herein. It should be noted that the specific embodiments are not intended as an exhaustive description or as a limitation to the broader aspects discussed herein. One aspect described in conjunction with a particular embodiment is not necessarily limited to that embodiment and can be practiced with any other embodiment(s). Reference throughout this specification to "one embodiment," "an embodiment," "an example embodiment," or other similar language means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the invention described herein. Thus, appearances of the phrases "in one embodiment," "in an embodiment," "an example embodiment," or other similar language in various places throughout this specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, or characteristics may be combined in any suitable manner in one or more embodiment(s), as would be apparent to a person having ordinary skill in the art and the benefit of this disclosure. Furthermore, while some embodiments described herein include some, but not other, features included in other embodiments, combinations of features of different embodiments are meant to be within the scope of the invention. For example, in the appended claims, any of the claimed embodiments can be used in any combination.

The following examples are presented to illustrate the present disclosure. The examples are not intended to be limiting in any manner.

Example 1 is a wearable system, comprising: at least one acoustic sensor system configured to transmit at least one acoustic signal, to receive at least one reflected acoustic signal, and to output the at least one transmitted acoustic signal and the at least one received reflected acoustic signal; a processor that receives the at least one transmitted acoustic signal and the at least one received reflected acoustic signal from each of the at least one acoustic sensor system; and a physical memory, the physical memory comprising instructions that when executed by the processor cause the processor to: calculate a profile associated with the at least one transmitted acoustic signal and the at least one received reflected acoustic signal; assign a deformation to the profile, the assigned deformation having a predetermined degree of correspondence to a selected one of a plurality of deformations in a model when compared to the profile; and communicate a facial output based on the assigned deformation.

Example 2 includes the subject matter of Example 1, wherein the at least one acoustic sensor system comprises at least one speaker and at least one microphone.

Example 3 includes the subject matter of Examples 1 and 2, wherein the at least one acoustic sensor system comprises a Bluetooth low energy ("BLE") module to output the at least one transmitted acoustic signal and the at least one received reflected acoustic signal.

Example 4 includes the subject matter of any of Examples 1-3, wherein the at least one transmitted acoustic signal is a frequency modulation continuous wave (“FMCW”) signal.

Example 5 includes the subject matter of any of Examples 1-4, wherein the profile is a differential echo profile.

Example 6 includes the subject matter of any of Examples 1-5, wherein the deformation is a skin deformation.

Example 7 includes the subject matter of any of Examples 1-6, wherein the communicated facial output comprises one of a facial movement, an avatar movement associated with the facial movement, a speech recognition, text associated with the speech recognition, a physical activity, an emotional status, a two-dimensional visualization of a facial expression, a three-dimensional visualization of a facial expression, an emoji associated with a facial expression, or an avatar image associated with a facial expression.

Example 8 includes the subject matter of any of Examples 1-7, the at least one acoustic sensor system positioned on the wearable system to transmit acoustic signals towards a first facial feature on a first side of a sagittal plane of a wearer and to transmit acoustic signals towards a second facial feature on a second side of the sagittal plane of a wearer.

Example 9 includes the subject matter of any of Examples 1-8, the at least one acoustic sensor system positioned on the wearable system to transmit acoustic signals towards an underside contour of a chin of a wearer.

Example 10 includes the subject matter of any of Examples 1-9, the wearable system comprising ear buds, ear pods, in-the-ear (ITE) headphones, over-the-ear headphones, or outside-the-ear (OTE) headphones to which the at least one acoustic sensor system is attached.

Example 11 includes the subject matter of any of Examples 1-10, the wearable system comprising glasses, smart glasses, a visor, a hat, a helmet, headgear, a virtual reality headset, or another head-borne device to which the at least one acoustic sensor system is attached.

Example 12 includes the subject matter of any of Examples 1-11, the wearable system comprising a necklace, a neckband, or a garment-attachable system to which the at least one acoustic sensor system is attached.

Example 13 includes the subject matter of any of Examples 1-12, further comprising a computing device that receives and displays the communicated facial output.

Example 14 includes the subject matter of any of Examples 1-13, wherein the model is trained using machine learning.

Example 15 includes the subject matter of any of Examples 1-14, the training comprising: receiving one or more frontal view facial image(s) of a subject, each of the frontal view facial images corresponding to a deformation of a plurality of deformations of the subject; receiving one or more transmitted acoustic signal(s) and one or more corresponding reflected acoustic signal(s) associated with the subject, each of the one or more transmitted acoustic signal(s) and the one or more corresponding reflected acoustic signal(s) from the at least one acoustic sensor system also corresponding to a deformation of the plurality of deformations of the subject; and correlating, for each of the deformations, the one or more transmitted acoustic signal(s) and the one or more corresponding reflected acoustic signal(s) from the at least one acoustic sensor system corresponding to a particular deformation to the one or more frontal view facial image(s) corresponding to the particular deformation.

What is claimed is:

1. A wearable system, comprising:

at least one imaging sensor configured to capture an image of a facial feature of a wearer of the wearable system and to output image data corresponding to the image;

a processor that receives the image data from each of the at least one imaging sensor; and

a physical memory, the physical memory comprising instructions that when executed by the processor cause the processor to:

compare the image data from each of the at least one imaging sensor to a model;

assign a contour to the image data, the assigned contour having a predetermined degree of correspondence to a selected one of a plurality of contours in the model when compared to the image data; and

communicate a facial expression image based on the assigned contour;

wherein the model is trained using machine learning; and wherein the training comprises:

receiving one or more frontal view facial image(s) of a subject, each of the frontal view facial images corresponding to a facial expression of a plurality of facial expressions of the subject;

receiving one or more image(s) of the subject from the at least one imaging sensor, each of the images from the at least one imaging sensor also corresponding to a facial expression of the plurality of facial expressions of the subject; and

correlating, for each of the facial expressions, the one or more image(s) from the at least one imaging sensor corresponding to a particular facial expression to the one or more frontal view facial image(s) corresponding to the particular facial expression.

2. The wearable system of claim 1, wherein the at least one imaging sensor comprises an acoustic camera or an acoustic sensor configured to:

transmit signals towards a wearer's face; and

detect reflected signals from the wearer's face, wherein the transmitted signals are reflected differently based on different facial expressions or movements of the wearer.

3. The wearable system of claim 1, wherein the communicated facial expression image comprises one of a two-dimensional visualization of the facial expression image, a three-dimensional visualization of the facial expression image, an emoji associated with the facial expression image, or an avatar image associated with the facial expression image.

4. The wearable system of claim 1, wherein comparing the image data from each of the at least one imaging sensor to the model comprises processing the image data to at least one derivative image data set and comparing the at least one derivative image data set to the model.

5. The wearable system of claim 1, the at least one imaging sensor positioned on the wearable system to capture image data of a first facial feature on a first side of a sagittal plane of the wearer and to capture image data of a second facial feature on a second side of the sagittal plane of the wearer.

6. The wearable system of claim 1, the at least one imaging sensor positioned on the wearable system to capture image data of an underside contour of a chin of the wearer.

7. The wearable system of claim 1, wherein correlating for each of the facial expressions comprises:

extracting a plurality of landmark positions from each of the one or more frontal view facial image(s) of the subject;

41

extracting a plurality of landmark positions from each of the one or more image(s) of the subject from the at least one imaging sensor;

correlating the landmark positions for a particular frontal view facial image with a particular facial expression; and

correlating the landmark positions for a particular one of the images of the subject from the at least one imaging sensor with the particular facial expression.

8. The wearable system of claim 1, the at least one imaging sensor comprising a first imaging sensor and a second imaging sensor, the first imaging sensor positioned on the wearable system to capture first image data of a first facial feature of the wearer, and the second imaging sensor positioned on the wearable system to capture second image data of a second facial feature of the wearer,

wherein comparing the image data from each of the at least one imaging sensor to the model comprises comparing the first image data of the first facial feature of the wearer to the model and comparing the second image data of the second facial feature of the wearer to the model,

wherein assigning a contour to the image data comprises assigning a first contour to the first image data and a second contour to the second image data, and

wherein the facial expression image being based on the assigned first contour and the assigned second contour.

9. The wearable system of claim 8, the first imaging sensor positioned on the wearable system to capture image data of the first facial feature on a first side of a sagittal plane of the wearer, and the second imaging sensor positioned on the wearable system to capture an image of the second facial feature on a second side of the sagittal plane of the wearer.

10. The wearable system of claim 8, the first facial feature comprising a left side contour of a face of the wearer, and the second facial feature comprising a right side contour of the face of the wearer.

11. The wearable system of claim 8, the first facial feature comprising a left underside contour of a chin of the wearer, and the second facial feature comprising a right underside contour of the chin of the wearer.

12. The wearable system of claim 1, the wearable system comprising ear buds, ear pods, in-the-ear (ITE) headphones, over-the-ear headphones, or outside-the-ear (OTE) headphones to which the at least one imaging sensor is attached.

13. The wearable system of claim 1, the wearable system comprising glasses, smart glasses, a visor, a hat, a helmet, headgear, a virtual reality headset, or another head-borne device to which the at least one imaging sensor is attached.

14. The wearable system of claim 1, the wearable system comprising a necklace, a neckband, or a garment-attachable system to which the at least one imaging sensor is attached.

15. The wearable system of claim 1, the physical memory comprising further instructions that when executed by the processor cause the processor to:

compare the assigned contour to a second model; and assign the facial expression image based on the assigned contour, the assigned facial expression image having a predetermined degree of correspondence to a selected one of a plurality of contours in the second model when compared to the assigned contour.

16. The wearable system of claim 15, wherein the second model is trained using machine learning.

17. The wearable system of claim 16, wherein the training of the second model comprises:

42

receiving one or more frontal view facial image(s) of a subject, each of the frontal view facial images corresponding to a facial expression of a plurality of facial expressions of the subject;

receiving one or more image(s) of the subject from the at least one imaging sensor, each of the images from the at least one imaging sensor also corresponding to a facial expression of the plurality of facial expressions of the subject; and

correlating, for each of the facial expressions, the one or more image(s) from the at least one imaging sensor corresponding to a particular facial expression to the one or more frontal view facial image(s) corresponding to the particular facial expression.

18. The wearable system of claim 17, wherein correlating for each of the facial expressions in the training of the second model comprises:

extracting a plurality of landmark positions from each of the one or more frontal view facial image(s) of the subject;

extracting a plurality of landmark positions from each of the one or more image(s) of the subject from the at least one imaging sensor;

correlating the landmark positions for a particular frontal view facial image with a particular facial expression; and

correlating the landmark positions for a particular one of the images of the subject from the at least one imaging sensor with the particular facial expression.

19. The wearable system of claim 1, further comprising a computing device that receives and displays the communicated facial expression image.

20. The wearable system of claim 1, wherein the at least one imaging sensor comprises a depth camera, a red-green-blue ("RGB") camera, an infrared ("IR") sensor, an acoustic camera, an acoustic sensor, a thermal imaging sensor, a charge-coupled device ("CCD"), a complementary metal oxide semiconductor ("CMOS") device, an active-pixel sensor, or a radar imaging sensor.

21. The wearable system of claim 1, the physical memory comprising further instructions that when executed by the processor cause the processor to:

compare the facial expression image to a second model; assign at least one of a letter or a word to the facial expression image based on a predetermined degree of correspondence of the assigned at least one of a letter or a word to a selected one of a plurality of letters or words in the second model when compared to the facial expression image, and

output text or speech corresponding to the assigned at least one of a letter or a word.

22. The wearable system of claim 1, wherein the captured image further captures an object in a vicinity of the captured facial feature of the wearer; and wherein the predetermined degree of correspondence of the image data to the selected one of the contours in the model includes correspondence based on the captured object in the image data.

23. A method to determine a facial expression using a wearable system, comprising:

positioning a first imaging sensor of a wearable system to image a first facial feature of a wearer;

imaging the first facial feature of the wearer using the first imaging sensor;

comparing, via a processor, the imaged first facial feature to a first model; and

43

assigning, via the processor, a facial expression corresponding to the imaged first facial feature selected from one of a plurality of facial expressions in the first model;

wherein the first model is trained using machine learning; 5
and

wherein the training comprises:

receiving one or more frontal view facial image(s), each of the frontal view facial images corresponding to a facial expression of a plurality of facial expressions; 10

receiving one or more image(s) from the first imaging sensor, each of the images from the first imaging sensor also corresponding to a facial expression of the plurality of facial expressions; and

correlating, for each of the facial expressions, the one or more image(s) from the first imaging sensor corresponding to a particular facial expression to the one or more frontal view facial image(s) corresponding to the particular facial expression. 15

24. The method of claim **23**, wherein the act of positioning the first imaging sensor of the wearable system comprises positioning the first imaging sensor in a vicinity of an ear of the wearer facing forward to include within a field of view at least one of a buccal region, a zygomatic region, 20
and/or a temporal region of one side of the wearer's face.

25. A method to determine a facial expression using a wearable system, comprising:

positioning a first imaging sensor of a wearable system to image a first facial feature of a wearer;

44

positioning a second imaging sensor of the wearable system to image a second facial feature of the wearer; imaging the first facial feature of the wearer using the first imaging sensor;

imaging the second facial feature of the wearer using the second imaging sensor;

comparing, via a processor, the imaged first facial feature and the imaged second facial feature to a model; and

assigning, via the processor, a facial expression selected from one of a plurality of facial expressions in the model corresponding to the imaged first facial feature and the imaged second facial feature;

wherein the model is trained using machine learning; and wherein the training comprises:

receiving one or more frontal view facial image(s), each of the frontal view facial images corresponding to a facial expression of a plurality of facial expressions;

receiving one or more image(s) from at least one of the first imaging sensor and the second imaging sensor, each of the images from the at least one of the first imaging sensor and the second imaging sensor also corresponding to a facial expression of the plurality of facial expressions; and

correlating, for each of the facial expressions, the one or more image(s) from the at least one of the first imaging sensor and the second imaging sensor corresponding to a particular facial expression to the one or more frontal view facial image(s) corresponding to the particular facial expression.

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